Grinstead and Snell's Introduction to Probability

The CHANCE Project[1]

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this question for dimensions one, two, and three. He established the remarkable result that the answer is *yes* in one and two dimensions and *no* in three dimensions.

(c) Write a program to simulate a random walk in three dimensions and see whether, from this simulation and the results of (a) and (b), you could have guessed Pólya's result.

1.2 Discrete Probability Distributions

In this book we shall study many different experiments from a probabilistic point of view. What is involved in this study will become evident as the theory is developed and examples are analyzed. However, the overall idea can be described and illustrated as follows: to each experiment that we consider there will be associated a random variable, which represents the outcome of any particular experiment. The set of possible outcomes is called the *sample space*. In the first part of this section, we will consider the case where the experiment has only finitely many possible outcomes, i.e., the sample space is finite. We will then generalize to the case that the sample space is either finite or countably infinite. This leads us to the following definition.

Random Variables and Sample Spaces

Definition 1.1 Suppose we have an experiment whose outcome depends on chance. We represent the outcome of the experiment by a capital Roman letter, such as X, called a *random variable*. The *sample space* of the experiment is the set of all possible outcomes. If the sample space is either finite or countably infinite, the random variable is said to be *discrete*. □

We generally denote a sample space by the capital Greek letter Ω. As stated above, in the correspondence between an experiment and the mathematical theory by which it is studied, the sample space Ω corresponds to the set of possible outcomes of the experiment.

We now make two additional definitions. These are subsidiary to the definition of sample space and serve to make precise some of the common terminology used in conjunction with sample spaces. First of all, we define the elements of a sample space to be *outcomes*. Second, each subset of a sample space is defined to be an *event*. Normally, we shall denote outcomes by lower case letters and events by capital letters.

Example 1.6 A die is rolled once. We let X denote the outcome of this experiment. Then the sample space for this experiment is the 6-element set

$$\Omega = \{1, 2, 3, 4, 5, 6\} ,$$

where each outcome i, for $i = 1, \ldots, 6$, corresponds to the number of dots on the face which turns up. The event

$$E = \{2, 4, 6\}$$

corresponds to the statement that the result of the roll is an even number. The event E can also be described by saying that X is even. Unless there is reason to believe the die is loaded, the natural assumption is that every outcome is equally likely. Adopting this convention means that we assign a probability of $1/6$ to each of the six outcomes, i.e., $m(i) = 1/6$, for $1 \leq i \leq 6$. □

Distribution Functions

We next describe the assignment of probabilities. The definitions are motivated by the example above, in which we assigned to each outcome of the sample space a nonnegative number such that the sum of the numbers assigned is equal to 1.

Definition 1.2 Let X be a random variable which denotes the value of the outcome of a certain experiment, and assume that this experiment has only finitely many possible outcomes. Let Ω be the sample space of the experiment (i.e., the set of all possible values of X, or equivalently, the set of all possible outcomes of the experiment.) A *distribution function* for X is a real-valued function m whose domain is Ω and which satisfies:

1. $m(\omega) \geq 0$, for all $\omega \in \Omega$, and

2. $\sum_{\omega \in \Omega} m(\omega) = 1$.

For any subset E of Ω, we define the *probability* of E to be the number $P(E)$ given by

$$P(E) = \sum_{\omega \in E} m(\omega) \ .$$

□

Example 1.7 Consider an experiment in which a coin is tossed twice. Let X be the random variable which corresponds to this experiment. We note that there are several ways to record the outcomes of this experiment. We could, for example, record the two tosses, in the order in which they occurred. In this case, we have $\Omega = \{\text{HH,HT,TH,TT}\}$. We could also record the outcomes by simply noting the number of heads that appeared. In this case, we have $\Omega = \{0,1,2\}$. Finally, we could record the two outcomes, without regard to the order in which they occurred. In this case, we have $\Omega = \{\text{HH,HT,TT}\}$.

We will use, for the moment, the first of the sample spaces given above. We will assume that all four outcomes are equally likely, and define the distribution function $m(\omega)$ by

$$m(\text{HH}) = m(\text{HT}) = m(\text{TH}) = m(\text{TT}) = \frac{1}{4} \ .$$

Let $E = \{\text{HH,HT,TH}\}$ be the event that at least one head comes up. Then, the probability of E can be calculated as follows:

$$\begin{aligned} P(E) &= m(\text{HH}) + m(\text{HT}) + m(\text{TH}) \\ &= \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = \frac{3}{4} \ . \end{aligned}$$

Similarly, if $F = \{\text{HH,HT}\}$ is the event that heads comes up on the first toss, then we have

$$\begin{aligned} P(F) &= m(\text{HH}) + m(\text{HT}) \\ &= \frac{1}{4} + \frac{1}{4} = \frac{1}{2} \ . \end{aligned}$$

□

Example 1.8 (Example 1.6 continued) The sample space for the experiment in which the die is rolled is the 6-element set $\Omega = \{1, 2, 3, 4, 5, 6\}$. We assumed that the die was fair, and we chose the distribution function defined by

$$m(i) = \frac{1}{6}, \qquad \text{for } i = 1, \ldots, 6 \ .$$

If E is the event that the result of the roll is an even number, then $E = \{2, 4, 6\}$ and

$$\begin{aligned} P(E) &= m(2) + m(4) + m(6) \\ &= \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2} \ . \end{aligned}$$

□

Notice that it is an immediate consequence of the above definitions that, for every $\omega \in \Omega$,

$$P(\{\omega\}) = m(\omega) \ .$$

That is, the probability of the elementary event $\{\omega\}$, consisting of a single outcome ω, is equal to the value $m(\omega)$ assigned to the outcome ω by the distribution function.

Example 1.9 Three people, A, B, and C, are running for the same office, and we assume that one and only one of them wins. The sample space may be taken as the 3-element set $\Omega = \{\text{A,B,C}\}$ where each element corresponds to the outcome of that

candidate's winning. Suppose that A and B have the same chance of winning, but that C has only 1/2 the chance of A or B. Then we assign

$$m(\mathrm{A}) = m(\mathrm{B}) = 2m(\mathrm{C}) \ .$$

Since

$$m(\mathrm{A}) + m(\mathrm{B}) + m(\mathrm{C}) = 1 \ ,$$

we see that

$$2m(\mathrm{C}) + 2m(\mathrm{C}) + m(\mathrm{C}) = 1 \ ,$$

which implies that $5m(\mathrm{C}) = 1$. Hence,

$$m(\mathrm{A}) = \frac{2}{5} \ , \qquad m(\mathrm{B}) = \frac{2}{5} \ , \qquad m(\mathrm{C}) = \frac{1}{5} \ .$$

Let E be the event that either A or C wins. Then $E = \{\mathrm{A,C}\}$, and

$$P(E) = m(\mathrm{A}) + m(\mathrm{C}) = \frac{2}{5} + \frac{1}{5} = \frac{3}{5} \ .$$

□

In many cases, events can be described in terms of other events through the use of the standard constructions of set theory. We will briefly review the definitions of these constructions. The reader is referred to Figure 1.7 for Venn diagrams which illustrate these constructions.

Let A and B be two sets. Then the union of A and B is the set

$$A \cup B = \{x \,|\, x \in A \text{ or } x \in B\} \ .$$

The intersection of A and B is the set

$$A \cap B = \{x \,|\, x \in A \text{ and } x \in B\} \ .$$

The difference of A and B is the set

$$A - B = \{x \,|\, x \in A \text{ and } x \notin B\} \ .$$

The set A is a subset of B, written $A \subset B$, if every element of A is also an element of B. Finally, the complement of A is the set

$$\tilde{A} = \{x \,|\, x \in \Omega \text{ and } x \notin A\} \ .$$

The reason that these constructions are important is that it is typically the case that complicated events described in English can be broken down into simpler events using these constructions. For example, if A is the event that "it will snow tomorrow and it will rain the next day," B is the event that "it will snow tomorrow," and C is the event that "it will rain two days from now," then A is the intersection of the events B and C. Similarly, if D is the event that "it will snow tomorrow or it will rain the next day," then $D = B \cup C$. (Note that care must be taken here, because sometimes the word "or" in English means that exactly one of the two alternatives will occur. The meaning is usually clear from context. In this book, we will always use the word "or" in the inclusive sense, i.e., A or B means that at least one of the two events A, B is true.) The event $\tilde{B}$ is the event that "it will not snow tomorrow." Finally, if E is the event that "it will snow tomorrow but it will not rain the next day," then $E = B - C$.

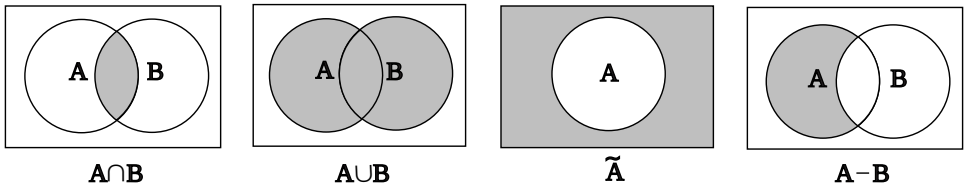


Figure 1.7: Basic set operations.

Properties

Theorem 1.1 The probabilities assigned to events by a distribution function on a sample space Ω satisfy the following properties:

1. $P(E) \geq 0$ for every $E \subset \Omega$.
2. $P(\Omega) = 1$.
3. If $E \subset F \subset \Omega$, then $P(E) \leq P(F)$.
4. If A and B are *disjoint* subsets of Ω, then $P(A \cup B) = P(A) + P(B)$.
5. $P(\tilde{A}) = 1 - P(A)$ for every $A \subset \Omega$.

Proof. For any event E the probability $P(E)$ is determined from the distribution m by

$$P(E) = \sum_{\omega \in E} m(\omega) \ ,$$

for every $E \subset \Omega$. Since the function m is nonnegative, it follows that $P(E)$ is also nonnegative. Thus, Property 1 is true.

Property 2 is proved by the equations

$$P(\Omega) = \sum_{\omega \in \Omega} m(\omega) = 1 \ .$$

Suppose that $E \subset F \subset \Omega$. Then every element ω that belongs to E also belongs to F. Therefore,

$$\sum_{\omega \in E} m(\omega) \leq \sum_{\omega \in F} m(\omega) \ ,$$

since each term in the left-hand sum is in the right-hand sum, and all the terms in both sums are non-negative. This implies that

$$P(E) \leq P(F) \ ,$$

and Property 3 is proved.

Suppose next that A and B are disjoint subsets of Ω. Then every element ω of $A \cup B$ lies either in A and not in B or in B and not in A. It follows that

$$\begin{aligned} P(A \cup B) &= \sum_{\omega \in A \cup B} m(\omega) = \sum_{\omega \in A} m(\omega) + \sum_{\omega \in B} m(\omega) \\ &= P(A) + P(B) \ , \end{aligned}$$

and Property 4 is proved.

Finally, to prove Property 5, consider the disjoint union

$$\Omega = A \cup \tilde{A} \ .$$

Since $P(\Omega) = 1$, the property of disjoint additivity (Property 4) implies that

$$1 = P(A) + P(\tilde{A}) \ ,$$

whence $P(\tilde{A}) = 1 - P(A)$. □

It is important to realize that Property 4 in Theorem 1.1 can be extended to more than two sets. The general finite additivity property is given by the following theorem.

Theorem 1.2 If $A_1, \ldots, A_n$ are pairwise disjoint subsets of Ω (i.e., no two of the A_i's have an element in common), then

$$P(A_1 \cup \cdots \cup A_n) = \sum_{i=1}^{n} P(A_i) \ .$$

Proof. Let ω be any element in the union

$$A_1 \cup \cdots \cup A_n \ .$$

Then $m(\omega)$ occurs exactly once on each side of the equality in the statement of the theorem. □

We shall often use the following consequence of the above theorem.

Theorem 1.3 Let $A_1, \ldots, A_n$ be pairwise disjoint events with $\Omega = A_1 \cup \cdots \cup A_n$, and let E be any event. Then

$$P(E) = \sum_{i=1}^{n} P(E \cap A_i) \ .$$

Proof. The sets $E \cap A_1, \ldots, E \cap A_n$ are pairwise disjoint, and their union is the set E. The result now follows from Theorem 1.2. □

Corollary 1.1 For any two events A and B,

$$P(A) = P(A \cap B) + P(A \cap \tilde{B}) \ .$$

□

Property 4 can be generalized in another way. Suppose that A and B are subsets of Ω which are not necessarily disjoint. Then:

Theorem 1.4 If A and B are subsets of Ω, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \ . \tag{1.1}$$

Proof. The left side of Equation 1.1 is the sum of $m(\omega)$ for ω in either A or B. We must show that the right side of Equation 1.1 also adds $m(\omega)$ for ω in A or B. If ω is in exactly one of the two sets, then it is counted in only one of the three terms on the right side of Equation 1.1. If it is in both A and B, it is added twice from the calculations of $P(A)$ and $P(B)$ and subtracted once for $P(A \cap B)$. Thus it is counted exactly once by the right side. Of course, if $A \cap B = \emptyset$, then Equation 1.1 reduces to Property 4. (Equation 1.1 can also be generalized; see Theorem 3.8.) □

Tree Diagrams

Example 1.10 Let us illustrate the properties of probabilities of events in terms of three tosses of a coin. When we have an experiment which takes place in stages such as this, we often find it convenient to represent the outcomes by a *tree diagram* as shown in Figure 1.8.

A *path* through the tree corresponds to a possible outcome of the experiment. For the case of three tosses of a coin, we have eight paths ω_1, ω_2, ..., ω_8 and, assuming each outcome to be equally likely, we assign equal weight, 1/8, to each path. Let E be the event "at least one head turns up." Then $\tilde{E}$ is the event "no heads turn up." This event occurs for only one outcome, namely, $\omega_8 = \text{TTT}$. Thus, $\tilde{E} = \{\text{TTT}\}$ and we have

$$P(\tilde{E}) = P(\{\text{TTT}\}) = m(\text{TTT}) = \frac{1}{8} \ .$$

By Property 5 of Theorem 1.1,

$$P(E) = 1 - P(\tilde{E}) = 1 - \frac{1}{8} = \frac{7}{8} \ .$$

Note that we shall often find it is easier to compute the probability that an event does not happen rather than the probability that it does. We then use Property 5 to obtain the desired probability.

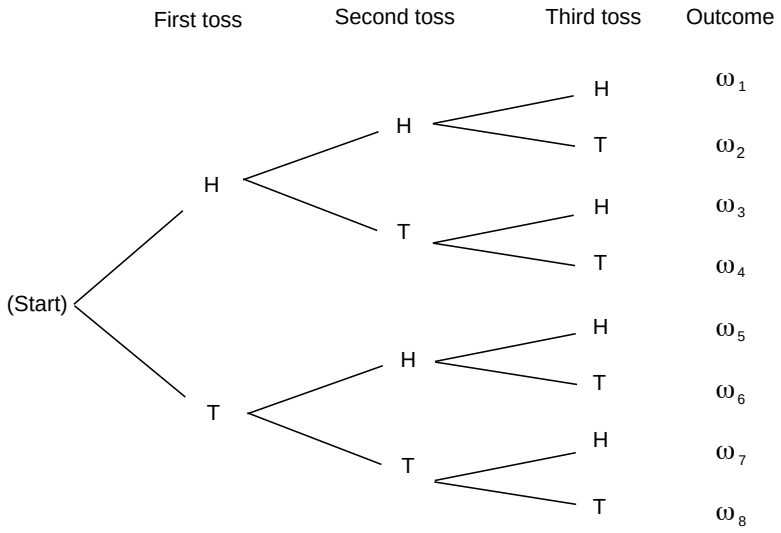


Figure 1.8: Tree diagram for three tosses of a coin.

Let A be the event "the first outcome is a head," and B the event "the second outcome is a tail." By looking at the paths in Figure 1.8, we see that

$$P(A) = P(B) = \frac{1}{2} \ .$$

Moreover, $A \cap B = \{\omega_3, \omega_4\}$, and so $P(A \cap B) = 1/4$. Using Theorem 1.4, we obtain

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= \frac{1}{2} + \frac{1}{2} - \frac{1}{4} = \frac{3}{4} \ . \end{aligned}$$

Since $A \cup B$ is the 6-element set,

$$A \cup B = \{\text{HHH,HHT,HTH,HTT,TTH,TTT}\} \ ,$$

we see that we obtain the same result by direct enumeration. □

In our coin tossing examples and in the die rolling example, we have assigned an equal probability to each possible outcome of the experiment. Corresponding to this method of assigning probabilities, we have the following definitions.

Uniform Distribution

Definition 1.3 The *uniform distribution* on a sample space Ω containing n elements is the function m defined by

$$m(\omega) = \frac{1}{n} \ ,$$

for every $\omega \in \Omega$. □

It is important to realize that when an experiment is analyzed to describe its possible outcomes, there is no single correct choice of sample space. For the experiment of tossing a coin twice in Example 1.2, we selected the 4-element set Ω ={HH,HT,TH,TT} as a sample space and assigned the uniform distribution function. These choices are certainly intuitively natural. On the other hand, for some purposes it may be more useful to consider the 3-element sample space $\bar{\Omega} = \{0, 1, 2\}$ in which 0 is the outcome "no heads turn up," 1 is the outcome "exactly one head turns up," and 2 is the outcome "two heads turn up." The distribution function $\bar{m}$ on $\bar{\Omega}$ defined by the equations

$$\bar{m}(0) = \frac{1}{4} , \qquad \bar{m}(1) = \frac{1}{2} , \qquad \bar{m}(2) = \frac{1}{4}$$

is the one corresponding to the uniform probability density on the original sample space Ω. Notice that it is perfectly possible to choose a different distribution function. For example, we may consider the uniform distribution function on $\bar{\Omega}$, which is the function $\bar{q}$ defined by

$$\bar{q}(0) = \bar{q}(1) = \bar{q}(2) = \frac{1}{3} .$$

Although $\bar{q}$ is a perfectly good distribution function, it is not consistent with observed data on coin tossing.

Example 1.11 Consider the experiment that consists of rolling a pair of dice. We take as the sample space Ω the set of all ordered pairs (i, j) of integers with $1 \leq i \leq 6$ and $1 \leq j \leq 6$. Thus,

$$\Omega = \{ (i, j) : 1 \leq i, j \leq 6 \} .$$

(There is at least one other "reasonable" choice for a sample space, namely the set of all unordered pairs of integers, each between 1 and 6. For a discussion of why we do not use this set, see Example 3.14.) To determine the size of Ω, we note that there are six choices for i, and for each choice of i there are six choices for j, leading to 36 different outcomes. Let us assume that the dice are not loaded. In mathematical terms, this means that we assume that each of the 36 outcomes is equally likely, or equivalently, that we adopt the uniform distribution function on Ω by setting

$$m((i, j)) = \frac{1}{36}, \qquad 1 \leq i, j \leq 6 .$$

What is the probability of getting a sum of 7 on the roll of two dice—or getting a sum of 11? The first event, denoted by E, is the subset

$$E = \{(1, 6), (6, 1), (2, 5), (5, 2), (3, 4), (4, 3)\} .$$

A sum of 11 is the subset F given by

$$F = \{(5, 6), (6, 5)\} .$$

Consequently,

$$P(E) = \sum_{\omega \in E} m(\omega) = 6 \cdot \tfrac{1}{36} = \tfrac{1}{6} ,$$

$$P(F) = \sum_{\omega \in F} m(\omega) = 2 \cdot \tfrac{1}{36} = \tfrac{1}{18} .$$

What is the probability of getting neither *snakeeyes* (double ones) nor *boxcars* (double sixes)? The event of getting either one of these two outcomes is the set

$$E = \{(1,1),(6,6)\} \ .$$

Hence, the probability of obtaining neither is given by

$$P(\tilde{E}) = 1 - P(E) = 1 - \frac{2}{36} = \frac{17}{18} \ .$$

□

In the above coin tossing and the dice rolling experiments, we have assigned an equal probability to each outcome. That is, in each example, we have chosen the uniform distribution function. These are the natural choices provided the coin is a fair one and the dice are not loaded. However, the decision as to which distribution function to select to describe an experiment is *not* a part of the basic mathematical theory of probability. The latter begins only when the sample space and the distribution function have already been defined.

Determination of Probabilities

It is important to consider ways in which probability distributions are determined in practice. One way is by *symmetry*. For the case of the toss of a coin, we do not see any physical difference between the two sides of a coin that should affect the chance of one side or the other turning up. Similarly, with an ordinary die there is no essential difference between any two sides of the die, and so by symmetry we assign the same probability for any possible outcome. In general, considerations of symmetry often suggest the uniform distribution function. Care must be used here. We should not always assume that, just because we do not know any reason to suggest that one outcome is more likely than another, it is appropriate to assign equal probabilities. For example, consider the experiment of guessing the sex of a newborn child. It has been observed that the proportion of newborn children who are boys is about .513. Thus, it is more appropriate to assign a distribution function which assigns probability .513 to the outcome *boy* and probability .487 to the outcome *girl* than to assign probability 1/2 to each outcome. This is an example where we use statistical observations to determine probabilities. Note that these probabilities may change with new studies and may vary from country to country. Genetic engineering might even allow an individual to influence this probability for a particular case.

Odds

Statistical estimates for probabilities are fine if the experiment under consideration can be repeated a number of times under similar circumstances. However, assume that, at the beginning of a football season, you want to assign a probability to the event that Dartmouth will beat Harvard. You really do not have data that relates to this year's football team. However, you can determine your own personal probability

by seeing what kind of a bet you would be willing to make. For example, suppose that you are willing to make a 1 dollar bet giving 2 to 1 odds that Dartmouth will win. Then you are willing to pay 2 dollars if Dartmouth loses in return for receiving 1 dollar if Dartmouth wins. This means that you think the appropriate probability for Dartmouth winning is 2/3.

Let us look more carefully at the relation between odds and probabilities. Suppose that we make a bet at r to 1 odds that an event E occurs. This means that we think that it is r times as likely that E will occur as that E will not occur. In general, r to s odds will be taken to mean the same thing as r/s to 1, i.e., the ratio between the two numbers is the only quantity of importance when stating odds.

Now if it is r times as likely that E will occur as that E will not occur, then the probability that E occurs must be $r/(r+1)$, since we have

$$P(E) = r\,P(\tilde{E})$$

and

$$P(E) + P(\tilde{E}) = 1\ .$$

In general, the statement that the odds are r to s in favor of an event E occurring is equivalent to the statement that

$$\begin{aligned} P(E) &= \frac{r/s}{(r/s)+1} \\ &= \frac{r}{r+s}\ . \end{aligned}$$

If we let $P(E) = p$, then the above equation can easily be solved for r/s in terms of p; we obtain $r/s = p/(1-p)$. We summarize the above discussion in the following definition.

Definition 1.4 If $P(E) = p$, the *odds* in favor of the event E occurring are $r : s$ (r to s) where $r/s = p/(1-p)$. If r and s are given, then p can be found by using the equation $p = r/(r+s)$. □

Example 1.12 (Example 1.9 continued) In Example 1.9 we assigned probability 1/5 to the event that candidate C wins the race. Thus the odds in favor of C winning are $1/5 : 4/5$. These odds could equally well have been written as $1 : 4$, $2 : 8$, and so forth. A bet that C wins is fair if we receive 4 dollars if C wins and pay 1 dollar if C loses. □

Infinite Sample Spaces

If a sample space has an infinite number of points, then the way that a distribution function is defined depends upon whether or not the sample space is countable. A sample space is *countably infinite* if the elements can be counted, i.e., can be put

in one-to-one correspondence with the positive integers, and *uncountably infinite* otherwise. Infinite sample spaces require new concepts in general (see Chapter 2), but countably infinite spaces do not. If

$$\Omega = \{\omega_1, \omega_2, \omega_3, \ldots\}$$

is a countably infinite sample space, then a distribution function is defined exactly as in Definition 1.2, except that the sum must now be a *convergent* infinite sum. Theorem 1.1 is still true, as are its extensions Theorems 1.2 and 1.4. One thing we cannot do on a countably infinite sample space that we could do on a finite sample space is to define a *uniform* distribution function as in Definition 1.3. You are asked in Exercise 20 to explain why this is not possible.

Example 1.13 A coin is tossed until the first time that a head turns up. Let the outcome of the experiment, ω, be the first time that a head turns up. Then the possible outcomes of our experiment are

$$\Omega = \{1, 2, 3, \ldots\} \ .$$

Note that even though the coin could come up tails every time we have not allowed for this possibility. We will explain why in a moment. The probability that heads comes up on the first toss is 1/2. The probability that tails comes up on the first toss and heads on the second is 1/4. The probability that we have two tails followed by a head is 1/8, and so forth. This suggests assigning the distribution function $m(n) = 1/2^n$ for $n = 1, 2, 3, \ldots$. To see that this is a distribution function we must show that

$$\sum_{\omega} m(\omega) = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots = 1 \ .$$

That this is true follows from the formula for the sum of a geometric series,

$$1 + r + r^2 + r^3 + \cdots = \frac{1}{1-r} \ ,$$

or

$$r + r^2 + r^3 + r^4 + \cdots = \frac{r}{1-r} \ , \tag{1.2}$$

for $-1 < r < 1$.

Putting $r = 1/2$, we see that we have a probability of 1 that the coin eventually turns up heads. The possible outcome of tails every time has to be assigned probability 0, so we omit it from our sample space of possible outcomes.

Let E be the event that the first time a head turns up is after an even number of tosses. Then

$$E = \{2, 4, 6, 8, \ldots\} \ ,$$

and

$$P(E) = \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \cdots \ .$$

Putting $r = 1/4$ in Equation 1.2 see that

$$P(E) = \frac{1/4}{1 - 1/4} = \frac{1}{3} \ .$$

Thus the probability that a head turns up for the first time after an even number of tosses is 1/3 and after an odd number of tosses is 2/3. □

Historical Remarks

An interesting question in the history of science is: Why was probability not developed until the sixteenth century? We know that in the sixteenth century problems in gambling and games of chance made people start to think about probability. But gambling and games of chance are almost as old as civilization itself. In ancient Egypt (at the time of the First Dynasty, ca. 3500 B.C.) a game now called "Hounds and Jackals" was played. In this game the movement of the hounds and jackals was based on the outcome of the roll of four-sided dice made out of animal bones called astragali. Six-sided dice made of a variety of materials date back to the sixteenth century B.C. Gambling was widespread in ancient Greece and Rome. Indeed, in the Roman Empire it was sometimes found necessary to invoke laws against gambling. Why, then, were probabilities not calculated until the sixteenth century?

Several explanations have been advanced for this late development. One is that the relevant mathematics was not developed and was not easy to develop. The ancient mathematical notation made numerical calculation complicated, and our familiar algebraic notation was not developed until the sixteenth century. However, as we shall see, many of the combinatorial ideas needed to calculate probabilities were discussed long before the sixteenth century. Since many of the chance events of those times had to do with lotteries relating to religious affairs, it has been suggested that there may have been religious barriers to the study of chance and gambling. Another suggestion is that a stronger incentive, such as the development of commerce, was necessary. However, none of these explanations seems completely satisfactory, and people still wonder why it took so long for probability to be studied seriously. An interesting discussion of this problem can be found in Hacking.[14]

The first person to calculate probabilities systematically was Gerolamo Cardano (1501–1576) in his book *Liber de Ludo Aleae.* This was translated from the Latin by Gould and appears in the book *Cardano: The Gambling Scholar* by Ore.[15] Ore provides a fascinating discussion of the life of this colorful scholar with accounts of his interests in many different fields, including medicine, astrology, and mathematics. You will also find there a detailed account of Cardano's famous battle with Tartaglia over the solution to the cubic equation.

In his book on probability Cardano dealt only with the special case that we have called the uniform distribution function. This restriction to equiprobable outcomes was to continue for a long time. In this case Cardano realized that the probability that an event occurs is the ratio of the number of favorable outcomes to the total number of outcomes.

Many of Cardano's examples dealt with rolling dice. Here he realized that the outcomes for two rolls should be taken to be the 36 ordered pairs (i, j) rather than the 21 unordered pairs. This is a subtle point that was still causing problems much

[14] I. Hacking, *The Emergence of Probability* (Cambridge: Cambridge University Press, 1975).
[15] O. Ore, *Cardano: The Gambling Scholar* (Princeton: Princeton University Press, 1953).

later for other writers on probability. For example, in the eighteenth century the famous French mathematician d'Alembert, author of several works on probability, claimed that when a coin is tossed twice the number of heads that turn up would be 0, 1, or 2, and hence we should assign equal probabilities for these three possible outcomes.[16] Cardano chose the correct sample space for his dice problems and calculated the correct probabilities for a variety of events.

Cardano's mathematical work is interspersed with a lot of advice to the potential gambler in short paragraphs, entitled, for example: "Who Should Play and When," "Why Gambling Was Condemned by Aristotle," "Do Those Who Teach Also Play Well?" and so forth. In a paragraph entitled "The Fundamental Principle of Gambling," Cardano writes:

> The most fundamental principle of all in gambling is simply equal conditions, e.g., of opponents, of bystanders, of money, of situation, of the dice box, and of the die itself. To the extent to which you depart from that equality, if it is in your opponent's favor, you are a fool, and if in your own, you are unjust.[17]

Cardano did make mistakes, and if he realized it later he did not go back and change his error. For example, for an event that is favorable in three out of four cases, Cardano assigned the correct odds $3:1$ that the event will occur. But then he assigned odds by squaring these numbers (i.e., $9:1$) for the event to happen twice in a row. Later, by considering the case where the odds are $1:1$, he realized that this cannot be correct and was led to the correct result that when f out of n outcomes are favorable, the odds for a favorable outcome twice in a row are $f^2 : n^2 - f^2$. Ore points out that this is equivalent to the realization that if the probability that an event happens in one experiment is p, the probability that it happens twice is p^2. Cardano proceeded to establish that for three successes the formula should be p^3 and for four successes p^4, making it clear that he understood that the probability is p^n for n successes in n independent repetitions of such an experiment. This will follow from the concept of independence that we introduce in Section 4.1.

Cardano's work was a remarkable first attempt at writing down the laws of probability, but it was not the spark that started a systematic study of the subject. This came from a famous series of letters between Pascal and Fermat. This correspondence was initiated by Pascal to consult Fermat about problems he had been given by Chevalier de Méré, a well-known writer, a prominent figure at the court of Louis XIV, and an ardent gambler.

The first problem de Méré posed was a dice problem. The story goes that he had been betting that at least one six would turn up in four rolls of a die and winning too often, so he then bet that a pair of sixes would turn up in 24 rolls of a pair of dice. The probability of a six with one die is $1/6$ and, by the product law for independent experiments, the probability of two sixes when a pair of dice is thrown is $(1/6)(1/6) = 1/36$. Ore[18] claims that a gambling rule of the time suggested that,

[16] J. d'Alembert, "Croix ou Pile," in *L'Encyclopédie,* ed. Diderot, vol. 4 (Paris, 1754).

[17] O. Ore, op. cit., p. 189.

[18] O. Ore, "Pascal and the Invention of Probability Theory," *American Mathematics Monthly*,

since four repetitions was favorable for the occurrence of an event with probability 1/6, for an event six times as unlikely, $6 \cdot 4 = 24$ repetitions would be sufficient for a favorable bet. Pascal showed, by exact calculation, that 25 rolls are required for a favorable bet for a pair of sixes.

The second problem was a much harder one: it was an old problem and concerned the determination of a fair division of the stakes in a tournament when the series, for some reason, is interrupted before it is completed. This problem is now referred to as the problem of points. The problem had been a standard problem in mathematical texts; it appeared in Fra Luca Paccioli's book *summa de Arithmetica, Geometria, Proportioni et Proportionalità*, printed in Venice in 1494,[19] in the form:

> A team plays ball such that a total of 60 points are required to win the game, and each inning counts 10 points. The stakes are 10 ducats. By some incident they cannot finish the game and one side has 50 points and the other 20. One wants to know what share of the prize money belongs to each side. In this case I have found that opinions differ from one to another but all seem to me insufficient in their arguments, but I shall state the truth and give the correct way.

Reasonable solutions, such as dividing the stakes according to the ratio of games won by each player, had been proposed, but no correct solution had been found at the time of the Pascal-Fermat correspondence. The letters deal mainly with the attempts of Pascal and Fermat to solve this problem. Blaise Pascal (1623–1662) was a child prodigy, having published his treatise on conic sections at age sixteen, and having invented a calculating machine at age eighteen. At the time of the letters, his demonstration of the weight of the atmosphere had already established his position at the forefront of contemporary physicists. Pierre de Fermat (1601–1665) was a learned jurist in Toulouse, who studied mathematics in his spare time. He has been called by some the prince of amateurs and one of the greatest pure mathematicians of all times.

The letters, translated by Maxine Merrington, appear in Florence David's fascinating historical account of probability, *Games, Gods and Gambling*.[20] In a letter dated Wednesday, 29th July, 1654, Pascal writes to Fermat:

> Sir,
>
> Like you, I am equally impatient, and although I am again ill in bed, I cannot help telling you that yesterday evening I received from M. de Carcavi your letter on the problem of points, which I admire more than I can possibly say. I have not the leisure to write at length, but, in a word, you have solved the two problems of points, one with dice and the other with sets of games with perfect justness; I am entirely satisfied with it for I do not doubt that I was in the wrong, seeing the admirable agreement in which I find myself with you now...

vol. 67 (1960), pp. 409–419.

[19] ibid., p. 414.

[20] F. N. David, *Games, Gods and Gambling* (London: G. Griffin, 1962), p. 230 ff.

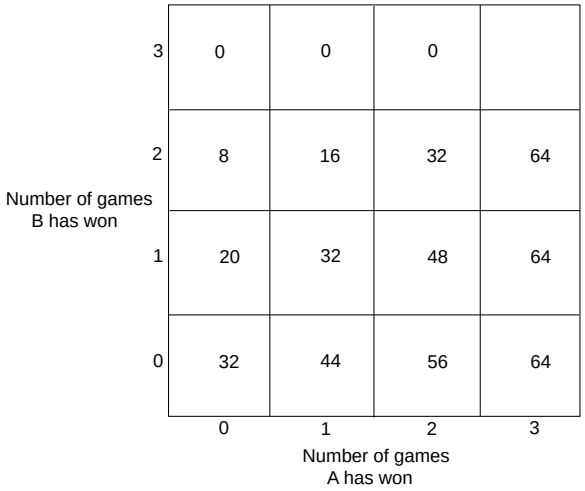


Figure 1.9: Pascal's table.

> Your method is very sound and is the one which first came to my mind in this research; but because the labour of the combination is excessive, I have found a short cut and indeed another method which is much quicker and neater, which I would like to tell you here in a few words: for henceforth I would like to open my heart to you, if I may, as I am so overjoyed with our agreement. I see that truth is the same in Toulouse as in Paris.
>
> Here, more or less, is what I do to show the fair value of each game, when two opponents play, for example, in three games and each person has staked 32 pistoles.
>
> Let us say that the first man had won twice and the other once; now they play another game, in which the conditions are that, if the first wins, he takes all the stakes; that is 64 pistoles; if the other wins it, then they have each won two games, and therefore, if they wish to stop playing, they must each take back their own stake, that is, 32 pistoles each.
>
> Then consider, Sir, if the first man wins, he gets 64 pistoles; if he loses he gets 32. Thus if they do not wish to risk this last game but wish to separate without playing it, the first man must say: 'I am certain to get 32 pistoles, even if I lost I still get them; but as for the other 32, perhaps I will get them, perhaps you will get them, the chances are equal. Let us then divide these 32 pistoles in half and give one half to me as well as my 32 which are mine for sure.' He will then have 48 pistoles and the other 16...

Pascal's argument produces the table illustrated in Figure 1.9 for the amount due player A at any quitting point.

Each entry in the table is the average of the numbers just above and to the right of the number. This fact, together with the known values when the tournament is completed, determines all the values in this table. If player A wins the first game, then he needs two games to win and B needs three games to win; and so, if the tounament is called off, A should receive 44 pistoles.

The letter in which Fermat presented his solution has been lost; but fortunately, Pascal describes Fermat's method in a letter dated Monday, 24th August, 1654. From Pascal's letter:[21]

> This is your procedure when there are two players: If two players, playing several games, find themselves in that position when the first man needs *two* games and second needs *three*, then to find the fair division of stakes, you say that one must know in how many games the play will be absolutely decided.
>
> It is easy to calculate that this will be in *four* games, from which you can conclude that it is necessary to see in how many ways four games can be arranged between two players, and one must see how many combinations would make the first man win and how many the second and to share out the stakes in this proportion. I would have found it difficult to understand this if I had not known it myself already; in fact you had explained it with this idea in mind.

Fermat realized that the number of ways that the game might be finished may not be equally likely. For example, if A needs two more games and B needs three to win, two possible ways that the tournament might go for A to win are WLW and LWLW. These two sequences do not have the same chance of occurring. To avoid this difficulty, Fermat extended the play, adding fictitious plays, so that all the ways that the games might go have the same length, namely four. He was shrewd enough to realize that this extension would not change the winner and that he now could simply count the number of sequences favorable to each player since he had made them all equally likely. If we list all possible ways that the extended game of four plays might go, we obtain the following 16 possible outcomes of the play:

WWWW	WLWW	LWWW	LLWW
WWWL	WLWL	LWWL	LLWL
WWLW	WLLW	LWLW	LLLW
WWLL	WLLL	LWLL	LLLL .

Player A wins in the cases where there are at least two wins (the 11 underlined cases), and B wins in the cases where there are at least three losses (the other 5 cases). Since A wins in 11 of the 16 possible cases Fermat argued that the probability that A wins is 11/16. If the stakes are 64 pistoles, A should receive 44 pistoles in agreement with Pascal's result. Pascal and Fermat developed more systematic methods for counting the number of favorable outcomes for problems

[21] ibid., p. 239ff.

like this, and this will be one of our central problems. Such counting methods fall under the subject of *combinatorics*, which is the topic of Chapter 3.

We see that these two mathematicians arrived at two very different ways to solve the problem of points. Pascal's method was to develop an algorithm and use it to calculate the fair division. This method is easy to implement on a computer and easy to generalize. Fermat's method, on the other hand, was to change the problem into an equivalent problem for which he could use counting or combinatorial methods. We will see in Chapter 3 that, in fact, Fermat used what has become known as Pascal's triangle! In our study of probability today we shall find that both the algorithmic approach and the combinatorial approach share equal billing, just as they did 300 years ago when probability got its start.

Exercises

1 Let $\Omega = \{a, b, c\}$ be a sample space. Let $m(a) = 1/2$, $m(b) = 1/3$, and $m(c) = 1/6$. Find the probabilities for all eight subsets of Ω.

2 Give a possible sample space Ω for each of the following experiments:

(a) An election decides between two candidates A and B.

(b) A two-sided coin is tossed.

(c) A student is asked for the month of the year and the day of the week on which her birthday falls.

(d) A student is chosen at random from a class of ten students.

(e) You receive a grade in this course.

3 For which of the cases in Exercise 2 would it be reasonable to assign the uniform distribution function?

4 Describe in words the events specified by the following subsets of

$$\Omega = \{HHH,\ HHT,\ HTH,\ HTT,\ THH,\ THT,\ TTH,\ TTT\}$$

(see Example 1.6).

(a) $E = \{$HHH,HHT,HTH,HTT$\}$.

(b) $E = \{$HHH,TTT$\}$.

(c) $E = \{$HHT,HTH,THH$\}$.

(d) $E = \{$HHT,HTH,HTT,THH,THT,TTH,TTT$\}$.

5 What are the probabilities of the events described in Exercise 4?

6 A die is loaded in such a way that the probability of each face turning up is proportional to the number of dots on that face. (For example, a six is three times as probable as a two.) What is the probability of getting an even number in one throw?

7 Let A and B be events such that $P(A \cap B) = 1/4$, $P(\tilde{A}) = 1/3$, and $P(B) = 1/2$. What is $P(A \cup B)$?

8 A student must choose one of the subjects, art, geology, or psychology, as an elective. She is equally likely to choose art or psychology and twice as likely to choose geology. What are the respective probabilities that she chooses art, geology, and psychology?

9 A student must choose exactly two out of three electives: art, French, and mathematics. He chooses art with probability 5/8, French with probability 5/8, and art and French together with probability 1/4. What is the probability that he chooses mathematics? What is the probability that he chooses either art or French?

10 For a bill to come before the president of the United States, it must be passed by both the House of Representatives and the Senate. Assume that, of the bills presented to these two bodies, 60 percent pass the House, 80 percent pass the Senate, and 90 percent pass at least one of the two. Calculate the probability that the next bill presented to the two groups will come before the president.

11 What odds should a person give in favor of the following events?

(a) A card chosen at random from a 52-card deck is an ace.

(b) Two heads will turn up when a coin is tossed twice.

(c) Boxcars (two sixes) will turn up when two dice are rolled.

12 You offer 3 : 1 odds that your friend Smith will be elected mayor of your city. What probability are you assigning to the event that Smith wins?

13 In a horse race, the odds that Romance will win are listed as 2 : 3 and that Downhill will win are 1 : 2. What odds should be given for the event that either Romance or Downhill wins?

14 Let X be a random variable with distribution function $m_X(x)$ defined by

$$m_X(-1) = 1/5, \quad m_X(0) = 1/5, \quad m_X(1) = 2/5, \quad m_X(2) = 1/5 \ .$$

(a) Let Y be the random variable defined by the equation $Y = X + 3$. Find the distribution function $m_Y(y)$ of Y.

(b) Let Z be the random variable defined by the equation $Z = X^2$. Find the distribution function $m_Z(z)$ of Z.

***15** John and Mary are taking a mathematics course. The course has only three grades: A, B, and C. The probability that John gets a B is .3. The probability that Mary gets a B is .4. The probability that neither gets an A but at least one gets a B is .1. What is the probability that at least one gets a B but neither gets a C?

16 In a fierce battle, not less than 70 percent of the soldiers lost one eye, not less than 75 percent lost one ear, not less than 80 percent lost one hand, and not less than 85 percent lost one leg. What is the minimal possible percentage of those who simultaneously lost one ear, one eye, one hand, and one leg?[22]

***17** Assume that the probability of a "success" on a single experiment with n outcomes is $1/n$. Let m be the number of experiments necessary to make it a favorable bet that at least one success will occur (see Exercise 1.1.5).

(a) Show that the probability that, in m trials, there are no successes is $(1 - 1/n)^m$.

(b) (de Moivre) Show that if $m = n \log 2$ then

$$\lim_{n\to\infty} \left(1 - \frac{1}{n}\right)^m = \frac{1}{2} .$$

Hint:

$$\lim_{n\to\infty} \left(1 - \frac{1}{n}\right)^n = e^{-1} .$$

Hence for large n we should choose m to be about $n \log 2$.

(c) Would DeMoivre have been led to the correct answer for de Méré's two bets if he had used his approximation?

18 (a) For events $A_1, \ldots, A_n$, prove that

$$P(A_1 \cup \cdots \cup A_n) \leq P(A_1) + \cdots + P(A_n) .$$

(b) For events A and B, prove that

$$P(A \cap B) \geq P(A) + P(B) - 1.$$

19 If A, B, and C are any three events, show that

$$\begin{aligned} P(A \cup B \cup C) &= P(A) + P(B) + P(C) \\ &\quad - P(A \cap B) - P(B \cap C) - P(C \cap A) \\ &\quad + P(A \cap B \cap C) . \end{aligned}$$

20 Explain why it is not possible to define a uniform distribution function (see Definition 1.3) on a countably infinite sample space. *Hint*: Assume $m(\omega) = a$ for all ω, where $0 \leq a \leq 1$. Does $m(\omega)$ have all the properties of a distribution function?

21 In Example 1.13 find the probability that the coin turns up heads for the first time on the tenth, eleventh, or twelfth toss.

[22] See Knot X, in Lewis Carroll, *Mathematical Recreations,* vol. 2 (Dover, 1958).

22 A die is rolled until the first time that a six turns up. We shall see that the probability that this occurs on the nth roll is $(5/6)^{n-1} \cdot (1/6)$. Using this fact, describe the appropriate infinite sample space and distribution function for the experiment of rolling a die until a six turns up for the first time. Verify that for your distribution function $\sum_\omega m(\omega) = 1$.

23 Let Ω be the sample space

$$\Omega = \{0, 1, 2, \ldots\} ,$$

and define a distribution function by

$$m(j) = (1 - r)^j r ,$$

for some fixed r, $0 < r < 1$, and for $j = 0, 1, 2, \ldots$. Show that this is a distribution function for Ω.

24 Our calendar has a 400-year cycle. B. H. Brown noticed that the number of times the thirteenth of the month falls on each of the days of the week in the 4800 months of a cycle is as follows:

Sunday 687

Monday 685

Tuesday 685

Wednesday 687

Thursday 684

Friday 688

Saturday 684

From this he deduced that the thirteenth was more likely to fall on Friday than on any other day. Explain what he meant by this.

25 Tversky and Kahneman[23] asked a group of subjects to carry out the following task. They are told that:

> Linda is 31, single, outspoken, and very bright. She majored in philosophy in college. As a student, she was deeply concerned with racial discrimination and other social issues, and participated in anti-nuclear demonstrations.

The subjects are then asked to rank the likelihood of various alternatives, such as:
(1) Linda is active in the feminist movement.
(2) Linda is a bank teller.
(3) Linda is a bank teller and active in the feminist movement.

[23] K. McKean, "Decisions, Decisions," pp. 22–31.

Tversky and Kahneman found that between 85 and 90 percent of the subjects rated alternative (1) most likely, but alternative (3) more likely than alternative (2). Is it? They call this phenomenon the *conjunction fallacy,* and note that it appears to be unaffected by prior training in probability or statistics. Is this phenomenon a fallacy? If so, why? Can you give a possible explanation for the subjects' choices?

26 Two cards are drawn successively from a deck of 52 cards. Find the probability that the second card is higher in rank than the first card. *Hint*: Show that $1 = P(\text{higher}) + P(\text{lower}) + P(\text{same})$ and use the fact that $P(\text{higher}) = P(\text{lower})$.

27 A *life table* is a table that lists for a given number of births the estimated number of people who will live to a given age. In Appendix C we give a life table based upon 100,000 births for ages from 0 to 85, both for women and for men. Show how from this table you can estimate the probability $m(x)$ that a person born in 1981 would live to age x. Write a program to plot $m(x)$ both for men and for women, and comment on the differences that you see in the two cases.

***28** Here is an attempt to get around the fact that we cannot choose a "random integer."

(a) What, intuitively, is the probability that a "randomly chosen" positive integer is a multiple of 3?

(b) Let $P_3(N)$ be the probability that an integer, chosen at random between 1 and N, is a multiple of 3 (since the sample space is finite, this is a legitimate probability). Show that the limit

$$P_3 = \lim_{N\to\infty} P_3(N)$$

exists and equals 1/3. This formalizes the intuition in (a), and gives us a way to assign "probabilities" to certain events that are infinite subsets of the positive integers.

(c) If A is any set of positive integers, let $A(N)$ mean the number of elements of A which are less than or equal to N. Then define the "probability" of A as

$$P(A) = \lim_{N\to\infty} A(N)/N \ ,$$

provided this limit exists. Show that this definition would assign probability 0 to any finite set and probability 1 to the set of all positive integers. Thus, the probability of the set of all integers is not the sum of the probabilities of the individual integers in this set. This means that the definition of probability given here is not a completely satisfactory definition.

(d) Let A be the set of all positive integers with an odd number of digits. Show that $P(A)$ does not exist. This shows that under the above definition of probability, not all sets have probabilities.

29 (from Sholander[24]) In a standard clover-leaf interchange, there are four ramps for making right-hand turns, and inside these four ramps, there are four more ramps for making left-hand turns. Your car approaches the interchange from the south. A mechanism has been installed so that at each point where there exists a choice of directions, the car turns to the right with fixed probability r.

(a) If $r = 1/2$, what is your chance of emerging from the interchange going west?

(b) Find the value of r that maximizes your chance of a westward departure from the interchange.

30 (from Benkoski[25]) Consider a "pure" cloverleaf interchange in which there are no ramps for right-hand turns, but only the two intersecting straight highways with cloverleaves for left-hand turns. (Thus, to turn right in such an interchange, one must make three left-hand turns.) As in the preceding problem, your car approaches the interchange from the south. What is the value of r that maximizes your chances of an eastward departure from the interchange?

31 (from vos Savant[26]) A reader of Marilyn vos Savant's column wrote in with the following question:

> My dad heard this story on the radio. At Duke University, two students had received A's in chemistry all semester. But on the night before the final exam, they were partying in another state and didn't get back to Duke until it was over. Their excuse to the professor was that they had a flat tire, and they asked if they could take a make-up test. The professor agreed, wrote out a test and sent the two to separate rooms to take it. The first question (on one side of the paper) was worth 5 points, and they answered it easily. Then they flipped the paper over and found the second question, worth 95 points: 'Which tire was it?' What was the probability that both students would say the same thing? My dad and I think it's 1 in 16. Is that right?"

(a) Is the answer 1/16?

(b) The following question was asked of a class of students. "I was driving to school today, and one of my tires went flat. Which tire do you think it was?" The responses were as follows: right front, 58%, left front, 11%, right rear, 18%, left rear, 13%. Suppose that this distribution holds in the general population, and assume that the two test-takers are randomly chosen from the general population. What is the probability that they will give the same answer to the second question?

[24]M. Sholander, Problem #1034, *Mathematics Magazine,* vol. 52, no. 3 (May 1979), p. 183.
[25]S. Benkoski, Comment on Problem #1034, *Mathematics Magazine,* vol. 52, no. 3 (May 1979), pp. 183-184.
[26]M. vos Savant, *Parade Magazine*, 3 March 1996, p. 14.

Chapter 3

Combinatorics

3.1 Permutations

Many problems in probability theory require that we count the number of ways that a particular event can occur. For this, we study the topics of *permutations* and *combinations.* We consider permutations in this section and combinations in the next section.

Before discussing permutations, it is useful to introduce a general counting technique that will enable us to solve a variety of counting problems, including the problem of counting the number of possible permutations of n objects.

Counting Problems

Consider an experiment that takes place in several stages and is such that the number of outcomes m at the nth stage is independent of the outcomes of the previous stages. The number m may be different for different stages. We want to count the number of ways that the entire experiment can be carried out.

Example 3.1 You are eating at Émile's restaurant and the waiter informs you that you have (a) two choices for appetizers: soup or juice; (b) three for the main course: a meat, fish, or vegetable dish; and (c) two for dessert: ice cream or cake. How many possible choices do you have for your complete meal? We illustrate the possible meals by a tree diagram shown in Figure 3.1. Your menu is decided in three stages—at each stage the number of possible choices does not depend on what is chosen in the previous stages: two choices at the first stage, three at the second, and two at the third. From the tree diagram we see that the total number of choices is the product of the number of choices at each stage. In this examples we have $2 \cdot 3 \cdot 2 = 12$ possible menus. Our menu example is an example of the following general counting technique. □

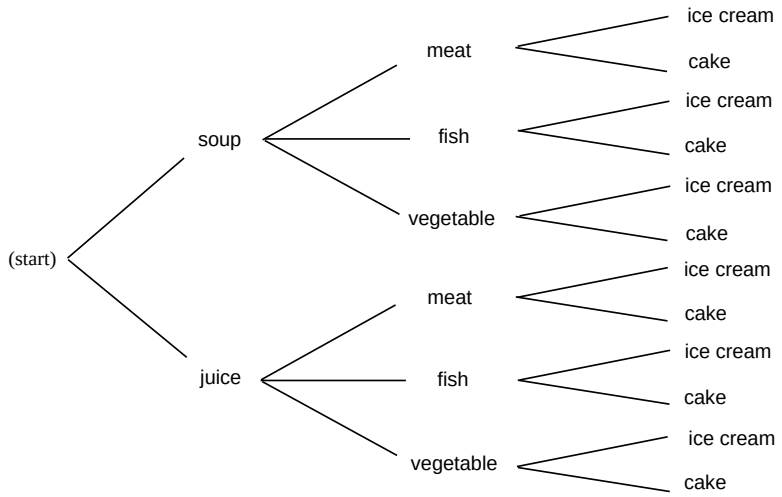


Figure 3.1: Tree for your menu.

A Counting Technique

A task is to be carried out in a sequence of r stages. There are n_1 ways to carry out the first stage; for each of these n_1 ways, there are n_2 ways to carry out the second stage; for each of these n_2 ways, there are n_3 ways to carry out the third stage, and so forth. Then the total number of ways in which the entire task can be accomplished is given by the product $N = n_1 \cdot n_2 \cdot \ldots \cdot n_r$.

Tree Diagrams

It will often be useful to use a tree diagram when studying probabilities of events relating to experiments that take place in stages and for which we are given the probabilities for the outcomes at each stage. For example, assume that the owner of Émile's restaurant has observed that 80 percent of his customers choose the soup for an appetizer and 20 percent choose juice. Of those who choose soup, 50 percent choose meat, 30 percent choose fish, and 20 percent choose the vegetable dish. Of those who choose juice for an appetizer, 30 percent choose meat, 40 percent choose fish, and 30 percent choose the vegetable dish. We can use this to estimate the probabilities at the first two stages as indicated on the tree diagram of Figure 3.2.

We choose for our sample space the set Ω of all possible paths $\omega = \omega_1$, ω_2, $\ldots$, ω_6 through the tree. How should we assign our probability distribution? For example, what probability should we assign to the customer choosing soup and then the meat? If $8/10$ of the customers choose soup and then $1/2$ of these choose meat, a proportion $8/10 \cdot 1/2 = 4/10$ of the customers choose soup and then meat. This suggests choosing our probability distribution for each path through the tree to be the *product* of the probabilities at each of the stages along the path. This results in the probability distribution for the sample points ω indicated in Figure 3.2. (Note that $m(\omega_1) + \cdots + m(\omega_6) = 1$.) From this we see, for example, that the probability

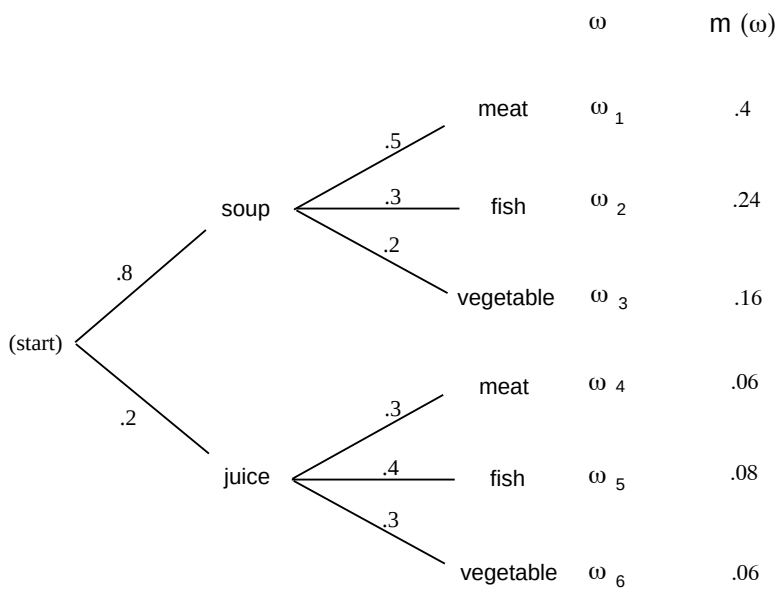


Figure 3.2: Two-stage probability assignment.

that a customer chooses meat is $m(\omega_1) + m(\omega_4) = .46$.

We shall say more about these tree measures when we discuss the concept of conditional probability in Chapter 4. We return now to more counting problems.

Example 3.2 We can show that there are at least two people in Columbus, Ohio, who have the same three initials. Assuming that each person has three initials, there are 26 possibilities for a person's first initial, 26 for the second, and 26 for the third. Therefore, there are $26^3 = 17{,}576$ possible sets of initials. This number is smaller than the number of people living in Columbus, Ohio; hence, there must be at least two people with the same three initials. □

We consider next the celebrated birthday problem—often used to show that naive intuition cannot always be trusted in probability.

Birthday Problem

Example 3.3 How many people do we need to have in a room to make it a favorable bet (probability of success greater than 1/2) that two people in the room will have the same birthday?

Since there are 365 possible birthdays, it is tempting to guess that we would need about 1/2 this number, or 183. You would surely win this bet. In fact, the number required for a favorable bet is only 23. To show this, we find the probability

Number of people	Probability that all birthdays are different
20	.5885616
21	.5563117
22	.5243047
23	.4927028
24	.4616557
25	.4313003

Table 3.1: Birthday problem.

p_r that, in a room with r people, there is no duplication of birthdays; we will have a favorable bet if this probability is less than one half.

Assume that there are 365 possible birthdays for each person (we ignore leap years). Order the people from 1 to r. For a sample point ω, we choose a possible sequence of length r of birthdays each chosen as one of the 365 possible dates. There are 365 possibilities for the first element of the sequence, and for each of these choices there are 365 for the second, and so forth, making 365^r possible sequences of birthdays. We must find the number of these sequences that have no duplication of birthdays. For such a sequence, we can choose any of the 365 days for the first element, then any of the remaining 364 for the second, 363 for the third, and so forth, until we make r choices. For the rth choice, there will be $365 - r + 1$ possibilities. Hence, the total number of sequences with no duplications is

$$365 \cdot 364 \cdot 363 \cdot \ldots \cdot (365 - r + 1) \ .$$

Thus, assuming that each sequence is equally likely,

$$p_r = \frac{365 \cdot 364 \cdot \ldots \cdot (365 - r + 1)}{365^r} \ .$$

We denote the product

$$(n)(n-1)\cdots(n-r+1)$$

by $(n)_r$ (read "n down r," or "n lower r"). Thus,

$$p_r = \frac{(365)_r}{(365)^r} \ .$$

The program **Birthday** carries out this computation and prints the probabilities for $r = 20$ to 25. Running this program, we get the results shown in Table 3.1. As we asserted above, the probability for no duplication changes from greater than one half to less than one half as we move from 22 to 23 people. To see how unlikely it is that we would lose our bet for larger numbers of people, we have run the program again, printing out values from $r = 10$ to $r = 100$ in steps of 10. We see that in a room of 40 people the odds already heavily favor a duplication, and in a room of 100 the odds are overwhelmingly in favor of a duplication. We have assumed that birthdays are equally likely to fall on any particular day. Statistical evidence suggests that this is not true. However, it is intuitively clear (but not easy to prove)

Number of people	Probability that all birthdays are different
10	.8830518
20	.5885616
30	.2936838
40	.1087682
50	.0296264
60	.0058773
70	.0008404
80	.0000857
90	.0000062
100	.0000003

Table 3.2: Birthday problem.

that this makes it even more likely to have a duplication with a group of 23 people. (See Exercise 19 to find out what happens on planets with more or fewer than 365 days per year.) □

We now turn to the topic of permutations.

Permutations

Definition 3.1 Let A be any finite set. A *permutation of* A is a one-to-one mapping of A onto itself. □

To specify a particular permutation we list the elements of A and, under them, show where each element is sent by the one-to-one mapping. For example, if $A = \{a, b, c\}$ a possible permutation σ would be

$$\sigma = \begin{pmatrix} a & b & c \\ b & c & a \end{pmatrix}.$$

By the permutation σ, a is sent to b, b is sent to c, and c is sent to a. The condition that the mapping be one-to-one means that no two elements of A are sent, by the mapping, into the same element of A.

We can put the elements of our set in some order and rename them 1, 2, ..., n. Then, a typical permutation of the set $A = \{a_1, a_2, a_3, a_4\}$ can be written in the form

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix},$$

indicating that a_1 went to a_2, a_2 to a_1, a_3 to a_4, and a_4 to a_3.

If we always choose the top row to be 1 2 3 4 then, to prescribe the permutation, we need only give the bottom row, with the understanding that this tells us where 1 goes, 2 goes, and so forth, under the mapping. When this is done, the permutation

is often called a *rearrangement* of the n objects 1, 2, 3, ..., n. For example, all possible permutations, or rearrangements, of the numbers $A = \{1, 2, 3\}$ are:

$$123,\ 132,\ 213,\ 231,\ 312,\ 321\ .$$

It is an easy matter to count the number of possible permutations of n objects. By our general counting principle, there are n ways to assign the first element, for each of these we have $n - 1$ ways to assign the second object, $n - 2$ for the third, and so forth. This proves the following theorem.

Theorem 3.1 The total number of permutations of a set A of n elements is given by $n \cdot (n\ -\ 1) \cdot (n-2) \cdot \ldots \cdot 1$. □

It is sometimes helpful to consider orderings of subsets of a given set. This prompts the following definition.

Definition 3.2 Let A be an n-element set, and let k be an integer between 0 and n. Then a k-permutation of A is an ordered listing of a subset of A of size k. □

Using the same techniques as in the last theorem, the following result is easily proved.

Theorem 3.2 The total number of k-permutations of a set A of n elements is given by $n \cdot (n-1) \cdot (n-2) \cdot \ldots \cdot (n-k+1)$. □

Factorials

The number given in Theorem 3.1 is called n *factorial,* and is denoted by $n!$. The expression 0! is defined to be 1 to make certain formulas come out simpler. The first few values of this function are shown in Table 3.3. The reader will note that this function grows very rapidly.

The expression $n!$ will enter into many of our calculations, and we shall need to have some estimate of its magnitude when n is large. It is clearly not practical to make exact calculations in this case. We shall instead use a result called *Stirling's formula.* Before stating this formula we need a definition.

Definition 3.3 Let a_n and b_n be two sequences of numbers. We say that a_n is *asymptotically equal to* b_n, and write $a_n \sim b_n$, if

$$\lim_{n\to\infty} \frac{a_n}{b_n} = 1\ .$$

□

n	$n!$
0	1
1	1
2	2
3	6
4	24
5	120
6	720
7	5040
8	40320
9	362880
10	3628800

Table 3.3: Values of the factorial function.

Example 3.4 If $a_n = n + \sqrt{n}$ and $b_n = n$ then, since $a_n/b_n = 1 + 1/\sqrt{n}$ and this ratio tends to 1 as n tends to infinity, we have $a_n \sim b_n$. □

Theorem 3.3 (Stirling's Formula) The sequence $n!$ is asymptotically equal to

$$n^n e^{-n} \sqrt{2\pi n} \ .$$

□

The proof of Stirling's formula may be found in most analysis texts. Let us verify this approximation by using the computer. The program **StirlingApproximations** prints $n!$, the Stirling approximation, and, finally, the ratio of these two numbers. Sample output of this program is shown in Table 3.4. Note that, while the ratio of the numbers is getting closer to 1, the difference between the exact value and the approximation is increasing, and indeed, this difference will tend to infinity as n tends to infinity, even though the ratio tends to 1. (This was also true in our Example 3.4 where $n + \sqrt{n} \sim n$, but the difference is $\sqrt{n}$.)

Generating Random Permutations

We now consider the question of generating a random permutation of the integers between 1 and n. Consider the following experiment. We start with a deck of n cards, labelled 1 through n. We choose a random card out of the deck, note its label, and put the card aside. We repeat this process until all n cards have been chosen. It is clear that each permutation of the integers from 1 to n can occur as a sequence of labels in this experiment, and that each sequence of labels is equally likely to occur. In our implementations of the computer algorithms, the above procedure is called **RandomPermutation**.

n	$n!$	Approximation	Ratio
1	1	.922	1.084
2	2	1.919	1.042
3	6	5.836	1.028
4	24	23.506	1.021
5	120	118.019	1.016
6	720	710.078	1.013
7	5040	4980.396	1.011
8	40320	39902.395	1.010
9	362880	359536.873	1.009
10	3628800	3598696.619	1.008

Table 3.4: Stirling approximations to the factorial function.

Number of fixed points	Fraction of permutations		
	n = 10	n = 20	n = 30
0	.362	.370	.358
1	.368	.396	.358
2	.202	.164	.192
3	.052	.060	.070
4	.012	.008	.020
5	.004	.002	.002
Average number of fixed points	.996	.948	1.042

Table 3.5: Fixed point distributions.

Fixed Points

There are many interesting problems that relate to properties of a permutation chosen at random from the set of all permutations of a given finite set. For example, since a permutation is a one-to-one mapping of the set onto itself, it is interesting to ask how many points are mapped onto themselves. We call such points *fixed points* of the mapping.

Let $p_k(n)$ be the probability that a random permutation of the set $\{1, 2, \ldots, n\}$ has exactly k fixed points. We will attempt to learn something about these probabilities using simulation. The program **FixedPoints** uses the procedure **RandomPermutation** to generate random permutations and count fixed points. The program prints the proportion of times that there are k fixed points as well as the average number of fixed points. The results of this program for 500 simulations for the cases $n = 10$, 20, and 30 are shown in Table 3.5. Notice the rather surprising fact that our estimates for the probabilities do not seem to depend very heavily on the number of elements in the permutation. For example, the probability that there are no fixed points, when $n = 10,\ 20$, or 30 is estimated to be between .35 and .37. We shall see later (see Example 3.12) that for $n \geq 10$ the exact probabilities $p_n(0)$ are, to six decimal place accuracy, equal to $1/e \approx .367879$. Thus, for all practical purposes, after $n = 10$ the probability that a random permutation of the set

Date	Snowfall in inches
1974	75
1975	88
1976	72
1977	110
1978	85
1979	30
1980	55
1981	86
1982	51
1983	64

Table 3.6: Snowfall in Hanover.

$\{1, 2, \ldots, n\}$ has no fixed points does not depend upon n. These simulations also suggest that the average number of fixed points is close to 1. It can be shown (see Example 6.8) that the average is exactly equal to 1 for all n.

More picturesque versions of the fixed-point problem are: You have arranged the books on your book shelf in alphabetical order by author and they get returned to your shelf at random; what is the probability that exactly k of the books end up in their correct position? (The library problem.) In a restaurant n hats are checked and they are hopelessly scrambled; what is the probability that no one gets his own hat back? (The hat check problem.) In the Historical Remarks at the end of this section, we give one method for solving the hat check problem exactly. Another method is given in Example 3.12.

Records

Here is another interesting probability problem that involves permutations. Estimates for the amount of measured snow in inches in Hanover, New Hampshire, in the ten years from 1974 to 1983 are shown in Table 3.6. Suppose we have started keeping records in 1974. Then our first year's snowfall could be considered a record snowfall starting from this year. A new record was established in 1975; the next record was established in 1977, and there were no new records established after this year. Thus, in this ten-year period, there were three records established: 1974, 1975, and 1977. The question that we ask is: How many records should we expect to be established in such a ten-year period? We can count the number of records in terms of a permutation as follows: We number the years from 1 to 10. The actual amounts of snowfall are not important but their relative sizes are. We can, therefore, change the numbers measuring snowfalls to numbers 1 to 10 by replacing the smallest number by 1, the next smallest by 2, and so forth. (We assume that there are no ties.) For our example, we obtain the data shown in Table 3.7.

This gives us a permutation of the numbers from 1 to 10 and, from this permutation, we can read off the records; they are in years 1, 2, and 4. Thus we can define records for a permutation as follows:

Year	1	2	3	4	5	6	7	8	9	10
Ranking	6	9	5	10	7	1	3	8	2	4

Table 3.7: Ranking of total snowfall.

Definition 3.4 Let σ be a permutation of the set $\{1, 2, \ldots, n\}$. Then i is a *record* of σ if either $i = 1$ or $\sigma(j) < \sigma(i)$ for every $j = 1, \ldots, i - 1$. □

Now if we regard all rankings of snowfalls over an n-year period to be equally likely (and allow no ties), we can estimate the probability that there will be k records in n years as well as the average number of records by simulation.

We have written a program **Records** that counts the number of records in randomly chosen permutations. We have run this program for the cases $n = 10$, 20, 30. For $n = 10$ the average number of records is 2.968, for 20 it is 3.656, and for 30 it is 3.960. We see now that the averages increase, but very slowly. We shall see later (see Example 6.11) that the average number is approximately $\log n$. Since $\log 10 = 2.3$, $\log 20 = 3$, and $\log 30 = 3.4$, this is consistent with the results of our simulations.

As remarked earlier, we shall be able to obtain formulas for exact results of certain problems of the above type. However, only minor changes in the problem make this impossible. The power of simulation is that minor changes in a problem do not make the simulation much more difficult. (See Exercise 20 for an interesting variation of the hat check problem.)

List of Permutations

Another method to solve problems that is not sensitive to small changes in the problem is to have the computer simply list all possible permutations and count the fraction that have the desired property. The program **AllPermutations** produces a list of all of the permutations of n. When we try running this program, we run into a limitation on the use of the computer. The number of permutations of n increases so rapidly that even to list all permutations of 20 objects is impractical.

Historical Remarks

Our basic counting principle stated that if you can do one thing in r ways and for each of these another thing in s ways, then you can do the pair in rs ways. This is such a self-evident result that you might expect that it occurred very early in mathematics. N. L. Biggs suggests that we might trace an example of this principle as follows: First, he relates a popular nursery rhyme dating back to at least 1730:

As I was going to St. Ives,
I met a man with seven wives,
Each wife had seven sacks,
Each sack had seven cats,
Each cat had seven kits.
Kits, cats, sacks and wives,
How many were going to St. Ives?

(You need our principle only if you are not clever enough to realize that you are supposed to answer *one,* since only the narrator is going to St. Ives; the others are going in the other direction!)

He also gives a problem appearing on one of the oldest surviving mathematical manuscripts of about 1650 B.C., roughly translated as:

Houses	7
Cats	49
Mice	343
Wheat	2401
Hekat	16807
	19607

The following interpretation has been suggested: there are seven houses, each with seven cats; each cat kills seven mice; each mouse would have eaten seven heads of wheat, each of which would have produced seven hekat measures of grain. With this interpretation, the table answers the question of how many hekat measures were saved by the cats' actions. It is not clear why the writer of the table wanted to add the numbers together.[1]

One of the earliest uses of factorials occurred in Euclid's proof that there are infinitely many prime numbers. Euclid argued that there must be a prime number between n and $n! + 1$ as follows: $n!$ and $n! + 1$ cannot have common factors. Either $n! + 1$ is prime or it has a proper factor. In the latter case, this factor cannot divide $n!$ and hence must be between n and $n! + 1$. If this factor is not prime, then it has a factor that, by the same argument, must be bigger than n. In this way, we eventually reach a prime bigger than n, and this holds for all n.

The "$n!$" rule for the number of permutations seems to have occurred first in India. Examples have been found as early as 300 B.C., and by the eleventh century the general formula seems to have been well known in India and then in the Arab countries.

The *hat check problem* is found in an early probability book written by de Montmort and first printed in 1708.[2] It appears in the form of a game called *Treize.* In a simplified version of this game considered by de Montmort one turns over cards numbered 1 to 13, calling out 1, 2, . . . , 13 as the cards are examined. De Montmort asked for the probability that no card that is turned up agrees with the number called out.

[1] N. L. Biggs, "The Roots of Combinatorics," *Historia Mathematica,* vol. 6 (1979), pp. 109–136.
[2] P. R. de Montmort, *Essay d'Analyse sur des Jeux de Hazard,* 2d ed. (Paris: Quillau, 1713).

This probability is the same as the probability that a random permutation of 13 elements has no fixed point. De Montmort solved this problem by the use of a recursion relation as follows: let w_n be the number of permutations of n elements with no fixed point (such permutations are called *derangements*). Then $w_1 = 0$ and $w_2 = 1$.

Now assume that $n \geq 3$ and choose a derangement of the integers between 1 and n. Let k be the integer in the first position in this derangement. By the definition of derangement, we have $k \neq 1$. There are two possibilities of interest concerning the position of 1 in the derangement: either 1 is in the kth position or it is elsewhere. In the first case, the $n-2$ remaining integers can be positioned in w_{n-2} ways without resulting in any fixed points. In the second case, we consider the set of integers $\{1, 2, \ldots, k-1, k+1, \ldots, n\}$. The numbers in this set must occupy the positions $\{2, 3, \ldots, n\}$ so that none of the numbers other than 1 in this set are fixed, and also so that 1 is not in position k. The number of ways of achieving this kind of arrangement is just w_{n-1}. Since there are $n-1$ possible values of k, we see that

$$w_n = (n-1)w_{n-1} + (n-1)w_{n-2}$$

for $n \geq 3$. One might conjecture from this last equation that the sequence $\{w_n\}$ grows like the sequence $\{n!\}$.

In fact, it is easy to prove by induction that

$$w_n = nw_{n-1} + (-1)^n \ .$$

Then $p_i = w_i/i!$ satisfies

$$p_i - p_{i-1} = \frac{(-1)^i}{i!} \ .$$

If we sum from $i = 2$ to n, and use the fact that $p_1 = 0$, we obtain

$$p_n = \frac{1}{2!} - \frac{1}{3!} + \cdots + \frac{(-1)^n}{n!} \ .$$

This agrees with the first $n+1$ terms of the expansion for e^x for $x = -1$ and hence for large n is approximately $e^{-1} \approx .368$. David remarks that this was possibly the first use of the exponential function in probability.[3] We shall see another way to derive de Montmort's result in the next section, using a method known as the Inclusion-Exclusion method.

Recently, a related problem appeared in a column of Marilyn vos Savant.[4] Charles Price wrote to ask about his experience playing a certain form of solitaire, sometimes called "frustration solitaire." In this particular game, a deck of cards is shuffled, and then dealt out, one card at a time. As the cards are being dealt, the player counts from 1 to 13, and then starts again at 1. (Thus, each number is counted four times.) If a number that is being counted coincides with the rank of the card that is being turned up, then the player loses the game. Price found that he rarely won and wondered how often he should win. Vos Savant remarked that the expected number of matches is 4 so it should be difficult to win the game.

[3] F. N. David, *Games, Gods and Gambling* (London: Griffin, 1962), p. 146.

[4] M. vos Savant, Ask Marilyn, *Parade Magazine, Boston Globe*, 21 August 1994.

Finding the chance of winning is a harder problem than the one that de Montmort solved because, when one goes through the entire deck, there are different patterns for the matches that might occur. For example matches may occur for two cards of the same rank, say two aces, or for two different ranks, say a two and a three.

A discussion of this problem can be found in Riordan.[5] In this book, it is shown that as $n \to \infty$, the probability of no matches tends to $1/e^4$.

The original game of Treize is more difficult to analyze than frustration solitaire. The game of Treize is played as follows. One person is chosen as dealer and the others are players. Each player, other than the dealer, puts up a stake. The dealer shuffles the cards and turns them up one at a time calling out, "Ace, two, three,..., king," just as in frustration solitaire. If the dealer goes through the 13 cards without a match he pays the players an amount equal to their stake, and the deal passes to someone else. If there is a match the dealer collects the players' stakes; the players put up new stakes, and the dealer continues through the deck, calling out, "Ace, two, three, ..." If the dealer runs out of cards he reshuffles and continues the count where he left off. He continues until there is a run of 13 without a match and then a new dealer is chosen.

The question at this point is how much money can the dealer expect to win from each player. De Montmort found that if each player puts up a stake of 1, say, then the dealer will win approximately .801 from each player.

Peter Doyle calculated the exact amount that the dealer can expect to win. The answer is:

$$
\begin{array}{l}
26516072156010218582227607912734182784642120482136091446715371962089931 \\
52311343541724554334912870541440299239251607694113500080775917818512013 \\
82176876653563173852874555859367254632009477403727395572807459384342747 \\
87664965076063990538261189388143513547366316017004945507201764278828306 \\
60117107953633142734382477922709835281753299035988581413688367655833113 \\
24476153310720627474169719301806649152698704084383914217907906954976036 \\
28528211590140316202120601549126920880824913325553882692055427830810368 \\
57818861208758248800680978640438118582834877542560955550662878927123048 \\
26997601700116233592793308297533642193505074540268925683193887821301442 \\
70519791882/ \\
33036929133582592220117220713156071114975101149831063364072138969878007 \\
99647204708825303387525892236581323015628005621143427290625658974433971 \\
65719454122908007086289841306087561302818991167357863623756067184986491 \\
35535553622197448890223267101158801016285931351979294387223277033396967 \\
79797069933475802423676949873661605184031477561560393380257070970711959 \\
69641268242455013319879747054693517809383750593488858698672364846950539 \\
88868628582609905586271001318150621134407056983214740221851567706672080 \\
94586589378459432799868706334161812988630496327287254818458879353024498 \\
00322425586446741048147720934108061350613503856973048971213063937040515
\end{array}
$$

[5] J. Riordan, *An Introduction to Combinatorial Analysis,* (New York: John Wiley & Sons, 1958).

59533731591.

This is .803 to 3 decimal places. A description of the algorithm used to find this answer can be found on his Web page.[6] A discussion of this problem and other problems can be found in Doyle et al.[7]

The *birthday problem* does not seem to have a very old history. Problems of this type were first discussed by von Mises.[8] It was made popular in the 1950s by Feller's book.[9]

Stirling presented his formula

$$n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$$

in his work *Methodus Differentialis* published in 1730.[10] This approximation was used by de Moivre in establishing his celebrated central limit theorem that we will study in Chapter 9. De Moivre himself had independently established this approximation, but without identifying the constant π. Having established the approximation

$$\frac{2B}{\sqrt{n}}$$

for the central term of the binomial distribution, where the constant B was determined by an infinite series, de Moivre writes:

> ... my worthy and learned Friend, Mr. James Stirling, who had applied himself after me to that inquiry, found that the Quantity B did denote the Square-root of the Circumference of a Circle whose Radius is Unity, so that if that Circumference be called c the Ratio of the middle Term to the Sum of all Terms will be expressed by $2/\sqrt{nc}$....[11]

Exercises

1 Four people are to be arranged in a row to have their picture taken. In how many ways can this be done?

2 An automobile manufacturer has four colors available for automobile exteriors and three for interiors. How many different color combinations can he produce?

3 In a digital computer, a *bit* is one of the integers {0,1}, and a *word* is any string of 32 bits. How many different words are possible?

---

[6] P. Doyle, "Solution to Montmort's Probleme du Treize," http://math.ucsd.edu/~doyle/.

[7] P. Doyle, C. Grinstead, and J. Snell, "Frustration Solitaire," *UMAP Journal*, vol. 16, no. 2 (1995), pp. 137-145.

[8] R. von Mises, "Über Aufteilungs- und Besetzungs-Wahrscheinlichkeiten," *Revue de la Faculté des Sciences de l'Université d'Istanbul, N. S.* vol. 4 (1938-39), pp. 145-163.

[9] W. Feller, *Introduction to Probability Theory and Its Applications,* vol. 1, 3rd ed. (New York: John Wiley & Sons, 1968).

[10] J. Stirling, *Methodus Differentialis,* (London: Bowyer, 1730).

[11] A. de Moivre, *The Doctrine of Chances,* 3rd ed. (London: Millar, 1756).

4 What is the probability that at least 2 of the presidents of the United States have died on the same day of the year? If you bet this has happened, would you win your bet?

5 There are three different routes connecting city A to city B. How many ways can a round trip be made from A to B and back? How many ways if it is desired to take a different route on the way back?

6 In arranging people around a circular table, we take into account their seats relative to each other, not the actual position of any one person. Show that n people can be arranged around a circular table in $(n-1)!$ ways.

7 Five people get on an elevator that stops at five floors. Assuming that each has an equal probability of going to any one floor, find the probability that they all get off at different floors.

8 A finite set Ω has n elements. Show that if we count the empty set and Ω as subsets, there are 2^n subsets of Ω.

9 A more refined inequality for approximating $n!$ is given by

$$\sqrt{2\pi n}\left(\frac{n}{e}\right)^n e^{1/(12n+1)} < n! < \sqrt{2\pi n}\left(\frac{n}{e}\right)^n e^{1/(12n)} \ .$$

Write a computer program to illustrate this inequality for $n = 1$ to 9.

10 A deck of ordinary cards is shuffled and 13 cards are dealt. What is the probability that the last card dealt is an ace?

11 There are n applicants for the director of computing. The applicants are interviewed independently by each member of the three-person search committee and ranked from 1 to n. A candidate will be hired if he or she is ranked first by at least two of the three interviewers. Find the probability that a candidate will be accepted if the members of the committee really have no ability at all to judge the candidates and just rank the candidates randomly. In particular, compare this probability for the case of three candidates and the case of ten candidates.

12 A symphony orchestra has in its repertoire 30 Haydn symphonies, 15 modern works, and 9 Beethoven symphonies. Its program always consists of a Haydn symphony followed by a modern work, and then a Beethoven symphony.

(a) How many different programs can it play?

(b) How many different programs are there if the three pieces can be played in any order?

(c) How many different three-piece programs are there if more than one piece from the same category can be played and they can be played in any order?

13 A certain state has license plates showing three numbers and three letters. How many different license plates are possible

(a) if the numbers must come before the letters?

(b) if there is no restriction on where the letters and numbers appear?

14 The door on the computer center has a lock which has five buttons numbered from 1 to 5. The combination of numbers that opens the lock is a sequence of five numbers and is reset every week.

(a) How many combinations are possible if every button must be used once?

(b) Assume that the lock can also have combinations that require you to push two buttons simultaneously and then the other three one at a time. How many more combinations does this permit?

15 A computing center has 3 processors that receive n jobs, with the jobs assigned to the processors purely at random so that all of the 3^n possible assignments are equally likely. Find the probability that exactly one processor has no jobs.

16 Prove that at least two people in Atlanta, Georgia, have the same initials, assuming no one has more than four initials.

17 Find a formula for the probability that among a set of n people, at least two have their birthdays in the same month of the year (assuming the months are equally likely for birthdays).

18 Consider the problem of finding the probability of more than one coincidence of birthdays in a group of n people. These include, for example, three people with the same birthday, or two pairs of people with the same birthday, or larger coincidences. Show how you could compute this probability, and write a computer program to carry out this computation. Use your program to find the smallest number of people for which it would be a favorable bet that there would be more than one coincidence of birthdays.

***19** Suppose that on planet Zorg a year has n days, and that the lifeforms there are equally likely to have hatched on any day of the year. We would like to estimate d, which is the minimum number of lifeforms needed so that the probability of at least two sharing a birthday exceeds 1/2.

(a) In Example 3.3, it was shown that in a set of d lifeforms, the probability that no two life forms share a birthday is

$$\frac{(n)_d}{n^d} ,$$

where $(n)_d = (n)(n-1)\cdots(n-d+1)$. Thus, we would like to set this equal to 1/2 and solve for d.

(b) Using Stirling's Formula, show that

$$\frac{(n)_d}{n^d} \sim \left(1 + \frac{d}{n-d}\right)^{n-d+1/2} e^{-d} \ .$$

(c) Now take the logarithm of the right-hand expression, and use the fact that for small values of x, we have

$$\log(1+x) \sim x - \frac{x^2}{2} \ .$$

(We are implicitly using the fact that d is of smaller order of magnitude than n. We will also use this fact in part (d).)

(d) Set the expression found in part (c) equal to $-\log(2)$, and solve for d as a function of n, thereby showing that

$$d \sim \sqrt{2(\log 2)\, n} \ .$$

Hint: If all three summands in the expression found in part (b) are used, one obtains a cubic equation in d. If the smallest of the three terms is thrown away, one obtains a quadratic equation in d.

(e) Use a computer to calculate the exact values of d for various values of n. Compare these values with the approximate values obtained by using the answer to part d).

20 At a mathematical conference, ten participants are randomly seated around a circular table for meals. Using simulation, estimate the probability that no two people sit next to each other at both lunch and dinner. Can you make an intelligent conjecture for the case of n participants when n is large?

21 Modify the program **AllPermutations** to count the number of permutations of n objects that have exactly j fixed points for $j = 0, 1, 2, \ldots, n$. Run your program for $n = 2$ to 6. Make a conjecture for the relation between the number that have 0 fixed points and the number that have exactly 1 fixed point. A proof of the correct conjecture can be found in Wilf.[12]

22 Mr. Wimply Dimple, one of London's most prestigious watch makers, has come to Sherlock Holmes in a panic, having discovered that someone has been producing and selling crude counterfeits of his best selling watch. The 16 counterfeits so far discovered bear stamped numbers, all of which fall between 1 and 56, and Dimple is anxious to know the extent of the forger's work. All present agree that it seems reasonable to assume that the counterfeits thus far produced bear consecutive numbers from 1 to whatever the total number is.

"Chin up, Dimple," opines Dr. Watson. "I shouldn't worry overly much if I were you; the Maximum Likelihood Principle, which estimates the total

[12]H. S. Wilf, "A Bijection in the Theory of Derangements," *Mathematics Magazine,* vol. 57, no. 1 (1984), pp. 37–40.

number as precisely that which gives the highest probability for the series of numbers found, suggests that we guess 56 itself as the total. Thus, your forgers are not a big operation, and we shall have them safely behind bars before your business suffers significantly."

"Stuff, nonsense, and bother your fancy principles, Watson," counters Holmes. "Anyone can see that, of course, there must be quite a few more than 56 watches—why the odds of our having discovered precisely the highest numbered watch made are laughably negligible. A much better guess would be *twice* 56."

(a) Show that Watson is correct that the Maximum Likelihood Principle gives 56.

(b) Write a computer program to compare Holmes's and Watson's guessing strategies as follows: fix a total N and choose 16 integers randomly between 1 and N. Let m denote the largest of these. Then Watson's guess for N is m, while Holmes's is $2m$. See which of these is closer to N. Repeat this experiment (with N still fixed) a hundred or more times, and determine the proportion of times that each comes closer. Whose seems to be the better strategy?

23 Barbara Smith is interviewing candidates to be her secretary. As she interviews the candidates, she can determine the relative rank of the candidates but not the true rank. Thus, if there are six candidates and their true rank is 6, 1, 4, 2, 3, 5, (where 1 is best) then after she had interviewed the first three candidates she would rank them 3, 1, 2. As she interviews each candidate, she must either accept or reject the candidate. If she does not accept the candidate after the interview, the candidate is lost to her. She wants to decide on a strategy for deciding when to stop and accept a candidate that will maximize the probability of getting the best candidate. Assume that there are n candidates and they arrive in a random rank order.

(a) What is the probability that Barbara gets the best candidate if she interviews all of the candidates? What is it if she chooses the first candidate?

(b) Assume that Barbara decides to interview the first half of the candidates and then continue interviewing until getting a candidate better than any candidate seen so far. Show that she has a better than 25 percent chance of ending up with the best candidate.

24 For the task described in Exercise 23, it can be shown[13] that the best strategy is to pass over the first $k-1$ candidates where k is the smallest integer for which

$$\frac{1}{k} + \frac{1}{k+1} + \cdots + \frac{1}{n-1} \leq 1 \ .$$

[13] E. B. Dynkin and A. A. Yushkevich, *Markov Processes: Theorems and Problems,* trans. J. S. Wood (New York: Plenum, 1969).

Using this strategy the probability of getting the best candidate is approximately $1/e = .368$. Write a program to simulate Barbara Smith's interviewing if she uses this optimal strategy, using $n = 10$, and see if you can verify that the probability of success is approximately $1/e$.

3.2 Combinations

Having mastered permutations, we now consider combinations. Let U be a set with n elements; we want to count the number of distinct subsets of the set U that have exactly j elements. The empty set and the set U are considered to be subsets of U. The empty set is usually denoted by ϕ.

Example 3.5 Let $U = \{a, b, c\}$. The subsets of U are

$$\phi,\ \{a\},\ \{b\},\ \{c\},\ \{a, b\},\ \{a, c\},\ \{b, c\},\ \{a, b, c\}\ .$$

□

Binomial Coefficients

The number of distinct subsets with j elements that can be chosen from a set with n elements is denoted by $\binom{n}{j}$, and is pronounced "n choose j." The number $\binom{n}{j}$ is called a *binomial coefficient.* This terminology comes from an application to algebra which will be discussed later in this section.

In the above example, there is one subset with no elements, three subsets with exactly 1 element, three subsets with exactly 2 elements, and one subset with exactly 3 elements. Thus, $\binom{3}{0} = 1$, $\binom{3}{1} = 3$, $\binom{3}{2} = 3$, and $\binom{3}{3} = 1$. Note that there are $2^3 = 8$ subsets in all. (We have already seen that a set with n elements has 2^n subsets; see Exercise 3.1.8.) It follows that

$$\binom{3}{0} + \binom{3}{1} + \binom{3}{2} + \binom{3}{3} = 2^3 = 8\ ,$$

$$\binom{n}{0} = \binom{n}{n} = 1\ .$$

Assume that $n > 0$. Then, since there is only one way to choose a set with no elements and only one way to choose a set with n elements, the remaining values of $\binom{n}{j}$ are determined by the following *recurrence relation*:

Theorem 3.4 For integers n and j, with $0 < j < n$, the binomial coefficients satisfy:

$$\binom{n}{j} = \binom{n-1}{j} + \binom{n-1}{j-1}\ . \tag{3.1}$$

	j = 0	1	2	3	4	5	6	7	8	9	10
n = 0	1										
1	1	1									
2	1	2	1								
3	1	3	3	1							
4	1	4	6	4	1						
5	1	5	10	10	5	1					
6	1	6	15	20	15	6	1				
7	1	7	21	35	35	21	7	1			
8	1	8	28	56	70	56	28	8	1		
9	1	9	36	84	126	126	84	36	9	1	
10	1	10	45	120	210	252	210	120	45	10	1

Figure 3.3: Pascal's triangle.

Proof. We wish to choose a subset of j elements. Choose an element u of U. Assume first that we do not want u in the subset. Then we must choose the j elements from a set of $n-1$ elements; this can be done in $\binom{n-1}{j}$ ways. On the other hand, assume that we do want u in the subset. Then we must choose the other $j-1$ elements from the remaining $n-1$ elements of U; this can be done in $\binom{n-1}{j-1}$ ways. Since u is either in our subset or not, the number of ways that we can choose a subset of j elements is the sum of the number of subsets of j elements which have u as a member and the number which do not—this is what Equation 3.1 states. □

The binomial coefficient $\binom{n}{j}$ is defined to be 0, if $j < 0$ or if $j > n$. With this definition, the restrictions on j in Theorem 3.4 are unnecessary.

Pascal's Triangle

The relation 3.1, together with the knowledge that

$$\binom{n}{0} = \binom{n}{n} = 1 \ ,$$

determines completely the numbers $\binom{n}{j}$. We can use these relations to determine the famous *triangle of Pascal,* which exhibits all these numbers in matrix form (see Figure 3.3).

The nth row of this triangle has the entries $\binom{n}{0}$, $\binom{n}{1}$,..., $\binom{n}{n}$. We know that the first and last of these numbers are 1. The remaining numbers are determined by the recurrence relation Equation 3.1; that is, the entry $\binom{n}{j}$ for $0 < j < n$ in the nth row of Pascal's triangle is the *sum* of the entry immediately above and the one immediately to its left in the $(n-1)$st row. For example, $\binom{5}{2} = 6 + 4 = 10$.

This algorithm for constructing Pascal's triangle can be used to write a computer program to compute the binomial coefficients. You are asked to do this in Exercise 4.

While Pascal's triangle provides a way to construct recursively the binomial coefficients, it is also possible to give a formula for $\binom{n}{j}$.

Theorem 3.5 The binomial coefficients are given by the formula

$$\binom{n}{j} = \frac{(n)_j}{j!} \ . \tag{3.2}$$

Proof. Each subset of size j of a set of size n can be ordered in $j!$ ways. Each of these orderings is a j-permutation of the set of size n. The number of j-permutations is $(n)_j$, so the number of subsets of size j is

$$\frac{(n)_j}{j!} \ .$$

This completes the proof. □

The above formula can be rewritten in the form

$$\binom{n}{j} = \frac{n!}{j!(n-j)!} \ .$$

This immediately shows that

$$\binom{n}{j} = \binom{n}{n-j} \ .$$

When using Equation 3.2 in the calculation of $\binom{n}{j}$, if one alternates the multiplications and divisions, then all of the intermediate values in the calculation are integers. Furthermore, none of these intermediate values exceed the final value. (See Exercise 40.)

Another point that should be made concerning Equation 3.2 is that if it is used to *define* the binomial coefficients, then it is no longer necessary to require n to be a positive integer. The variable j must still be a non-negative integer under this definition. This idea is useful when extending the Binomial Theorem to general exponents. (The Binomial Theorem for non-negative integer exponents is given below as Theorem 3.7.)

Poker Hands

Example 3.6 Poker players sometimes wonder why a *four of a kind* beats a *full house*. A poker hand is a random subset of 5 elements from a deck of 52 cards. A hand has four of a kind if it has four cards with the same value—for example, four sixes or four kings. It is a full house if it has three of one value and two of a second—for example, three twos and two queens. Let us see which hand is more likely. How many hands have four of a kind? There are 13 ways that we can specify the value for the four cards. For each of these, there are 48 possibilities for the fifth card. Thus, the number of four-of-a-kind hands is $13 \cdot 48 = 624$. Since the total number of possible hands is $\binom{52}{5} = 2598960$, the probability of a hand with four of a kind is $624/2598960 = .00024$.

Now consider the case of a full house; how many such hands are there? There are 13 choices for the value which occurs three times; for each of these there are $\binom{4}{3} = 4$ choices for the particular three cards of this value that are in the hand. Having picked these three cards, there are 12 possibilities for the value which occurs twice; for each of these there are $\binom{4}{2} = 6$ possibilities for the particular pair of this value. Thus, the number of full houses is $13 \cdot 4 \cdot 12 \cdot 6 = 3744$, and the probability of obtaining a hand with a full house is $3744/2598960 = .0014$. Thus, while both types of hands are unlikely, you are six times more likely to obtain a full house than four of a kind. □

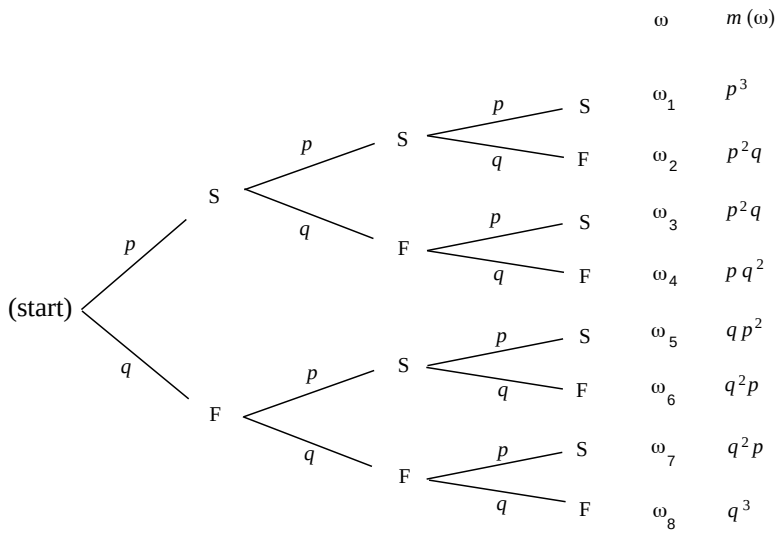


Figure 3.4: Tree diagram of three Bernoulli trials.

Bernoulli Trials

Our principal use of the binomial coefficients will occur in the study of one of the important chance processes called *Bernoulli trials.*

Definition 3.5 A *Bernoulli trials process* is a sequence of n chance experiments such that

1. Each experiment has two possible outcomes, which we may call *success* and *failure.*

2. The probability p of success on each experiment is the same for each experiment, and this probability is not affected by any knowledge of previous outcomes. The probability q of failure is given by $q = 1 - p$.

□

Example 3.7 The following are Bernoulli trials processes:

1. A coin is tossed ten times. The two possible outcomes are heads and tails. The probability of heads on any one toss is 1/2.

2. An opinion poll is carried out by asking 1000 people, randomly chosen from the population, if they favor the Equal Rights Amendment—the two outcomes being yes and no. The probability p of a yes answer (i.e., a success) indicates the proportion of people in the entire population that favor this amendment.

3. A gambler makes a sequence of 1-dollar bets, betting each time on black at roulette at Las Vegas. Here a success is winning 1 dollar and a failure is losing 1 dollar. Since in American roulette the gambler wins if the ball stops on one of 18 out of 38 positions and loses otherwise, the probability of winning is $p = 18/38 = .474$.

□

To analyze a Bernoulli trials process, we choose as our sample space a binary tree and assign a probability distribution to the paths in this tree. Suppose, for example, that we have three Bernoulli trials. The possible outcomes are indicated in the tree diagram shown in Figure 3.4. We define X to be the random variable which represents the outcome of the process, i.e., an ordered triple of S's and F's. The probabilities assigned to the branches of the tree represent the probability for each individual trial. Let the outcome of the ith trial be denoted by the random variable X_i, with distribution function m_i. Since we have assumed that outcomes on any one trial do not affect those on another, we assign the same probabilities at each level of the tree. An outcome ω for the entire experiment will be a path through the tree. For example, ω_3 represents the outcomes SFS. Our frequency interpretation of probability would lead us to expect a fraction p of successes on the first experiment; of these, a fraction q of failures on the second; and, of these, a fraction p of successes on the third experiment. This suggests assigning probability pqp to the outcome ω_3. More generally, we assign a distribution function $m(\omega)$ for paths ω by defining $m(\omega)$ to be the product of the branch probabilities along the path ω. Thus, the probability that the three events S on the first trial, F on the second trial, and S on the third trial occur is the product of the probabilities for the individual events. We shall see in the next chapter that this means that the events involved are *independent* in the sense that the knowledge of one event does not affect our prediction for the occurrences of the other events.

Binomial Probabilities

We shall be particularly interested in the probability that in n Bernoulli trials there are exactly j successes. We denote this probability by $b(n, p, j)$. Let us calculate the particular value $b(3, p, 2)$ from our tree measure. We see that there are three paths which have exactly two successes and one failure, namely ω_2, ω_3, and ω_5. Each of these paths has the same probability p^2q. Thus $b(3, p, 2) = 3p^2q$. Considering all possible numbers of successes we have

$$\begin{aligned} b(3,p,0) &= q^3 \ , \\ b(3,p,1) &= 3pq^2 \ , \\ b(3,p,2) &= 3p^2q \ , \\ b(3,p,3) &= p^3 \ . \end{aligned}$$

We can, in the same manner, carry out a tree measure for n experiments and determine $b(n, p, j)$ for the general case of n Bernoulli trials.

Theorem 3.6 Given n Bernoulli trials with probability p of success on each experiment, the probability of exactly j successes is

$$b(n,p,j) = \binom{n}{j} p^j q^{n-j}$$

where $q = 1 - p$.

Proof. We construct a tree measure as described above. We want to find the sum of the probabilities for all paths which have exactly j successes and $n - j$ failures. Each such path is assigned a probability $p^j q^{n-j}$. How many such paths are there? To specify a path, we have to pick, from the n possible trials, a subset of j to be successes, with the remaining $n - j$ outcomes being failures. We can do this in $\binom{n}{j}$ ways. Thus the sum of the probabilities is

$$b(n,p,j) = \binom{n}{j} p^j q^{n-j} \ .$$

□

Example 3.8 A fair coin is tossed six times. What is the probability that exactly three heads turn up? The answer is

$$b(6,.5,3) = \binom{6}{3} \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^3 = 20 \cdot \frac{1}{64} = .3125 \ .$$

□

Example 3.9 A die is rolled four times. What is the probability that we obtain exactly one 6? We treat this as Bernoulli trials with *success* = "rolling a 6" and *failure* = "rolling some number other than a 6." Then $p = 1/6$, and the probability of exactly one success in four trials is

$$b(4,1/6,1) = \binom{4}{1} \left(\frac{1}{6}\right)^1 \left(\frac{5}{6}\right)^3 = .386 \ .$$

□

To compute binomial probabilities using the computer, multiply the function choose(n,k) by $p^k q^{n-k}$. The program **BinomialProbabilities** prints out the binomial probabilities $b(n,p,k)$ for k between $kmin$ and $kmax$, and the sum of these probabilities. We have run this program for $n = 100$, $p = 1/2$, $kmin = 45$, and $kmax = 55$; the output is shown in Table 3.8. Note that the individual probabilities are quite small. The probability of exactly 50 heads in 100 tosses of a coin is about .08. Our intuition tells us that this is the most likely outcome, which is correct; but, all the same, it is not a very likely outcome.

k	$b(n,p,k)$
45	.0485
46	.0580
47	.0666
48	.0735
49	.0780
50	.0796
51	.0780
52	.0735
53	.0666
54	.0580
55	.0485

Table 3.8: Binomial probabilities for $n = 100,\ p = 1/2$.

Binomial Distributions

Definition 3.6 Let n be a positive integer, and let p be a real number between 0 and 1. Let B be the random variable which counts the number of successes in a Bernoulli trials process with parameters n and p. Then the distribution $b(n,p,k)$ of B is called the *binomial distribution*. □

We can get a better idea about the binomial distribution by graphing this distribution for different values of n and p (see Figure 3.5). The plots in this figure were generated using the program **BinomialPlot**.

We have run this program for $p = .5$ and $p = .3$. Note that even for $p = .3$ the graphs are quite symmetric. We shall have an explanation for this in Chapter 9. We also note that the highest probability occurs around the value np, but that these highest probabilities get smaller as n increases. We shall see in Chapter 6 that np is the *mean* or *expected* value of the binomial distribution $b(n,p,k)$.

The following example gives a nice way to see the binomial distribution, when $p = 1/2$.

Example 3.10 A *Galton board* is a board in which a large number of BB-shots are dropped from a chute at the top of the board and deflected off a number of pins on their way down to the bottom of the board. The final position of each slot is the result of a number of random deflections either to the left or the right. We have written a program **GaltonBoard** to simulate this experiment.

We have run the program for the case of 20 rows of pins and 10,000 shots being dropped. We show the result of this simulation in Figure 3.6.

Note that if we write 0 every time the shot is deflected to the left, and 1 every time it is deflected to the right, then the path of the shot can be described by a sequence of 0's and 1's of length n, just as for the n-fold coin toss.

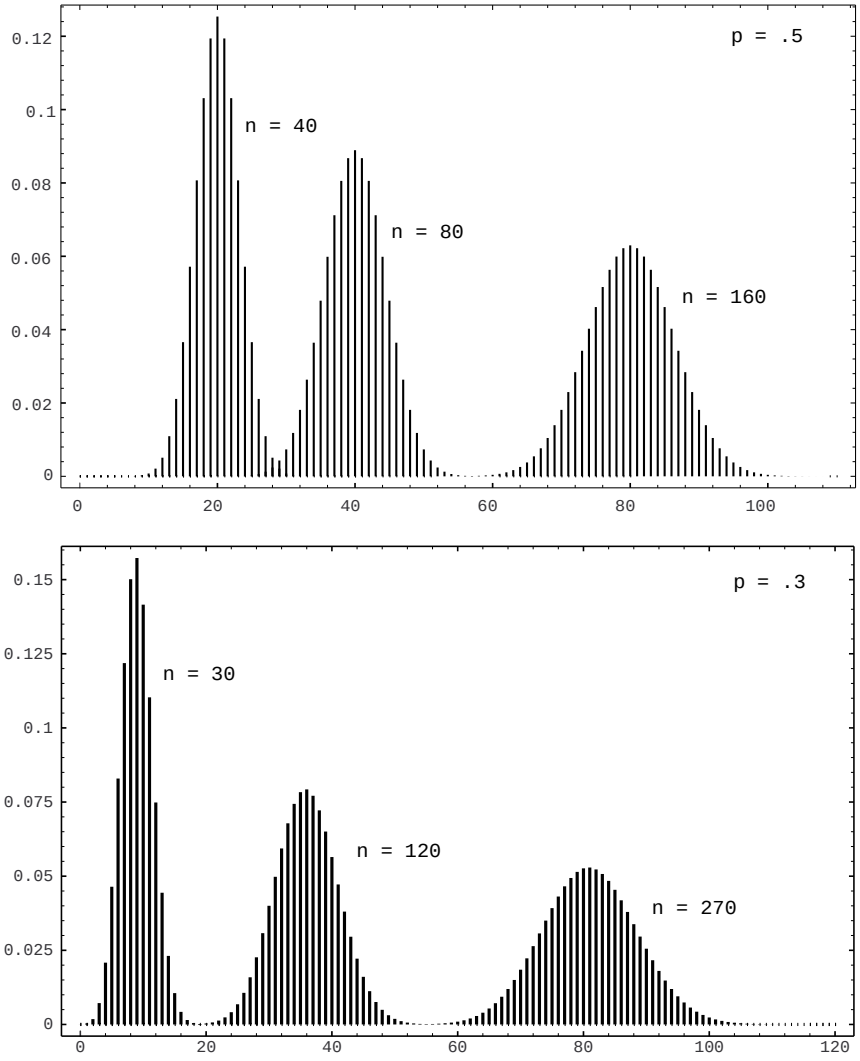


Figure 3.5: Binomial distributions.

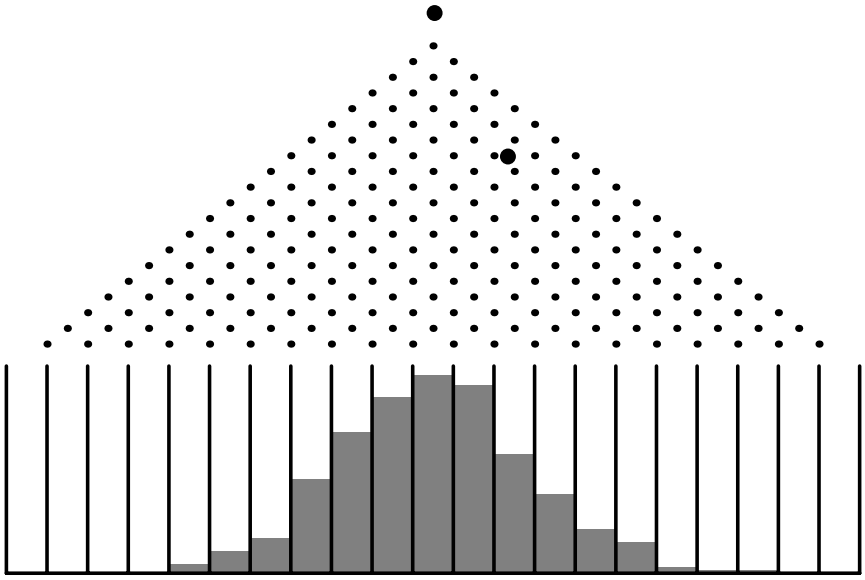


Figure 3.6: Simulation of the Galton board.

The distribution shown in Figure 3.6 is an example of an empirical distribution, in the sense that it comes about by means of a sequence of experiments. As expected, this empirical distribution resembles the corresponding binomial distribution with parameters $n = 20$ and $p = 1/2$. □

Hypothesis Testing

Example 3.11 Suppose that ordinary aspirin has been found effective against headaches 60 percent of the time, and that a drug company claims that its new aspirin with a special headache additive is more effective. We can test this claim as follows: we call their claim the *alternate hypothesis,* and its negation, that the additive has no appreciable effect, the *null hypothesis.* Thus the null hypothesis is that $p = .6$, and the alternate hypothesis is that $p > .6$, where p is the probability that the new aspirin is effective.

We give the aspirin to n people to take when they have a headache. We want to find a number m, called the *critical value* for our experiment, such that we reject the null hypothesis if at least m people are cured, and otherwise we accept it. How should we determine this critical value?

First note that we can make two kinds of errors. The first, often called a *type 1 error* in statistics, is to reject the null hypothesis when in fact it is true. The second, called a *type 2 error,* is to accept the null hypothesis when it is false. To determine the probability of both these types of errors we introduce a function $\alpha(p)$, defined to be the probability that we reject the null hypothesis, where this probability is

calculated under the assumption that the null hypothesis is true. In the present case, we have

$$\alpha(p) = \sum_{m \le k \le n} b(n, p, k) \ .$$

Note that $\alpha(.6)$ is the probability of a type 1 error, since this is the probability of a high number of successes for an ineffective additive. So for a given n we want to choose m so as to make $\alpha(.6)$ quite small, to reduce the likelihood of a type 1 error. But as m increases above the most probable value $np = .6n$, $\alpha(.6)$, being the upper tail of a binomial distribution, approaches 0. Thus *increasing* m makes a type 1 error less likely.

Now suppose that the additive really is effective, so that p is appreciably greater than .6; say $p = .8$. (This alternative value of p is chosen arbitrarily; the following calculations depend on this choice.) Then choosing m well below $np = .8n$ will increase $\alpha(.8)$, since now $\alpha(.8)$ is all but the lower tail of a binomial distribution. Indeed, if we put $\beta(.8) = 1 - \alpha(.8)$, then $\beta(.8)$ gives us the probability of a type 2 error, and so *decreasing* m makes a type 2 error less likely.

The manufacturer would like to guard against a type 2 error, since if such an error is made, then the test does not show that the new drug is better, when in fact it is. If the alternative value of p is chosen closer to the value of p given in the null hypothesis (in this case $p = .6$), then for a given test population, the value of β will increase. So, if the manufacturer's statistician chooses an alternative value for p which is close to the value in the null hypothesis, then it will be an expensive proposition (i.e., the test population will have to be large) to reject the null hypothesis with a small value of β.

What we hope to do then, for a given test population n, is to choose a value of m, if possible, which makes both these probabilities small. If we make a type 1 error we end up buying a lot of essentially ordinary aspirin at an inflated price; a type 2 error means we miss a bargain on a superior medication. Let us say that we want our critical number m to make each of these undesirable cases less than 5 percent probable.

We write a program **PowerCurve** to plot, for $n = 100$ and selected values of m, the function $\alpha(p)$, for p ranging from .4 to 1. The result is shown in Figure 3.7. We include in our graph a box (in dotted lines) from .6 to .8, with bottom and top at heights .05 and .95. Then a value for m satisfies our requirements if and only if the graph of α enters the box from the bottom, and leaves from the top (why?—which is the type 1 and which is the type 2 criterion?). As m increases, the graph of α moves to the right. A few experiments have shown us that $m = 69$ is the smallest value for m that thwarts a type 1 error, while $m = 73$ is the largest which thwarts a type 2. So we may choose our critical value between 69 and 73. If we're more intent on avoiding a type 1 error we favor 73, and similarly we favor 69 if we regard a type 2 error as worse. Of course, the drug company may not be happy with having as much as a 5 percent chance of an error. They might insist on having a 1 percent chance of an error. For this we would have to increase the number n of trials (see Exercise 28). □

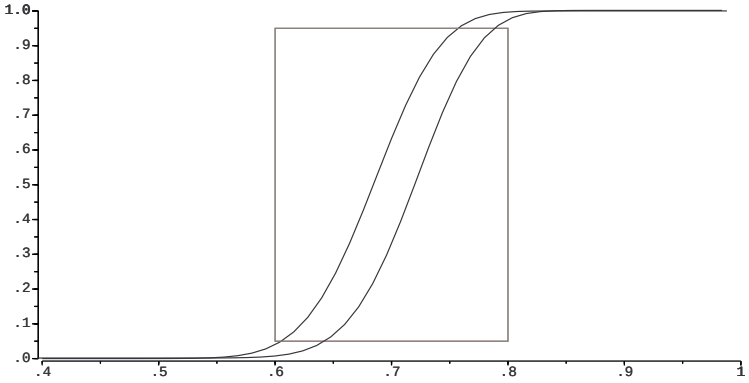


Figure 3.7: The power curve.

Binomial Expansion

We next remind the reader of an application of the binomial coefficients to algebra. This is the *binomial expansion,* from which we get the term binomial coefficient.

Theorem 3.7 (Binomial Theorem) The quantity $(a+b)^n$ can be expressed in the form

$$(a+b)^n = \sum_{j=0}^{n} \binom{n}{j} a^j b^{n-j} \ .$$

Proof. To see that this expansion is correct, write

$$(a+b)^n = (a+b)(a+b)\cdots(a+b) \ .$$

When we multiply this out we will have a sum of terms each of which results from a choice of an a or b for each of n factors. When we choose j a's and $(n-j)$ b's, we obtain a term of the form $a^j b^{n-j}$. To determine such a term, we have to specify j of the n terms in the product from which we choose the a. This can be done in $\binom{n}{j}$ ways. Thus, collecting these terms in the sum contributes a term $\binom{n}{j}a^j b^{n-j}$. □

For example, we have

$$\begin{aligned}
(a+b)^0 &= 1 \\
(a+b)^1 &= a+b \\
(a+b)^2 &= a^2+2ab+b^2 \\
(a+b)^3 &= a^3+3a^2b+3ab^2+b^3 \ .
\end{aligned}$$

We see here that the coefficients of successive powers do indeed yield Pascal's triangle.

Corollary 3.1 The sum of the elements in the nth row of Pascal's triangle is 2^n. If the elements in the nth row of Pascal's triangle are added with alternating signs, the sum is 0.

Proof. The first statement in the corollary follows from the fact that

$$2^n = (1+1)^n = \binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n} ,$$

and the second from the fact that

$$0 = (1-1)^n = \binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \cdots + (-1)^n \binom{n}{n} .$$

□

The first statement of the corollary tells us that the number of subsets of a set of n elements is 2^n. We shall use the second statement in our next application of the binomial theorem.

We have seen that, when A and B are any two events (cf. Section 1.2),

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

We now extend this theorem to a more general version, which will enable us to find the probability that at least one of a number of events occurs.

Inclusion-Exclusion Principle

Theorem 3.8 Let P be a probability distribution on a sample space Ω, and let $\{A_1,\ A_2,\ \ldots,\ A_n\}$ be a finite set of events. Then

$$P(A_1 \cup A_2 \cup \cdots \cup A_n) = \sum_{i=1}^{n} P(A_i) \ - \sum_{1 \le i < j \le n} P(A_i \cap A_j) + \sum_{1 \le i < j < k \le n} P(A_i \cap A_j \cap A_k) - \cdots . \quad (3.3)$$

That is, to find the probability that at least one of n events A_i occurs, first add the probability of each event, then subtract the probabilities of all possible two-way intersections, add the probability of all three-way intersections, and so forth.

Proof. If the outcome ω occurs in at least one of the events A_i, its probability is added exactly once by the left side of Equation 3.3. We must show that it is added exactly once by the right side of Equation 3.3. Assume that ω is in exactly k of the sets. Then its probability is added k times in the first term, subtracted $\binom{k}{2}$ times in the second, added $\binom{k}{3}$ times in the third term, and so forth. Thus, the total number of times that it is added is

$$\binom{k}{1} - \binom{k}{2} + \binom{k}{3} - \cdots (-1)^{k-1} \binom{k}{k} .$$

But

$$0 = (1-1)^k = \sum_{j=0}^{k} \binom{k}{j} (-1)^j = \binom{k}{0} - \sum_{j=1}^{k} \binom{k}{j} (-1)^{j-1} \ .$$

Hence,

$$1 = \binom{k}{0} = \sum_{j=1}^{k} \binom{k}{j} (-1)^{j-1} \ .$$

If the outcome ω is not in any of the events A_i, then it is not counted on either side of the equation. □

Hat Check Problem

Example 3.12 We return to the hat check problem discussed in Section 3.1, that is, the problem of finding the probability that a random permutation contains at least one fixed point. Recall that a permutation is a one-to-one map of a set $A = \{a_1, a_2, \ldots, a_n\}$ onto itself. Let A_i be the event that the ith element a_i remains fixed under this map. If we require that a_i is fixed, then the map of the remaining $n-1$ elements provides an arbitrary permutation of $(n-1)$ objects. Since there are $(n-1)!$ such permutations, $P(A_i) = (n-1)!/n! = 1/n$. Since there are n choices for a_i, the first term of Equation 3.3 is 1. In the same way, to have a particular pair (a_i, a_j) fixed, we can choose any permutation of the remaining $n-2$ elements; there are $(n-2)!$ such choices and thus

$$P(A_i \cap A_j) = \frac{(n-2)!}{n!} = \frac{1}{n(n-1)} \ .$$

The number of terms of this form in the right side of Equation 3.3 is

$$\binom{n}{2} = \frac{n(n-1)}{2!} \ .$$

Hence, the second term of Equation 3.3 is

$$-\frac{n(n-1)}{2!} \cdot \frac{1}{n(n-1)} = -\frac{1}{2!} \ .$$

Similarly, for any specific three events A_i, A_j, A_k,

$$P(A_i \cap A_j \cap A_k) = \frac{(n-3)!}{n!} = \frac{1}{n(n-1)(n-2)} \ ,$$

and the number of such terms is

$$\binom{n}{3} = \frac{n(n-1)(n-2)}{3!} \ ,$$

making the third term of Equation 3.3 equal to $1/3!$. Continuing in this way, we obtain

$$P(\text{at least one fixed point}) = 1 - \frac{1}{2!} + \frac{1}{3!} - \cdots (-1)^{n-1} \frac{1}{n!}$$

n	Probability that no one gets his own hat back
3	.333333
4	.375
5	.366667
6	.368056
7	.367857
8	.367882
9	.367879
10	.367879

Table 3.9: Hat check problem.

and

$$P(\text{no fixed point}) = \frac{1}{2!} - \frac{1}{3!} + \cdots (-1)^n \frac{1}{n!} \ .$$

From calculus we learn that

$$e^x = 1 + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \cdots + \frac{1}{n!}x^n + \cdots \ .$$

Thus, if $x = -1$, we have

$$\begin{aligned} e^{-1} &= \frac{1}{2!} - \frac{1}{3!} + \cdots + \frac{(-1)^n}{n!} + \cdots \\ &= .3678794 \ . \end{aligned}$$

Therefore, the probability that there is no fixed point, i.e., that none of the n people gets his own hat back, is equal to the sum of the first n terms in the expression for e^{-1}. This series converges very fast. Calculating the partial sums for $n = 3$ to 10 gives the data in Table 3.9.

After $n = 9$ the probabilities are essentially the same to six significant figures. Interestingly, the probability of no fixed point alternately increases and decreases as n increases. Finally, we note that our exact results are in good agreement with our simulations reported in the previous section. □

Choosing a Sample Space

We now have some of the tools needed to accurately describe sample spaces and to assign probability functions to those sample spaces. Nevertheless, in some cases, the description and assignment process is somewhat arbitrary. Of course, it is to be hoped that the description of the sample space and the subsequent assignment of a probability function will yield a model which accurately predicts what would happen if the experiment were actually carried out. As the following examples show, there are situations in which "reasonable" descriptions of the sample space do not produce a model which fits the data.

In Feller's book,[14] a pair of models is given which describe arrangements of certain kinds of elementary particles, such as photons and protons. It turns out that

[14]W. Feller, *Introduction to Probability Theory and Its Applications* vol. 1, 3rd ed. (New York: John Wiley and Sons, 1968), p. 41

experiments have shown that certain types of elementary particles exhibit behavior which is accurately described by one model, called *"Bose-Einstein statistics,"* while other types of elementary particles can be modelled using *"Fermi-Dirac statistics."* Feller says:

> We have here an instructive example of the impossibility of selecting or justifying probability models by *a priori* arguments. In fact, no pure reasoning could tell that photons and protons would not obey the same probability laws.

We now give some examples of this description and assignment process.

Example 3.13 In the quantum mechanical model of the helium atom, various parameters can be used to classify the energy states of the atom. In the triplet spin state ($S = 1$) with orbital angular momentum 1 ($L = 1$), there are three possibilities, 0, 1, or 2, for the total angular momentum (J). (It is not assumed that the reader knows what any of this means; in fact, the example is more illustrative if the reader does *not* know anything about quantum mechanics.) We would like to assign probabilities to the three possibilities for J. The reader is undoubtedly resisting the idea of assigning the probability of $1/3$ to each of these outcomes. She should now ask herself why she is resisting this assignment. The answer is probably because she does not have any "intuition" (i.e., experience) about the way in which helium atoms behave. In fact, in this example, the probabilities $1/9$, $3/9$, and $5/9$ are assigned by the theory. The theory gives these assignments because these frequencies were observed *in experiments* and further parameters were developed in the theory to allow these frequencies to be predicted. □

Example 3.14 Suppose two pennies are flipped once each. There are several "reasonable" ways to describe the sample space. One way is to count the number of heads in the outcome; in this case, the sample space can be written $\{0, 1, 2\}$. Another description of the sample space is the set of all ordered pairs of H's and T's, i.e.,

$$\{(H, H), (H, T), (T, H), (T, T)\}.$$

Both of these descriptions are accurate ones, but it is easy to see that (at most) one of these, if assigned a constant probability function, can claim to accurately model reality. In this case, as opposed to the preceding example, the reader will probably say that the second description, with each outcome being assigned a probability of $1/4$, is the "right" description. This conviction is due to experience; there is no proof that this is the way reality works. □

The reader is also referred to Exercise 26 for another example of this process.

1	1	1	1	1	1	1	1	1	1
1	2	3	4	5	6	7	8	9	
1	3	6	10	15	21	28	36		
1	4	10	20	35	56	84			
1	5	15	35	70	126				
1	6	21	56	126					
1	7	28	84						
1	8	36							
1	9								
1									

Table 3.10: Pascal's triangle.

Historical Remarks

The binomial coefficients have a long and colorful history leading up to Pascal's *Treatise on the Arithmetical Triangle,*[15] where Pascal developed many important properties of these numbers. This history is set forth in the book *Pascal's Arithmetical Triangle* by A. W. F. Edwards.[16] Pascal wrote his triangle in the form shown in Table 3.10.

Edwards traces three different ways that the binomial coefficients arose. He refers to these as the *figurate numbers,* the *combinatorial numbers,* and the *binomial numbers.* They are all names for the same thing (which we have called binomial coefficients) but that they are all the same was not appreciated until the sixteenth century.

The *figurate numbers* date back to the Pythagorean interest in number patterns around 540 BC. The Pythagoreans considered, for example, triangular patterns shown in Figure 3.8. The sequence of numbers

$$1, 3, 6, 10, \ldots$$

obtained as the number of points in each triangle are called *triangular numbers.* From the triangles it is clear that the nth triangular number is simply the sum of the first n integers. The *tetrahedral numbers* are the sums of the triangular numbers and were obtained by the Greek mathematicians Theon and Nicomachus at the beginning of the second century BC. The tetrahedral number 10, for example, has the geometric representation shown in Figure 3.9. The first three types of figurate numbers can be represented in tabular form as shown in Table 3.11.

These numbers provide the first four rows of Pascal's triangle, but the table was not to be completed in the West until the sixteenth century.

In the East, Hindu mathematicians began to encounter the binomial coefficients in combinatorial problems. Bhaskara in his *Lilavati* of 1150 gave a rule to find the number of medicinal preparations using 1, 2, 3, 4, 5, or 6 possible ingredients.[17] His

[15] B. Pascal, *Traité du Triangle Arithmétique* (Paris: Desprez, 1665).
[16] A. W. F. Edwards, *Pascal's Arithmetical Triangle* (London: Griffin, 1987).
[17] ibid., p. 27.

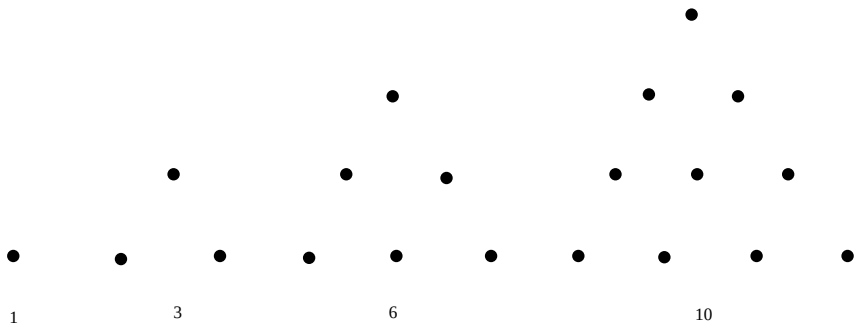


Figure 3.8: Pythagorean triangular patterns.

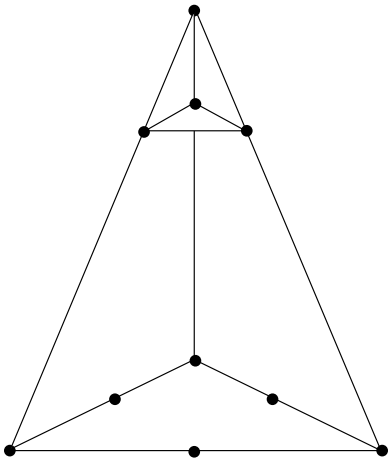


Figure 3.9: Geometric representation of the tetrahedral number 10.

natural numbers	1	2	3	4	5	6	7	8	9
triangular numbers	1	3	6	10	15	21	28	36	45
tetrahedral numbers	1	4	10	20	35	56	84	120	165

Table 3.11: Figurate numbers.

11					
12	22				
13	23	33			
14	24	34	44		
15	25	35	45	55	
16	26	36	46	56	66

Table 3.12: Outcomes for the roll of two dice.

rule is equivalent to our formula

$$\binom{n}{r} = \frac{(n)_r}{r!} \ .$$

The binomial numbers as coefficients of $(a+b)^n$ appeared in the works of mathematicians in China around 1100. There are references about this time to "the tabulation system for unlocking binomial coefficients." The triangle to provide the coefficients up to the eighth power is given by Chu Shih-chieh in a book written around 1303 (see Figure 3.10).[18] The original manuscript of Chu's book has been lost, but copies have survived. Edwards notes that there is an error in this copy of Chu's triangle. Can you find it? (*Hint*: Two numbers which should be equal are not.) Other copies do not show this error.

The first appearance of Pascal's triangle in the West seems to have come from calculations of Tartaglia in calculating the number of possible ways that n dice might turn up.[19] For one die the answer is clearly 6. For two dice the possibilities may be displayed as shown in Table 3.12.

Displaying them this way suggests the sixth triangular number $1+2+3+4+5+6=21$ for the throw of 2 dice. Tartaglia "on the first day of Lent, 1523, in Verona, having thought about the problem all night,"[20] realized that the extension of the figurate table gave the answers for n dice. The problem had suggested itself to Tartaglia from watching people casting their own horoscopes by means of a *Book of Fortune,* selecting verses by a process which included noting the numbers on the faces of three dice. The 56 ways that three dice can fall were set out on each page. The way the numbers were written in the book did not suggest the connection with figurate numbers, but a method of enumeration similar to the one we used for 2 dice does. Tartaglia's table was not published until 1556.

A table for the binomial coefficients was published in 1554 by the German mathematician Stifel.[21] Pascal's triangle appears also in Cardano's *Opus novum* of 1570.[22] Cardano was interested in the problem of finding the number of ways to choose r

[18]J. Needham, *Science and Civilization in China,* vol. 3 (New York: Cambridge University Press, 1959), p. 135.
[19]N. Tartaglia, *General Trattato di Numeri et Misure* (Vinegia, 1556).
[20]Quoted in Edwards, op. cit., p. 37.
[21]M. Stifel, *Arithmetica Integra* (Norimburgae, 1544).
[22]G. Cardano, *Opus Novum de Proportionibus Numerorum* (Basilea, 1570).

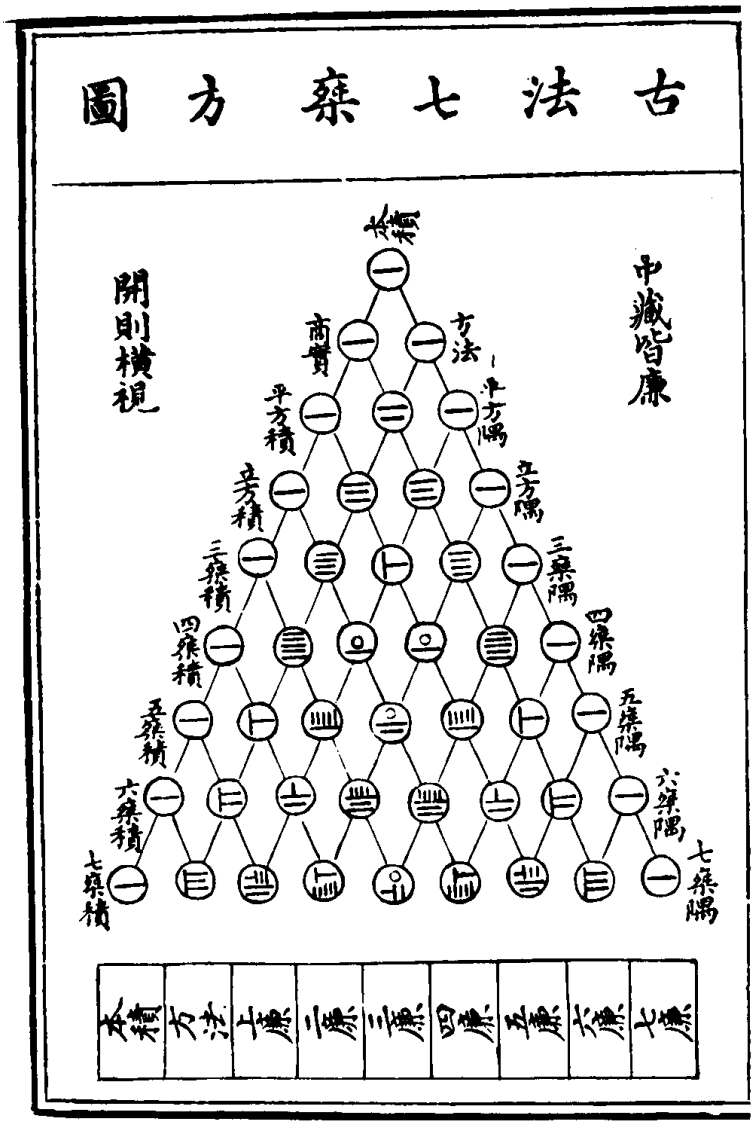


Figure 3.10: Chu Shih-chieh's triangle. [From J. Needham, *Science and Civilization in China,* vol. 3 (New York: Cambridge University Press, 1959), p. 135. Reprinted with permission.]

	if each one staken 256 in					
From my opponent's 256	6	5	4	3	2	1
positions I get, for the	games	games	games	games	games	games
1st game	63	70	80	96	128	256
2nd game	63	70	80	96	128	
3rd game	56	60	64	64		
4th game	42	40	32			
5th game	24	16				
6th game	8					

Table 3.13: Pascal's solution for the problem of points.

objects out of n. Thus by the time of Pascal's work, his triangle had appeared as a result of looking at the figurate numbers, the combinatorial numbers, and the binomial numbers, and the fact that all three were the same was presumably pretty well understood.

Pascal's interest in the binomial numbers came from his letters with Fermat concerning a problem known as the problem of points. This problem, and the correspondence between Pascal and Fermat, were discussed in Chapter 1. The reader will recall that this problem can be described as follows: Two players A and B are playing a sequence of games and the first player to win n games wins the match. It is desired to find the probability that A wins the match at a time when A has won a games and B has won b games. (See Exercises 4.1.40-4.1.42.)

Pascal solved the problem by backward induction, much the way we would do today in writing a computer program for its solution. He referred to the combinatorial method of Fermat which proceeds as follows: If A needs c games and B needs d games to win, we require that the players continue to play until they have played $c+d-1$ games. The winner in this extended series will be the same as the winner in the original series. The probability that A wins in the extended series and hence in the original series is

$$\sum_{r=c}^{c+d-1} \frac{1}{2^{c+d-1}} \binom{c+d-1}{r} .$$

Even at the time of the letters Pascal seemed to understand this formula.

Suppose that the first player to win n games wins the match, and suppose that each player has put up a stake of x. Pascal studied the value of winning a particular game. By this he meant the increase in the expected winnings of the winner of the particular game under consideration. He showed that the value of the first game is

$$\frac{1 \cdot 3 \cdot 5 \cdot \ldots \cdot (2n-1)}{2 \cdot 4 \cdot 6 \cdot \ldots \cdot (2n)} x .$$

His proof of this seems to use Fermat's formula and the fact that the above ratio of products of odd to products of even numbers is equal to the probability of exactly n heads in $2n$ tosses of a coin. (See Exercise 39.)

Pascal presented Fermat with the table shown in Table 3.13. He states:

> You will see as always, that the value of the first game is equal to that of the second which is easily shown by combinations. You will see, in the same way, that the numbers in the first line are always increasing; so also are those in the second; and those in the third. But those in the fourth line are decreasing, and those in the fifth, etc. This seems odd.[23]

The student can pursue this question further using the computer and Pascal's backward iteration method for computing the expected payoff at any point in the series.

In his treatise, Pascal gave a formal proof of Fermat's combinatorial formula as well as proofs of many other basic properties of binomial numbers. Many of his proofs involved induction and represent some of the first proofs by this method. His book brought together all the different aspects of the numbers in the Pascal triangle as known in 1654, and, as Edwards states, "That the Arithmetical Triangle should bear Pascal's name cannot be disputed."[24]

The first serious study of the binomial distribution was undertaken by James Bernoulli in his *Ars Conjectandi* published in 1713.[25] We shall return to this work in the historical remarks in Chapter 8.

Exercises

1 Compute the following:

(a) $\binom{6}{3}$
(b) $b(5, .2, 4)$
(c) $\binom{7}{2}$
(d) $\binom{26}{26}$
(e) $b(4, .2, 3)$
(f) $\binom{6}{2}$
(g) $\binom{10}{9}$
(h) $b(8, .3, 5)$

2 In how many ways can we choose five people from a group of ten to form a committee?

3 How many seven-element subsets are there in a set of nine elements?

4 Using the relation Equation 3.1 write a program to compute Pascal's triangle, putting the results in a matrix. Have your program print the triangle for $n = 10$.

5 Use the program **BinomialProbabilities** to find the probability that, in 100 tosses of a fair coin, the number of heads that turns up lies between 35 and 65, between 40 and 60, and between 45 and 55.

[23] F. N. David, op. cit., p. 235.
[24] A. W. F. Edwards, op. cit., p. ix.
[25] J. Bernoulli, *Ars Conjectandi* (Basil: Thurnisiorum, 1713).

6 Charles claims that he can distinguish between beer and ale 75 percent of the time. Ruth bets that he cannot and, in fact, just guesses. To settle this, a bet is made: Charles is to be given ten small glasses, each having been filled with beer or ale, chosen by tossing a fair coin. He wins the bet if he gets seven or more correct. Find the probability that Charles wins if he has the ability that he claims. Find the probability that Ruth wins if Charles is guessing.

7 Show that

$$b(n,p,j) = \frac{p}{q}\left(\frac{n-j+1}{j}\right) b(n,p,j-1) \ ,$$

for $j \geq 1$. Use this fact to determine the value or values of j which give $b(n,p,j)$ its greatest value. *Hint*: Consider the successive ratios as j increases.

8 A die is rolled 30 times. What is the probability that a 6 turns up exactly 5 times? What is the most probable number of times that a 6 will turn up?

9 Find integers n and r such that the following equation is true:

$$\binom{13}{5} + 2\binom{13}{6} + \binom{13}{7} = \binom{n}{r} \ .$$

10 In a ten-question true-false exam, find the probability that a student gets a grade of 70 percent or better by guessing. Answer the same question if the test has 30 questions, and if the test has 50 questions.

11 A restaurant offers apple and blueberry pies and stocks an equal number of each kind of pie. Each day ten customers request pie. They choose, with equal probabilities, one of the two kinds of pie. How many pieces of each kind of pie should the owner provide so that the probability is about .95 that each customer gets the pie of his or her own choice?

12 A poker hand is a set of 5 cards randomly chosen from a deck of 52 cards. Find the probability of a

(a) royal flush (ten, jack, queen, king, ace in a single suit).

(b) straight flush (five in a sequence in a single suit, but not a royal flush).

(c) four of a kind (four cards of the same face value).

(d) full house (one pair and one triple, each of the same face value).

(e) flush (five cards in a single suit but not a straight or royal flush).

(f) straight (five cards in a sequence, not all the same suit). (Note that in straights, an ace counts high or low.)

13 If a set has $2n$ elements, show that it has more subsets with n elements than with any other number of elements.

14 Let $b(2n, .5, n)$ be the probability that in $2n$ tosses of a fair coin exactly n heads turn up. Using Stirling's formula (Theorem 3.3), show that $b(2n, .5, n) \sim 1/\sqrt{\pi n}$. Use the program **BinomialProbabilities** to compare this with the exact value for $n = 10$ to 25.

15 A baseball player, Smith, has a batting average of .300 and in a typical game comes to bat three times. Assume that Smith's hits in a game can be considered to be a Bernoulli trials process with probability .3 for *success.* Find the probability that Smith gets 0, 1, 2, and 3 hits.

16 The Siwash University football team plays eight games in a season, winning three, losing three, and ending two in a tie. Show that the number of ways that this can happen is

$$\binom{8}{3}\binom{5}{3} = \frac{8!}{3!\,3!\,2!}\ .$$

17 Using the technique of Exercise 16, show that the number of ways that one can put n different objects into three boxes with a in the first, b in the second, and c in the third is $n!/(a!\,b!\,c!)$.

18 Baumgartner, Prosser, and Crowell are grading a calculus exam. There is a true-false question with ten parts. Baumgartner notices that one student has only two out of the ten correct and remarks, "The student was not even bright enough to have flipped a coin to determine his answers." "Not so clear," says Prosser. "With 340 students I bet that if they all flipped coins to determine their answers there would be at least one exam with two or fewer answers correct." Crowell says, "I'm with Prosser. In fact, I bet that we should expect at least one exam in which no answer is correct if everyone is just guessing." Who is right in all of this?

19 A gin hand consists of 10 cards from a deck of 52 cards. Find the probability that a gin hand has

(a) all 10 cards of the same suit.

(b) exactly 4 cards in one suit and 3 in two other suits.

(c) a 4, 3, 2, 1, distribution of suits.

20 A six-card hand is dealt from an ordinary deck of cards. Find the probability that:

(a) All six cards are hearts.

(b) There are three aces, two kings, and one queen.

(c) There are three cards of one suit and three of another suit.

21 A lady wishes to color her fingernails on one hand using at most two of the colors red, yellow, and blue. How many ways can she do this?

22 How many ways can six indistinguishable letters be put in three mail boxes? *Hint*: One representation of this is given by a sequence |LL|L|LLL| where the |'s represent the partitions for the boxes and the L's the letters. Any possible way can be so described. Note that we need two bars at the ends and the remaining two bars and the six L's can be put in any order.

23 Using the method for the hint in Exercise 22, show that r indistinguishable objects can be put in n boxes in

$$\binom{n+r-1}{n-1} = \binom{n+r-1}{r}$$

different ways.

24 A travel bureau estimates that when 20 tourists go to a resort with ten hotels they distribute themselves as if the bureau were putting 20 indistinguishable objects into ten distinguishable boxes. Assuming this model is correct, find the probability that no hotel is left vacant when the first group of 20 tourists arrives.

25 An elevator takes on six passengers and stops at ten floors. We can assign two different equiprobable measures for the ways that the passengers are discharged: (a) we consider the passengers to be distinguishable or (b) we consider them to be indistinguishable (see Exercise 23 for this case). For each case, calculate the probability that all the passengers get off at different floors.

26 You are playing *heads or tails* with Prosser but you suspect that his coin is unfair. Von Neumann suggested that you proceed as follows: Toss Prosser's coin twice. If the outcome is HT call the result *win.* if it is TH call the result *lose.* If it is TT or HH ignore the outcome and toss Prosser's coin twice again. Keep going until you get either an HT or a TH and call the result win or lose in a single play. Repeat this procedure for each play. Assume that Prosser's coin turns up heads with probability p.

(a) Find the probability of HT, TH, HH, TT with two tosses of Prosser's coin.

(b) Using part (a), show that the probability of a win on any one play is $1/2$, no matter what p is.

27 John claims that he has extrasensory powers and can tell which of two symbols is on a card turned face down (see Example 3.11). To test his ability he is asked to do this for a sequence of trials. Let the null hypothesis be that he is just guessing, so that the probability is $1/2$ of his getting it right each time, and let the alternative hypothesis be that he can name the symbol correctly more than half the time. Devise a test with the property that the probability of a type 1 error is less than .05 and the probability of a type 2 error is less than .05 if John can name the symbol correctly 75 percent of the time.

28 In Example 3.11 assume the alternative hypothesis is that $p = .8$ and that it is desired to have the probability of each type of error less than .01. Use the program **PowerCurve** to determine values of n and m that will achieve this. Choose n as small as possible.

29 A drug is assumed to be effective with an unknown probability p. To estimate p the drug is given to n patients. It is found to be effective for m patients.

The *method of maximum likelihood* for estimating p states that we should choose the value for p that gives the highest probability of getting what we got on the experiment. Assuming that the experiment can be considered as a Bernoulli trials process with probability p for success, show that the maximum likelihood estimate for p is the proportion m/n of successes.

30 Recall that in the World Series the first team to win four games wins the series. The series can go at most seven games. Assume that the Red Sox and the Mets are playing the series. Assume that the Mets win each game with probability p. Fermat observed that even though the series might not go seven games, the probability that the Mets win the series is the same as the probability that they win four or more game in a series that was forced to go seven games no matter who wins the individual games.

(a) Using the program **PowerCurve** of Example 3.11 find the probability that the Mets win the series for the cases $p = .5$, $p = .6$, $p = .7$.

(b) Assume that the Mets have probability .6 of winning each game. Use the program **PowerCurve** to find a value of n so that, if the series goes to the first team to win more than half the games, the Mets will have a 95 percent chance of winning the series. Choose n as small as possible.

31 Each of the four engines on an airplane functions correctly on a given flight with probability .99, and the engines function independently of each other. Assume that the plane can make a safe landing if at least two of its engines are functioning correctly. What is the probability that the engines will allow for a safe landing?

32 A small boy is lost coming down Mount Washington. The leader of the search team estimates that there is a probability p that he came down on the east side and a probability $1 - p$ that he came down on the west side. He has n people in his search team who will search independently and, if the boy is on the side being searched, each member will find the boy with probability u. Determine how he should divide the n people into two groups to search the two sides of the mountain so that he will have the highest probability of finding the boy. How does this depend on u?

***33** $2n$ balls are chosen at random from a total of $2n$ red balls and $2n$ blue balls. Find a combinatorial expression for the probability that the chosen balls are equally divided in color. Use Stirling's formula to estimate this probability. Using **BinomialProbabilities**, compare the exact value with Stirling's approximation for $n = 20$.

34 Assume that every time you buy a box of Wheaties, you receive one of the pictures of the n players on the New York Yankees. Over a period of time, you buy $m \geq n$ boxes of Wheaties.

(a) Use Theorem 3.8 to show that the probability that you get all n pictures is

$$1 - \binom{n}{1}\left(\frac{n-1}{n}\right)^m + \binom{n}{2}\left(\frac{n-2}{n}\right)^m - \cdots + (-1)^{n-1}\binom{n}{n-1}\left(\frac{1}{n}\right)^m.$$

Hint: Let E_k be the event that you do not get the kth player's picture.

(b) Write a computer program to compute this probability. Use this program to find, for given n, the smallest value of m which will give probability $\geq .5$ of getting all n pictures. Consider $n = 50$, 100, and 150 and show that $m = n\log n + n\log 2$ is a good estimate for the number of boxes needed. (For a derivation of this estimate, see Feller.[26])

***35** Prove the following *binomial identity*

$$\binom{2n}{n} = \sum_{j=0}^{n}\binom{n}{j}^2.$$

Hint: Consider an urn with n red balls and n blue balls inside. Show that each side of the equation equals the number of ways to choose n balls from the urn.

36 Let j and n be positive integers, with $j \leq n$. An experiment consists of choosing, at random, a j-tuple of *positive* integers whose sum is at most n.

(a) Find the size of the sample space. *Hint*: Consider n indistinguishable balls placed in a row. Place j markers between consecutive pairs of balls, with no two markers between the same pair of balls. (We also allow one of the n markers to be placed at the end of the row of balls.) Show that there is a 1-1 correspondence between the set of possible positions for the markers and the set of j-tuples whose size we are trying to count.

(b) Find the probability that the j-tuple selected contains at least one 1.

37 Let $n \pmod m$ denote the remainder when the integer n is divided by the integer m. Write a computer program to compute the numbers $\binom{n}{j} \pmod m$ where $\binom{n}{j}$ is a binomial coefficient and m is an integer. You can do this by using the recursion relations for generating binomial coefficients, doing all the arithmetic using the basic function $\mathrm{mod}(n, m)$. Try to write your program to make as large a table as possible. Run your program for the cases $m = 2$ to 7. Do you see any patterns? In particular, for the case $m = 2$ and n a power of 2, verify that all the entries in the $(n-1)$st row are 1. (The corresponding binomial numbers are odd.) Use your pictures to explain why this is true.

[26] W. Feller, *Introduction to Probability Theory and its Applications,* vol. I, 3rd ed. (New York: John Wiley & Sons, 1968), p. 106.

Chapter 4

Conditional Probability

4.1 Discrete Conditional Probability

Conditional Probability

In this section we ask and answer the following question. Suppose we assign a distribution function to a sample space and then learn that an event E has occurred. How should we change the probabilities of the remaining events? We shall call the new probability for an event F the *conditional probability of F given E* and denote it by $P(F|E)$.

Example 4.1 An experiment consists of rolling a die once. Let X be the outcome. Let F be the event $\{X = 6\}$, and let E be the event $\{X > 4\}$. We assign the distribution function $m(\omega) = 1/6$ for $\omega = 1, 2, \ldots, 6$. Thus, $P(F) = 1/6$. Now suppose that the die is rolled and we are told that the event E has occurred. This leaves only two possible outcomes: 5 and 6. In the absence of any other information, we would still regard these outcomes to be equally likely, so the probability of F becomes $1/2$, making $P(F|E) = 1/2$. □

Example 4.2 In the Life Table (see Appendix C), one finds that in a population of 100,000 females, 89.835% can expect to live to age 60, while 57.062% can expect to live to age 80. Given that a woman is 60, what is the probability that she lives to age 80?

This is an example of a conditional probability. In this case, the original sample space can be thought of as a set of 100,000 females. The events E and F are the subsets of the sample space consisting of all women who live at least 60 years, and at least 80 years, respectively. We consider E to be the new sample space, and note that F is a subset of E. Thus, the size of E is 89,835, and the size of F is 57,062. So, the probability in question equals 57,062/89,835 = .6352. Thus, a woman who is 60 has a 63.52% chance of living to age 80. □

Example 4.3 Consider our voting example from Section 1.2: three candidates A, B, and C are running for office. We decided that A and B have an equal chance of winning and C is only 1/2 as likely to win as A. Let A be the event "A wins," B that "B wins," and C that "C wins." Hence, we assigned probabilities $P(A) = 2/5$, $P(B) = 2/5$, and $P(C) = 1/5$.

Suppose that before the election is held, A drops out of the race. As in Example 4.1, it would be natural to assign new probabilities to the events B and C which are proportional to the original probabilities. Thus, we would have $P(B|\ A) = 2/3$, and $P(C|\ A) = 1/3$. It is important to note that any time we assign probabilities to real-life events, the resulting distribution is only useful if we take into account all relevant information. In this example, we may have knowledge that most voters who favor A will vote for C if A is no longer in the race. This will clearly make the probability that C wins greater than the value of 1/3 that was assigned above. □

In these examples we assigned a distribution function and then were given new information that determined a new sample space, consisting of the outcomes that are still possible, and caused us to assign a new distribution function to this space.

We want to make formal the procedure carried out in these examples. Let $\Omega = \{\omega_1, \omega_2, \ldots, \omega_r\}$ be the original sample space with distribution function $m(\omega_j)$ assigned. Suppose we learn that the event E has occurred. We want to assign a new distribution function $m(\omega_j|E)$ to Ω to reflect this fact. Clearly, if a sample point ω_j is not in E, we want $m(\omega_j|E) = 0$. Moreover, in the absence of information to the contrary, it is reasonable to assume that the probabilities for ω_k in E should have the same relative magnitudes that they had before we learned that E had occurred. For this we require that

$$m(\omega_k|E) = cm(\omega_k)$$

for all ω_k in E, with c some positive constant. But we must also have

$$\sum_E m(\omega_k|E) = c\sum_E m(\omega_k) = 1\ .$$

Thus,

$$c = \frac{1}{\sum_E m(\omega_k)} = \frac{1}{P(E)}\ .$$

(Note that this requires us to assume that $P(E) > 0$.) Thus, we will define

$$m(\omega_k|E) = \frac{m(\omega_k)}{P(E)}$$

for ω_k in E. We will call this new distribution the *conditional distribution* given E. For a general event F, this gives

$$P(F|E) = \sum_{F\cap E} m(\omega_k|E) = \sum_{F\cap E} \frac{m(\omega_k)}{P(E)} = \frac{P(F\cap E)}{P(E)}\ .$$

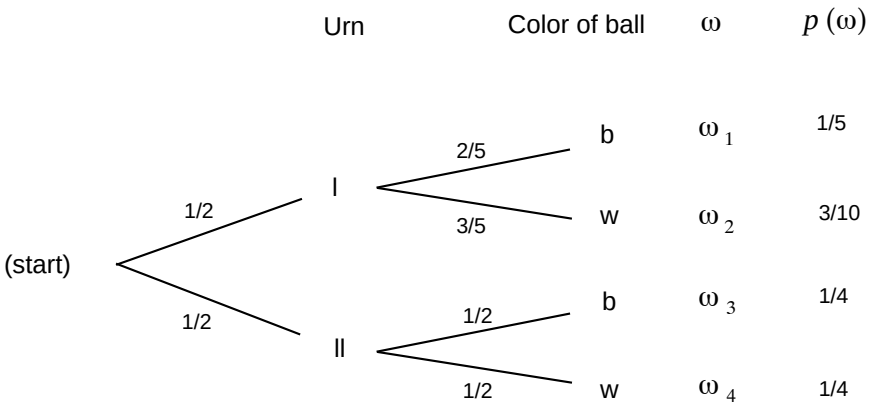


Figure 4.1: Tree diagram.

We call $P(F|E)$ the *conditional probability of F occurring given that E occurs,* and compute it using the formula

$$P(F|E) = \frac{P(F \cap E)}{P(E)} \ .$$

Example 4.4 (Example 4.1 continued) Let us return to the example of rolling a die. Recall that F is the event $X = 6$, and E is the event $X > 4$. Note that $E \cap F$ is the event F. So, the above formula gives

$$\begin{aligned} P(F|E) &= \frac{P(F \cap E)}{P(E)} \\ &= \frac{1/6}{1/3} \\ &= \frac{1}{2} \ , \end{aligned}$$

in agreement with the calculations performed earlier. □

Example 4.5 We have two urns, I and II. Urn I contains 2 black balls and 3 white balls. Urn II contains 1 black ball and 1 white ball. An urn is drawn at random and a ball is chosen at random from it. We can represent the sample space of this experiment as the paths through a tree as shown in Figure 4.1. The probabilities assigned to the paths are also shown.

Let B be the event "a black ball is drawn," and I the event "urn I is chosen." Then the branch weight 2/5, which is shown on one branch in the figure, can now be interpreted as the conditional probability $P(B|I)$.

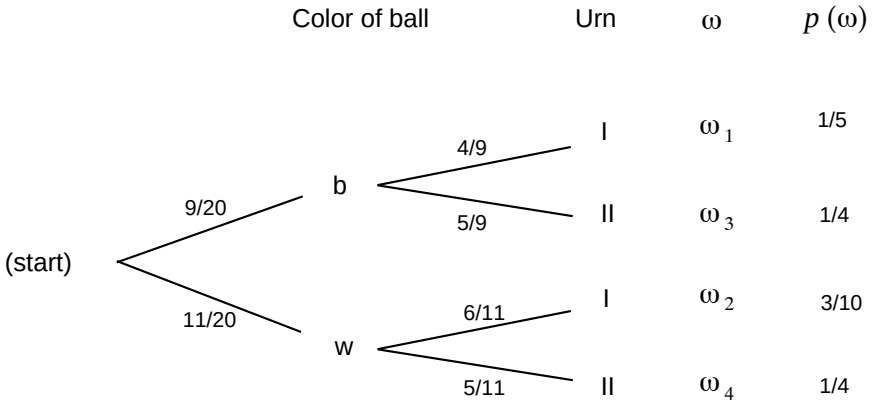


Figure 4.2: Reverse tree diagram.

Suppose we wish to calculate $P(I|B)$. Using the formula, we obtain

$$\begin{aligned} P(I|B) &= \frac{P(I \cap B)}{P(B)} \\ &= \frac{P(I \cap B)}{P(B \cap I) + P(B \cap II)} \\ &= \frac{1/5}{1/5 + 1/4} = \frac{4}{9} \, . \end{aligned}$$

□

Bayes Probabilities

Our original tree measure gave us the probabilities for drawing a ball of a given color, given the urn chosen. We have just calculated the *inverse probability* that a particular urn was chosen, given the color of the ball. Such an inverse probability is called a *Bayes probability* and may be obtained by a formula that we shall develop later. Bayes probabilities can also be obtained by simply constructing the tree measure for the two-stage experiment carried out in reverse order. We show this tree in Figure 4.2.

The paths through the reverse tree are in one-to-one correspondence with those in the forward tree, since they correspond to individual outcomes of the experiment, and so they are assigned the same probabilities. From the forward tree, we find that the probability of a black ball is

$$\frac{1}{2} \cdot \frac{2}{5} + \frac{1}{2} \cdot \frac{1}{2} = \frac{9}{20} \, .$$

The probabilities for the branches at the second level are found by simple division. For example, if x is the probability to be assigned to the top branch at the second level, we must have

$$\frac{9}{20} \cdot x = \frac{1}{5}$$

or $x = 4/9$. Thus, $P(I|B) = 4/9$, in agreement with our previous calculations. The reverse tree then displays all of the inverse, or Bayes, probabilities.

Example 4.6 We consider now a problem called the *Monty Hall* problem. This has long been a favorite problem but was revived by a letter from Craig Whitaker to Marilyn vos Savant for consideration in her column in *Parade Magazine*.[1] Craig wrote:

> Suppose you're on Monty Hall's *Let's Make a Deal!* You are given the choice of three doors, behind one door is a car, the others, goats. You pick a door, say 1, Monty opens another door, say 3, which has a goat. Monty says to you "Do you want to pick door 2?" Is it to your advantage to switch your choice of doors?

Marilyn gave a solution concluding that you should switch, and if you do, your probability of winning is 2/3. Several irate readers, some of whom identified themselves as having a PhD in mathematics, said that this is absurd since after Monty has ruled out one door there are only two possible doors and they should still each have the same probability 1/2 so there is no advantage to switching. Marilyn stuck to her solution and encouraged her readers to simulate the game and draw their own conclusions from this. We also encourage the reader to do this (see Exercise 11).

Other readers complained that Marilyn had not described the problem completely. In particular, the way in which certain decisions were made during a play of the game were not specified. This aspect of the problem will be discussed in Section 4.3. We will assume that the car was put behind a door by rolling a three-sided die which made all three choices equally likely. Monty knows where the car is, and always opens a door with a goat behind it. Finally, we assume that if Monty has a choice of doors (i.e., the contestant has picked the door with the car behind it), he chooses each door with probability 1/2. Marilyn clearly expected her readers to assume that the game was played in this manner.

As is the case with most apparent paradoxes, this one can be resolved through careful analysis. We begin by describing a simpler, related question. We say that a contestant is using the "stay" strategy if he picks a door, and, if offered a chance to switch to another door, declines to do so (i.e., he stays with his original choice). Similarly, we say that the contestant is using the "switch" strategy if he picks a door, and, if offered a chance to switch to another door, takes the offer. Now suppose that a contestant decides in advance to play the "stay" strategy. His only action in this case is to pick a door (and decline an invitation to switch, if one is offered). What is the probability that he wins a car? The same question can be asked about the "switch" strategy.

Using the "stay" strategy, a contestant will win the car with probability 1/3, since 1/3 of the time the door he picks will have the car behind it. On the other hand, if a contestant plays the "switch" strategy, then he will win whenever the

[1] Marilyn vos Savant, Ask Marilyn, *Parade Magazine*, 9 September; 2 December; 17 February 1990, reprinted in Marilyn vos Savant, *Ask Marilyn*, St. Martins, New York, 1992.

door he originally picked does not have the car behind it, which happens 2/3 of the time.

This very simple analysis, though correct, does not quite solve the problem that Craig posed. Craig asked for the conditional probability that you win if you switch, given that you have chosen door 1 and that Monty has chosen door 3. To solve this problem, we set up the problem before getting this information and then compute the conditional probability given this information. This is a process that takes place in several stages; the car is put behind a door, the contestant picks a door, and finally Monty opens a door. Thus it is natural to analyze this using a tree measure. Here we make an additional assumption that if Monty has a choice of doors (i.e., the contestant has picked the door with the car behind it) then he picks each door with probability 1/2. The assumptions we have made determine the branch probabilities and these in turn determine the tree measure. The resulting tree and tree measure are shown in Figure 4.3. It is tempting to reduce the tree's size by making certain assumptions such as: "Without loss of generality, we will assume that the contestant always picks door 1." We have chosen not to make any such assumptions, in the interest of clarity.

Now the given information, namely that the contestant chose door 1 and Monty chose door 3, means only two paths through the tree are possible (see Figure 4.4). For one of these paths, the car is behind door 1 and for the other it is behind door 2. The path with the car behind door 2 is twice as likely as the one with the car behind door 1. Thus the conditional probability is 2/3 that the car is behind door 2 and 1/3 that it is behind door 1, so if you switch you have a 2/3 chance of winning the car, as Marilyn claimed.

At this point, the reader may think that the two problems above are the same, since they have the same answers. Recall that we assumed in the original problem if the contestant chooses the door with the car, so that Monty has a choice of two doors, he chooses each of them with probability 1/2. Now suppose instead that in the case that he has a choice, he chooses the door with the larger number with probability 3/4. In the "switch" vs. "stay" problem, the probability of winning with the "switch" strategy is still 2/3. However, in the original problem, if the contestant switches, he wins with probability 4/7. The reader can check this by noting that the same two paths as before are the only two possible paths in the tree. The path leading to a win, if the contestant switches, has probability 1/3, while the path which leads to a loss, if the contestant switches, has probability 1/4. □

Independent Events

It often happens that the knowledge that a certain event E has occurred has no effect on the probability that some other event F has occurred, that is, that $P(F|E) = P(F)$. One would expect that in this case, the equation $P(E|F) = P(E)$ would also be true. In fact (see Exercise 1), each equation implies the other. If these equations are true, we might say the F is *independent* of E. For example, you would not expect the knowledge of the outcome of the first toss of a coin to change

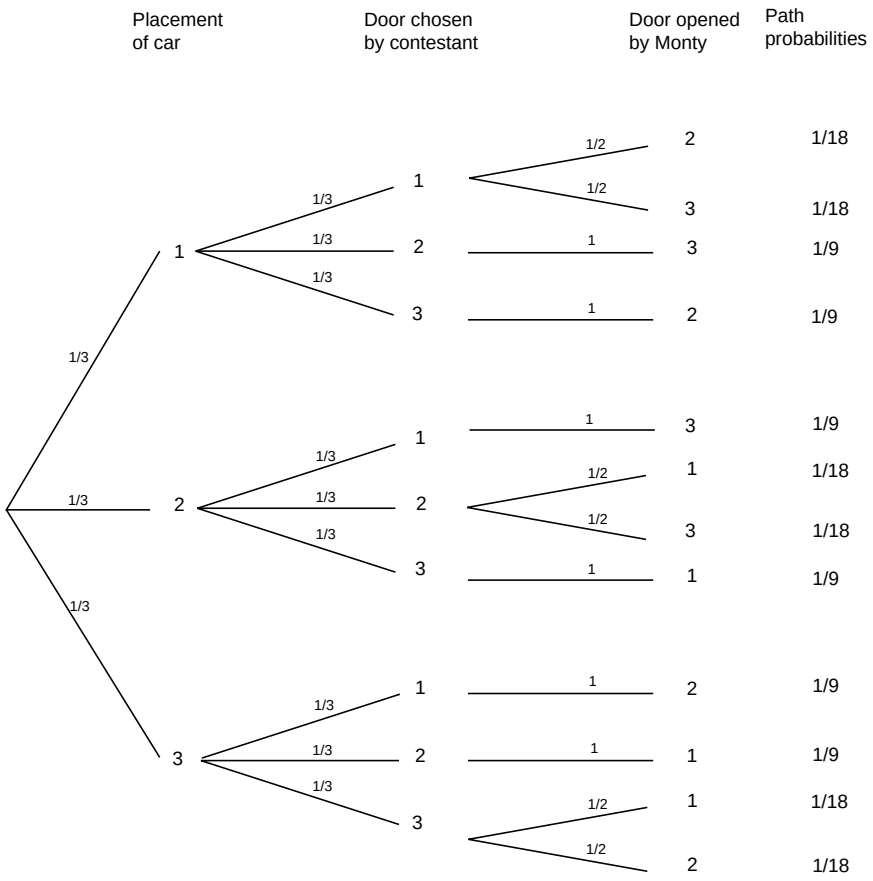


Figure 4.3: The Monty Hall problem.

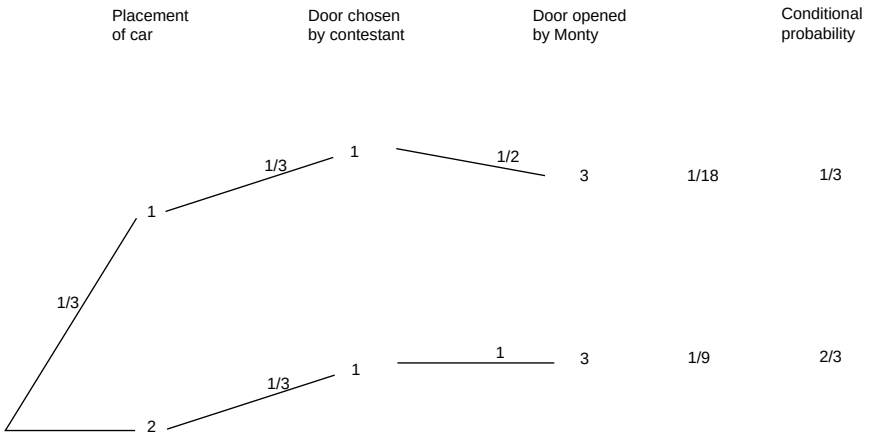


Figure 4.4: Conditional probabilities for the Monty Hall problem.

the probability that you would assign to the possible outcomes of the second toss, that is, you would not expect that the second toss depends on the first. This idea is formalized in the following definition of independent events.

Definition 4.1 Let E and F be two events. We say that they are *independent* if either 1) both events have positive probability and

$$P(E|F) = P(E) \text{ and } P(F|E) = P(F) \ ,$$

or 2) at least one of the events has probability 0. □

As noted above, if both $P(E)$ and $P(F)$ are positive, then each of the above equations imply the other, so that to see whether two events are independent, only one of these equations must be checked (see Exercise 1).

The following theorem provides another way to check for independence.

Theorem 4.1 Two events E and F are independent if and only if

$$P(E \cap F) = P(E)P(F) \ .$$

Proof. If either event has probability 0, then the two events are independent and the above equation is true, so the theorem is true in this case. Thus, we may assume that both events have positive probability in what follows. Assume that E and F are independent. Then $P(E|F) = P(E)$, and so

$$\begin{aligned} P(E \cap F) &= P(E|F)P(F) \\ &= P(E)P(F) \ . \end{aligned}$$

Assume next that $P(E \cap F) = P(E)P(F)$. Then

$$P(E|F) = \frac{P(E \cap F)}{P(F)} = P(E) \ .$$

Also,

$$P(F|E) = \frac{P(F \cap E)}{P(E)} = P(F) \ .$$

Therefore, E and F are independent. □

Example 4.7 Suppose that we have a coin which comes up heads with probability p, and tails with probability q. Now suppose that this coin is tossed twice. Using a frequency interpretation of probability, it is reasonable to assign to the outcome (H, H) the probability p^2, to the outcome (H, T) the probability pq, and so on. Let E be the event that heads turns up on the first toss and F the event that tails

turns up on the second toss. We will now check that with the above probability assignments, these two events are independent, as expected. We have $P(E) = p^2 + pq = p$, $P(F) = pq + q^2 = q$. Finally $P(E \cap F) = pq$, so $P(E \cap F) = P(E)P(F)$. □

Example 4.8 It is often, but not always, intuitively clear when two events are independent. In Example 4.7, let A be the event "the first toss is a head" and B the event "the two outcomes are the same." Then

$$P(B|A) = \frac{P(B \cap A)}{P(A)} = \frac{P\{\text{HH}\}}{P\{\text{HH,HT}\}} = \frac{1/4}{1/2} = \frac{1}{2} = P(B).$$

Therefore, A and B are independent, but the result was not so obvious. □

Example 4.9 Finally, let us give an example of two events that are not independent. In Example 4.7, let I be the event "heads on the first toss" and J the event "two heads turn up." Then $P(I) = 1/2$ and $P(J) = 1/4$. The event $I \cap J$ is the event "heads on both tosses" and has probability $1/4$. Thus, I and J are not independent since $P(I)P(J) = 1/8 \neq P(I \cap J)$. □

We can extend the concept of independence to any finite set of events A_1, A_2, $\ldots$, A_n.

Definition 4.2 A set of events $\{A_1, A_2, \ldots, A_n\}$ is said to be *mutually independent* if for any subset $\{A_i, A_j, \ldots, A_m\}$ of these events we have

$$P(A_i \cap A_j \cap \cdots \cap A_m) = P(A_i)P(A_j) \cdots P(A_m),$$

or equivalently, if for any sequence $\bar{A}_1$, $\bar{A}_2$, $\ldots$, $\bar{A}_n$ with $\bar{A}_j = A_j$ or $\tilde{A}_j$,

$$P(\bar{A}_1 \cap \bar{A}_2 \cap \cdots \cap \bar{A}_n) = P(\bar{A}_1)P(\bar{A}_2) \cdots P(\bar{A}_n).$$

(For a proof of the equivalence in the case $n = 3$, see Exercise 33.) □

Using this terminology, it is a fact that any sequence $(\text{S}, \text{S}, \text{F}, \text{F}, \text{S}, \ldots, \text{S})$ of possible outcomes of a Bernoulli trials process forms a sequence of mutually independent events.

It is natural to ask: If all pairs of a set of events are independent, is the whole set mutually independent? The answer is *not necessarily,* and an example is given in Exercise 7.

It is important to note that the statement

$$P(A_1 \cap A_2 \cap \cdots \cap A_n) = P(A_1)P(A_2) \cdots P(A_n)$$

does not imply that the events A_1, A_2, $\ldots$, A_n are mutually independent (see Exercise 8).

Joint Distribution Functions and Independence of Random Variables

It is frequently the case that when an experiment is performed, several different quantities concerning the outcomes are investigated.

Example 4.10 Suppose we toss a coin three times. The basic random variable $\bar{X}$ corresponding to this experiment has eight possible outcomes, which are the ordered triples consisting of H's and T's. We can also define the random variable X_i, for $i = 1, 2, 3$, to be the outcome of the ith toss. If the coin is fair, then we should assign the probability 1/8 to each of the eight possible outcomes. Thus, the distribution functions of X_1, X_2, and X_3 are identical; in each case they are defined by $m(H) = m(T) = 1/2$. □

If we have several random variables $X_1, X_2, \ldots, X_n$ which correspond to a given experiment, then we can consider the joint random variable $\bar{X} = (X_1, X_2, \ldots, X_n)$ defined by taking an outcome ω of the experiment, and writing, as an n-tuple, the corresponding n outcomes for the random variables $X_1, X_2, \ldots, X_n$. Thus, if the random variable X_i has, as its set of possible outcomes the set R_i, then the set of possible outcomes of the joint random variable $\bar{X}$ is the Cartesian product of the R_i's, i.e., the set of all n-tuples of possible outcomes of the X_i's.

Example 4.11 (Example 4.10 continued) In the coin-tossing example above, let X_i denote the outcome of the ith toss. Then the joint random variable $\bar{X} = (X_1, X_2, X_3)$ has eight possible outcomes.

Suppose that we now define Y_i, for $i = 1, 2, 3$, as the number of heads which occur in the first i tosses. Then Y_i has $\{0, 1, \ldots, i\}$ as possible outcomes, so at first glance, the set of possible outcomes of the joint random variable $\bar{Y} = (Y_1, Y_2, Y_3)$ should be the set

$$\{(a_1, a_2, a_3) \ : \ 0 \leq a_1 \leq 1, 0 \leq a_2 \leq 2, 0 \leq a_3 \leq 3\} \ .$$

However, the outcome $(1, 0, 1)$ cannot occur, since we must have $a_1 \leq a_2 \leq a_3$. The solution to this problem is to define the probability of the outcome $(1, 0, 1)$ to be 0. In addition, we must have $a_{i+1} - a_i \leq 1$ for $i = 1, 2$.

We now illustrate the assignment of probabilities to the various outcomes for the joint random variables $\bar{X}$ and $\bar{Y}$. In the first case, each of the eight outcomes should be assigned the probability 1/8, since we are assuming that we have a fair coin. In the second case, since Y_i has $i + 1$ possible outcomes, the set of possible outcomes has size 24. Only eight of these 24 outcomes can actually occur, namely the ones satisfying $a_1 \leq a_2 \leq a_3$. Each of these outcomes corresponds to exactly one of the outcomes of the random variable $\bar{X}$, so it is natural to assign probability 1/8 to each of these. We assign probability 0 to the other 16 outcomes. In each case, the probability function is called a joint distribution function. □

We collect the above ideas in a definition.

Definition 4.3 Let $X_1, X_2, \ldots, X_n$ be random variables associated with an experiment. Suppose that the sample space (i.e., the set of possible outcomes) of X_i is the set R_i. Then the joint random variable $\bar{X} = (X_1, X_2, \ldots, X_n)$ is defined to be the random variable whose outcomes consist of ordered n-tuples of outcomes, with the ith coordinate lying in the set R_i. The sample space Ω of $\bar{X}$ is the Cartesian product of the R_i's:

$$\Omega = R_1 \times R_2 \times \cdots \times R_n \ .$$

The joint distribution function of $\bar{X}$ is the function which gives the probability of each of the outcomes of $\bar{X}$. □

Example 4.12 (Example 4.10 continued) We now consider the assignment of probabilities in the above example. In the case of the random variable $\bar{X}$, the probability of any outcome (a_1, a_2, a_3) is just the product of the probabilities $P(X_i = a_i)$, for $i = 1, 2, 3$. However, in the case of $\bar{Y}$, the probability assigned to the outcome $(1, 1, 0)$ is not the product of the probabilities $P(Y_1 = 1)$, $P(Y_2 = 1)$, and $P(Y_3 = 0)$. The difference between these two situations is that the value of X_i does not affect the value of X_j, if $i \neq j$, while the values of Y_i and Y_j affect one another. For example, if $Y_1 = 1$, then Y_2 cannot equal 0. This prompts the next definition. □

Definition 4.4 The random variables X_1, X_2, ..., X_n are *mutually independent* if

$$\begin{aligned} &P(X_1 = r_1, X_2 = r_2, \ldots, X_n = r_n) \\ &\qquad = P(X_1 = r_1)P(X_2 = r_2) \cdots P(X_n = r_n) \end{aligned}$$

for any choice of $r_1, r_2, \ldots, r_n$. Thus, if $X_1,\ X_2, \ldots,\ X_n$ are mutually independent, then the joint distribution function of the random variable

$$\bar{X} = (X_1, X_2, \ldots, X_n)$$

is just the product of the individual distribution functions. When two random variables are mutually independent, we shall say more briefly that they are *independent.* □

Example 4.13 In a group of 60 people, the numbers who do or do not smoke and do or do not have cancer are reported as shown in Table 4.1. Let Ω be the sample space consisting of these 60 people. A person is chosen at random from the group. Let $C(\omega) = 1$ if this person has cancer and 0 if not, and $S(\omega) = 1$ if this person

	Not smoke	Smoke	Total
Not cancer	40	10	50
Cancer	7	3	10
Totals	47	13	60

Table 4.1: Smoking and cancer.

		S	
		0	1
C	0	40/60	10/60
	1	7/60	3/60

Table 4.2: Joint distribution.

smokes and 0 if not. Then the joint distribution of $\{C, S\}$ is given in Table 4.2. For example $P(C = 0, S = 0) = 40/60$, $P(C = 0, S = 1) = 10/60$, and so forth. The distributions of the individual random variables are called *marginal distributions.* The marginal distributions of C and S are:

$$p_C = \begin{pmatrix} 0 & 1 \\ 50/60 & 10/60 \end{pmatrix},$$

$$p_S = \begin{pmatrix} 0 & 1 \\ 47/60 & 13/60 \end{pmatrix}.$$

The random variables S and C are not independent, since

$$\begin{aligned} P(C = 1, S = 1) &= \frac{3}{60} = .05 \ , \\ P(C = 1)P(S = 1) &= \frac{10}{60} \cdot \frac{13}{60} = .036 \ . \end{aligned}$$

Note that we would also see this from the fact that

$$\begin{aligned} P(C = 1|S = 1) &= \frac{3}{13} = .23 \ , \\ P(C = 1) &= \frac{1}{6} = .167 \ . \end{aligned}$$

□

Independent Trials Processes

The study of random variables proceeds by considering special classes of random variables. One such class that we shall study is the class of *independent trials.*

Definition 4.5 A sequence of random variables $X_1, X_2, \ldots, X_n$ that are mutually independent and that have the same distribution is called a sequence of independent trials or an *independent trials process.*

Independent trials processes arise naturally in the following way. We have a single experiment with sample space $R = \{r_1, r_2, \ldots, r_s\}$ and a distribution function

$$m_X = \begin{pmatrix} r_1 & r_2 & \cdots & r_s \\ p_1 & p_2 & \cdots & p_s \end{pmatrix} .$$

We repeat this experiment n times. To describe this total experiment, we choose as sample space the space

$$\Omega = R \times R \times \cdots \times R,$$

consisting of all possible sequences $\omega = (\omega_1, \omega_2, \ldots, \omega_n)$ where the value of each ω_j is chosen from R. We assign a distribution function to be the *product distribution*

$$m(\omega) = m(\omega_1) \cdot \ \ldots \ \cdot m(\omega_n) \ ,$$

with $m(\omega_j) = p_k$ when $\omega_j = r_k$. Then we let X_j denote the jth coordinate of the outcome $(r_1, r_2, \ldots, r_n)$. The random variables X_1, $\ldots$, X_n form an independent trials process. □

Example 4.14 An experiment consists of rolling a die three times. Let X_i represent the outcome of the ith roll, for $i = 1, 2, 3$. The common distribution function is

$$m_i = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1/6 & 1/6 & 1/6 & 1/6 & 1/6 & 1/6 \end{pmatrix}.$$

The sample space is $R^3 = R \times R \times R$ with $R = \{1, 2, 3, 4, 5, 6\}$. If $\omega = (1, 3, 6)$, then $X_1(\omega) = 1$, $X_2(\omega) = 3$, and $X_3(\omega) = 6$ indicating that the first roll was a 1, the second was a 3, and the third was a 6. The probability assigned to any sample point is

$$m(\omega) = \frac{1}{6} \cdot \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{216} \ .$$

□

Example 4.15 Consider next a Bernoulli trials process with probability p for success on each experiment. Let $X_j(\omega) = 1$ if the jth outcome is success and $X_j(\omega) = 0$ if it is a failure. Then X_1, X_2, $\ldots$, X_n is an independent trials process. Each X_j has the same distribution function

$$m_j = \begin{pmatrix} 0 & 1 \\ q & p \end{pmatrix},$$

where $q = 1 - p$.

If $S_n = X_1 + X_2 + \cdots + X_n$, then

$$P(S_n = j) = \binom{n}{j} p^j q^{n-j} \ ,$$

and S_n has, as distribution, the binomial distribution $b(n, p, j)$. □

Bayes' Formula

In our examples, we have considered conditional probabilities of the following form: Given the outcome of the second stage of a two-stage experiment, find the probability for an outcome at the first stage. We have remarked that these probabilities are called *Bayes probabilities.*

We return now to the calculation of more general Bayes probabilities. Suppose we have a set of events H_1, H_2, ..., H_m that are pairwise disjoint and such that the sample space Ω satisfies the equation

$$\Omega = H_1 \cup H_2 \cup \cdots \cup H_m \ .$$

We call these events *hypotheses.* We also have an event E that gives us some information about which hypothesis is correct. We call this event *evidence.*

Before we receive the evidence, then, we have a set of *prior probabilities* $P(H_1)$, $P(H_2)$, ..., $P(H_m)$ for the hypotheses. If we know the correct hypothesis, we know the probability for the evidence. That is, we know $P(E|H_i)$ for all i. We want to find the probabilities for the hypotheses given the evidence. That is, we want to find the conditional probabilities $P(H_i|E)$. These probabilities are called the *posterior probabilities.*

To find these probabilities, we write them in the form

$$P(H_i|E) = \frac{P(H_i \cap E)}{P(E)} \ . \tag{4.1}$$

We can calculate the numerator from our given information by

$$P(H_i \cap E) = P(H_i)P(E|H_i) \ . \tag{4.2}$$

Since one and only one of the events H_1, H_2, ..., H_m can occur, we can write the probability of E as

$$P(E) = P(H_1 \cap E) + P(H_2 \cap E) + \cdots + P(H_m \cap E) \ .$$

Using Equation 4.2, the above expression can be seen to equal

$$P(H_1)P(E|H_1) + P(H_2)P(E|H_2) + \cdots + P(H_m)P(E|H_m) \ . \tag{4.3}$$

Using (4.1), (4.2), and (4.3) yields *Bayes' formula*:

$$P(H_i|E) = \frac{P(H_i)P(E|H_i)}{\sum_{k=1}^{m} P(H_k)P(E|H_k)} \ .$$

Although this is a very famous formula, we will rarely use it. If the number of hypotheses is small, a simple tree measure calculation is easily carried out, as we have done in our examples. If the number of hypotheses is large, then we should use a computer.

Bayes probabilities are particularly appropriate for medical diagnosis. A doctor is anxious to know which of several diseases a patient might have. She collects evidence in the form of the outcomes of certain tests. From statistical studies the

Disease	Number having this disease	The results			
		+ +	+ −	− +	− −
d_1	3215	2110	301	704	100
d_2	2125	396	132	1187	410
d_3	4660	510	3568	73	509
Total	10000				

Table 4.3: Diseases data.

	d_1	d_2	d_3
+ +	.700	.131	.169
+ −	.075	.033	.892
− +	.358	.604	.038
− −	.098	.403	.499

Table 4.4: Posterior probabilities.

doctor can find the prior probabilities of the various diseases before the tests, and the probabilities for specific test outcomes, given a particular disease. What the doctor wants to know is the posterior probability for the particular disease, given the outcomes of the tests.

Example 4.16 A doctor is trying to decide if a patient has one of three diseases d_1, d_2, or d_3. Two tests are to be carried out, each of which results in a positive $(+)$ or a negative $(-)$ outcome. There are four possible test patterns $++$, $+-$, $-+$, and $--$. National records have indicated that, for 10,000 people having one of these three diseases, the distribution of diseases and test results are as in Table 4.3.

From this data, we can estimate the prior probabilities for each of the diseases and, given a particular disease, the probability of a particular test outcome. For example, the prior probability of disease d_1 may be estimated to be $3215/10{,}000 = .3215$. The probability of the test result $+-$, given disease d_1, may be estimated to be $301/3215 = .094$.

We can now use Bayes' formula to compute various posterior probabilities. The computer program **Bayes** computes these posterior probabilities. The results for this example are shown in Table 4.4.

We note from the outcomes that, when the test result is $++$, the disease d_1 has a significantly higher probability than the other two. When the outcome is $+-$, this is true for disease d_3. When the outcome is $-+$, this is true for disease d_2. Note that these statements might have been guessed by looking at the data. If the outcome is $--$, the most probable cause is d_3, but the probability that a patient has d_2 is only slightly smaller. If one looks at the data in this case, one can see that it might be hard to guess which of the two diseases d_2 and d_3 is more likely. □

Our final example shows that one has to be careful when the prior probabilities are small.

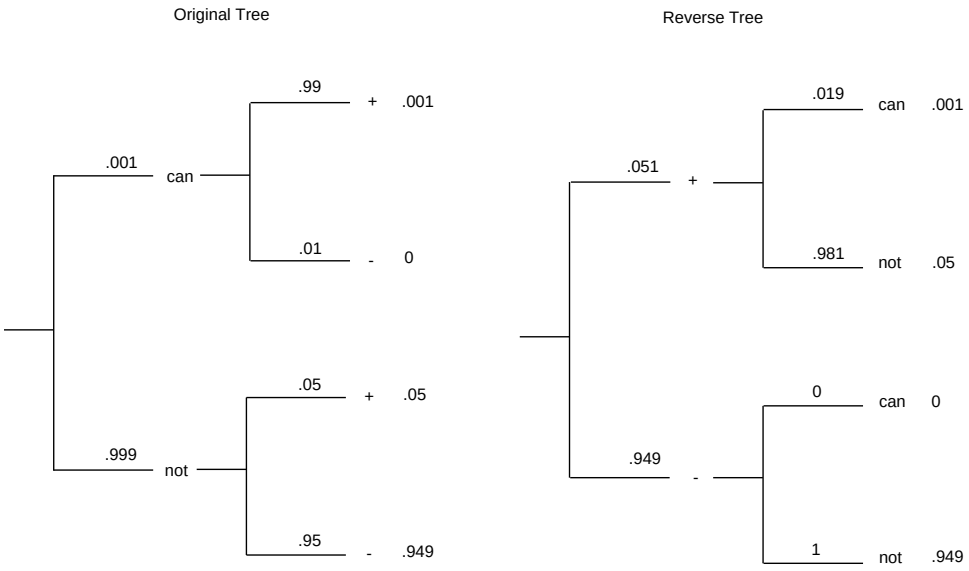


Figure 4.5: Forward and reverse tree diagrams.

Example 4.17 A doctor gives a patient a test for a particular cancer. Before the results of the test, the only evidence the doctor has to go on is that 1 woman in 1000 has this cancer. Experience has shown that, in 99 percent of the cases in which cancer is present, the test is positive; and in 95 percent of the cases in which it is not present, it is negative. If the test turns out to be positive, what probability should the doctor assign to the event that cancer is present? An alternative form of this question is to ask for the relative frequencies of false positives and cancers.

We are given that prior(cancer) = .001 and prior(not cancer) = .999. We know also that $P(+|\text{cancer}) = .99$, $P(-|\text{cancer}) = .01$, $P(+|\text{not cancer}) = .05$, and $P(-|\text{not cancer}) = .95$. Using this data gives the result shown in Figure 4.5.

We see now that the probability of cancer given a positive test has only increased from .001 to .019. While this is nearly a twenty-fold increase, the probability that the patient has the cancer is still small. Stated in another way, among the positive results, 98.1 percent are false positives, and 1.9 percent are cancers. When a group of second-year medical students was asked this question, over half of the students incorrectly guessed the probability to be greater than .5. □

Historical Remarks

Conditional probability was used long before it was formally defined. Pascal and Fermat considered the *problem of points*: given that team A has won m games and team B has won n games, what is the probability that A will win the series? (See Exercises 40–42.) This is clearly a conditional probability problem.

In his book, Huygens gave a number of problems, one of which was:

> Three gamblers, A, B and C, take 12 balls of which 4 are white and 8 black. They play with the rules that the drawer is blindfolded, A is to draw first, then B and then C, the winner to be the one who first draws a white ball. What is the ratio of their chances?[2]

From his answer it is clear that Huygens meant that each ball is replaced after drawing. However, John Hudde, the mayor of Amsterdam, assumed that he meant to sample without replacement and corresponded with Huygens about the difference in their answers. Hacking remarks that "Neither party can understand what the other is doing."[3]

By the time of de Moivre's book, *The Doctrine of Chances,* these distinctions were well understood. De Moivre defined independence and dependence as follows:

> Two Events are independent, when they have no connexion one with the other, and that the happening of one neither forwards nor obstructs the happening of the other.
>
> Two Events are dependent, when they are so connected together as that the Probability of either's happening is altered by the happening of the other.[4]

De Moivre used sampling with and without replacement to illustrate that the probability that two independent events both happen is the product of their probabilities, and for dependent events that:

> The Probability of the happening of two Events dependent, is the product of the Probability of the happening of one of them, by the Probability which the other will have of happening, when the first is considered as having happened; and the same Rule will extend to the happening of as many Events as may be assigned.[5]

The formula that we call Bayes' formula, and the idea of computing the probability of a hypothesis given evidence, originated in a famous essay of Thomas Bayes. Bayes was an ordained minister in Tunbridge Wells near London. His mathematical interests led him to be elected to the Royal Society in 1742, but none of his results were published within his lifetime. The work upon which his fame rests, "An Essay Toward Solving a Problem in the Doctrine of Chances," was published in 1763, three years after his death.[6] Bayes reviewed some of the basic concepts of probability and then considered a new kind of inverse probability problem requiring the use of conditional probability.

Bernoulli, in his study of processes that we now call Bernoulli trials, had proven his famous law of large numbers which we will study in Chapter 8. This theorem

[2]Quoted in F. N. David, *Games, Gods and Gambling* (London: Griffin, 1962), p. 119.

[3]I. Hacking, *The Emergence of Probability* (Cambridge: Cambridge University Press, 1975), p. 99.

[4]A. de Moivre, *The Doctrine of Chances,* 3rd ed. (New York: Chelsea, 1967), p. 6.

[5]ibid, p. 7.

[6]T. Bayes, "An Essay Toward Solving a Problem in the Doctrine of Chances," *Phil. Trans. Royal Soc. London,* vol. 53 (1763), pp. 370–418.

assured the experimenter that if he knew the probability p for success, he could predict that the proportion of successes would approach this value as he increased the number of experiments. Bernoulli himself realized that in most interesting cases you do not know the value of p and saw his theorem as an important step in showing that you could determine p by experimentation.

To study this problem further, Bayes started by assuming that the probability p for success is itself determined by a random experiment. He assumed in fact that this experiment was such that this value for p is equally likely to be any value between 0 and 1. Without knowing this value we carry out n experiments and observe m successes. Bayes proposed the problem of finding the conditional probability that the unknown probability p lies between a and b. He obtained the answer:

$$P(a \leq p < b | m \text{ successes in } n \text{ trials}) = \frac{\int_a^b x^m(1-x)^{n-m}\,dx}{\int_0^1 x^m(1-x)^{n-m}\,dx}\ .$$

We shall see in the next section how this result is obtained. Bayes clearly wanted to show that the conditional distribution function, given the outcomes of more and more experiments, becomes concentrated around the true value of p. Thus, Bayes was trying to solve an *inverse problem.* The computation of the integrals was too difficult for exact solution except for small values of j and n, and so Bayes tried approximate methods. His methods were not very satisfactory and it has been suggested that this discouraged him from publishing his results.

However, his paper was the first in a series of important studies carried out by Laplace, Gauss, and other great mathematicians to solve inverse problems. They studied this problem in terms of errors in measurements in astronomy. If an astronomer were to know the true value of a distance and the nature of the random errors caused by his measuring device he could predict the probabilistic nature of his measurements. In fact, however, he is presented with the inverse problem of knowing the nature of the random errors, and the values of the measurements, and wanting to make inferences about the unknown true value.

As Maistrov remarks, the formula that we have called Bayes' formula does not appear in his essay. Laplace gave it this name when he studied these inverse problems.[7] The computation of inverse probabilities is fundamental to statistics and has led to an important branch of statistics called Bayesian analysis, assuring Bayes eternal fame for his brief essay.

Exercises

1 Assume that E and F are two events with positive probabilities. Show that if $P(E|F) = P(E)$, then $P(F|E) = P(F)$.

2 A coin is tossed three times. What is the probability that exactly two heads occur, given that

(a) the first outcome was a head?

[7] L. E. Maistrov, *Probability Theory: A Historical Sketch,* trans. and ed. Samual Kotz (New York: Academic Press, 1974), p. 100.

(b) the first outcome was a tail?

(c) the first two outcomes were heads?

(d) the first two outcomes were tails?

(e) the first outcome was a head and the third outcome was a head?

3 A die is rolled twice. What is the probability that the sum of the faces is greater than 7, given that

(a) the first outcome was a 4?

(b) the first outcome was greater than 3?

(c) the first outcome was a 1?

(d) the first outcome was less than 5?

4 A card is drawn at random from a deck of cards. What is the probability that

(a) it is a heart, given that it is red?

(b) it is higher than a 10, given that it is a heart? (Interpret J, Q, K, A as 11, 12, 13, 14.)

(c) it is a jack, given that it is red?

5 A coin is tossed three times. Consider the following events
A: Heads on the first toss.
B: Tails on the second.
C: Heads on the third toss.
D: All three outcomes the same (HHH or TTT).
E: Exactly one head turns up.

(a) Which of the following pairs of these events are independent?
(1) A, B
(2) A, D
(3) A, E
(4) D, E

(b) Which of the following triples of these events are independent?
(1) A, B, C
(2) A, B, D
(3) C, D, E

6 From a deck of five cards numbered 2, 4, 6, 8, and 10, respectively, a card is drawn at random and replaced. This is done three times. What is the probability that the card numbered 2 was drawn exactly two times, given that the sum of the numbers on the three draws is 12?

7 A coin is tossed twice. Consider the following events.
A: Heads on the first toss.
B: Heads on the second toss.
C: The two tosses come out the same.

(a) Show that A, B, C are pairwise independent but not independent.

(b) Show that C is independent of A and B but not of $A \cap B$.

8 Let $\Omega = \{a, b, c, d, e, f\}$. Assume that $m(a) = m(b) = 1/8$ and $m(c) = m(d) = m(e) = m(f) = 3/16$. Let A, B, and C be the events $A = \{d, e, a\}$, $B = \{c, e, a\}$, $C = \{c, d, a\}$. Show that $P(A \cap B \cap C) = P(A)P(B)P(C)$ but no two of these events are independent.

9 What is the probability that a family of two children has

(a) two boys given that it has at least one boy?

(b) two boys given that the first child is a boy?

10 In Example 4.2, we used the Life Table (see Appendix C) to compute a conditional probability. The number 93,753 in the table, corresponding to 40-year-old males, means that of all the males born in the United States in 1950, 93.753% were alive in 1990. Is it reasonable to use this as an estimate for the probability of a male, born this year, surviving to age 40?

11 Simulate the Monty Hall problem. Carefully state any assumptions that you have made when writing the program. Which version of the problem do you think that you are simulating?

12 In Example 4.17, how large must the prior probability of cancer be to give a posterior probability of .5 for cancer given a positive test?

13 Two cards are drawn from a bridge deck. What is the probability that the second card drawn is red?

14 If $P(\tilde{B}) = 1/4$ and $P(A|B) = 1/2$, what is $P(A \cap B)$?

15 (a) What is the probability that your bridge partner has exactly two aces, given that she has at least one ace?

(b) What is the probability that your bridge partner has exactly two aces, given that she has the ace of spades?

16 Prove that for any three events A, B, C, each having positive probability, and with the property that $P(A \cap B) > 0$,

$$P(A \cap B \cap C) = P(A)P(B|A)P(C|A \cap B) \ .$$

17 Prove that if A and B are independent so are

(a) A and $\tilde{B}$.

(b) $\tilde{A}$ and $\tilde{B}$.

18 A doctor assumes that a patient has one of three diseases d_1, d_2, or d_3. Before any test, he assumes an equal probability for each disease. He carries out a test that will be positive with probability .8 if the patient has d_1, .6 if he has disease d_2, and .4 if he has disease d_3. Given that the outcome of the test was positive, what probabilities should the doctor now assign to the three possible diseases?

19 In a poker hand, John has a very strong hand and bets 5 dollars. The probability that Mary has a better hand is .04. If Mary had a better hand she would raise with probability .9, but with a poorer hand she would only raise with probability .1. If Mary raises, what is the probability that she has a better hand than John does?

20 The Polya urn model for contagion is as follows: We start with an urn which contains one white ball and one black ball. At each second we choose a ball at random from the urn and replace this ball and add one more of the color chosen. Write a program to simulate this model, and see if you can make any predictions about the proportion of white balls in the urn after a large number of draws. Is there a tendency to have a large fraction of balls of the same color in the long run?

21 It is desired to find the probability that in a bridge deal each player receives an ace. A student argues as follows. It does not matter where the first ace goes. The second ace must go to one of the other three players and this occurs with probability 3/4. Then the next must go to one of two, an event of probability 1/2, and finally the last ace must go to the player who does not have an ace. This occurs with probability 1/4. The probability that all these events occur is the product $(3/4)(1/2)(1/4) = 3/32$. Is this argument correct?

22 One coin in a collection of 65 has two heads. The rest are fair. If a coin, chosen at random from the lot and then tossed, turns up heads 6 times in a row, what is the probability that it is the two-headed coin?

23 You are given two urns and fifty balls. Half of the balls are white and half are black. You are asked to distribute the balls in the urns with no restriction placed on the number of either type in an urn. How should you distribute the balls in the urns to maximize the probability of obtaining a white ball if an urn is chosen at random and a ball drawn out at random? Justify your answer.

24 A fair coin is thrown n times. Show that the conditional probability of a head on any specified trial, given a total of k heads over the n trials, is k/n $(k > 0)$.

25 (Johnsonbough[8]) A coin with probability p for heads is tossed n times. Let E be the event "a head is obtained on the first toss' and F_k the event 'exactly k heads are obtained." For which pairs (n, k) are E and F_k independent?

[8]R. Johnsonbough, "Problem #103," *Two Year College Math Journal,* vol. 8 (1977), p. 292.

26 Suppose that A and B are events such that $P(A|B) = P(B|A)$ and $P(A \cup B) = 1$ and $P(A \cap B) > 0$. Prove that $P(A) > 1/2$.

27 (Chung[9]) In London, half of the days have some rain. The weather forecaster is correct 2/3 of the time, i.e., the probability that it rains, given that she has predicted rain, and the probability that it does not rain, given that she has predicted that it won't rain, are both equal to 2/3. When rain is forecast, Mr. Pickwick takes his umbrella. When rain is not forecast, he takes it with probability 1/3. Find

(a) the probability that Pickwick has no umbrella, given that it rains.

(b) the probability that he brings his umbrella, given that it doesn't rain.

28 Probability theory was used in a famous court case: *People v. Collins.*[10] In this case a purse was snatched from an elderly person in a Los Angeles suburb. A couple seen running from the scene were described as a black man with a beard and a mustache and a blond girl with hair in a ponytail. Witnesses said they drove off in a partly yellow car. Malcolm and Janet Collins were arrested. He was black and though clean shaven when arrested had evidence of recently having had a beard and a mustache. She was blond and usually wore her hair in a ponytail. They drove a partly yellow Lincoln. The prosecution called a professor of mathematics as a witness who suggested that a conservative set of probabilities for the characteristics noted by the witnesses would be as shown in Table 4.5.

man with mustache	1/4
girl with blond hair	1/3
girl with ponytail	1/10
black man with beard	1/10
interracial couple in a car	1/1000
partly yellow car	1/10

Table 4.5: Collins case probabilities.

The prosecution then argued that the probability that all of these characteristics are met by a randomly chosen couple is the product of the probabilities or 1/12,000,000, which is very small. He claimed this was proof beyond a reasonable doubt that the defendants were guilty. The jury agreed and handed down a verdict of guilty of second-degree robbery.

If you were the lawyer for the Collins couple how would you have countered the above argument? (The appeal of this case is discussed in Exercise 5.1.34.)

29 A student is applying to Harvard and Dartmouth. He estimates that he has a probability of .5 of being accepted at Dartmouth and .3 of being accepted

[9]K. L. Chung, *Elementary Probability Theory With Stochastic Processes, 3rd ed.* (New York: Springer-Verlag, 1979), p. 152.

[10]M. W. Gray, "Statistics and the Law," *Mathematics Magazine,* vol. 56 (1983), pp. 67–81.

at Harvard. He further estimates the probability that he will be accepted by both is .2. What is the probability that he is accepted by Dartmouth if he is accepted by Harvard? Is the event "accepted at Harvard" independent of the event "accepted at Dartmouth"?

30 Luxco, a wholesale lightbulb manufacturer, has two factories. Factory A sells bulbs in lots that consists of 1000 regular and 2000 *softglow* bulbs each. Random sampling has shown that on the average there tend to be about 2 bad regular bulbs and 11 bad softglow bulbs per lot. At factory B the lot size is reversed—there are 2000 regular and 1000 softglow per lot—and there tend to be 5 bad regular and 6 bad softglow bulbs per lot.

The manager of factory A asserts, "We're obviously the better producer; our bad bulb rates are .2 percent and .55 percent compared to B's .25 percent and .6 percent. We're better at both regular and softglow bulbs by half of a tenth of a percent each."

"Au contraire," counters the manager of B, "each of our 3000 bulb lots contains only 11 bad bulbs, while A's 3000 bulb lots contain 13. So our .37 percent bad bulb rate beats their .43 percent."

Who is right?

31 Using the Life Table for 1981 given in Appendix C, find the probability that a male of age 60 in 1981 lives to age 80. Find the same probability for a female.

32 (a) There has been a blizzard and Helen is trying to drive from Woodstock to Tunbridge, which are connected like the top graph in Figure 4.6. Here p and q are the probabilities that the two roads are passable. What is the probability that Helen can get from Woodstock to Tunbridge?

(b) Now suppose that Woodstock and Tunbridge are connected like the middle graph in Figure 4.6. What now is the probability that she can get from W to T? Note that if we think of the roads as being components of a system, then in (a) and (b) we have computed the *reliability* of a system whose components are (a) *in series* and (b) *in parallel.*

(c) Now suppose W and T are connected like the bottom graph in Figure 4.6. Find the probability of Helen's getting from W to T. *Hint*: If the road from C to D is impassable, it might as well not be there at all; if it is passable, then figure out how to use part (b) twice.

33 Let A_1, A_2, and A_3 be events, and let B_i represent either A_i or its complement $\tilde{A}_i$. Then there are eight possible choices for the triple (B_1, B_2, B_3). Prove that the events A_1, A_2, A_3 are independent if and only if

$$P(B_1 \cap B_2 \cap B_3) = P(B_1)P(B_2)P(B_3) \ ,$$

for all eight of the possible choices for the triple (B_1, B_2, B_3).

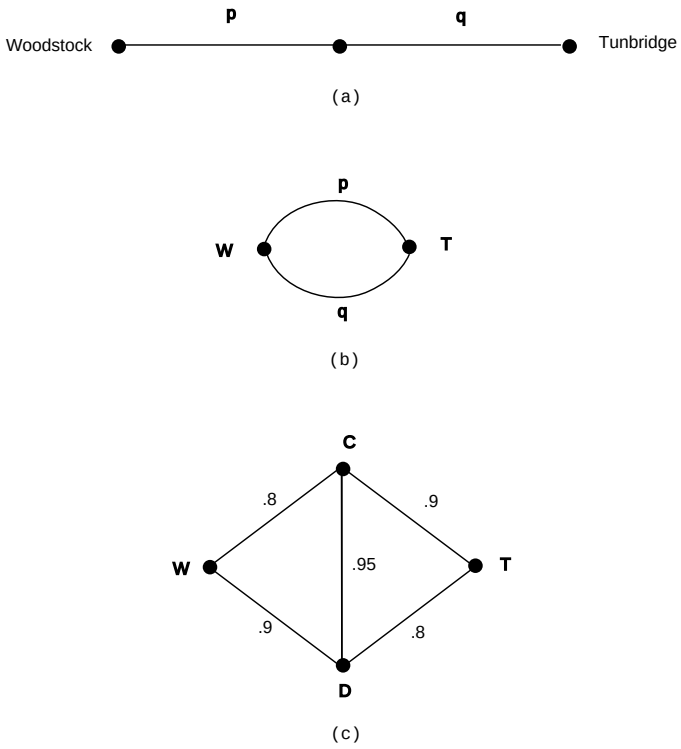


Figure 4.6: From Woodstock to Tunbridge.

		Y			
		-1	0	1	2
X	-1	0	1/36	1/6	1/12
	0	1/18	0	1/18	0
	1	0	1/36	1/6	1/12
	2	1/12	0	1/12	1/6

Table 4.6: Joint distribution.

34 Four women, A, B, C, and D, check their hats, and the hats are returned in a random manner. Let Ω be the set of all possible permutations of A, B, C, D. Let $X_j = 1$ if the jth woman gets her own hat back and 0 otherwise. What is the distribution of X_j? Are the X_i's mutually independent?

35 A box has numbers from 1 to 10. A number is drawn at random. Let X_1 be the number drawn. This number is replaced, and the ten numbers mixed. A second number X_2 is drawn. Find the distributions of X_1 and X_2. Are X_1 and X_2 independent? Answer the same questions if the first number is not replaced before the second is drawn.

36 A die is thrown twice. Let X_1 and X_2 denote the outcomes. Define $X = \min(X_1, X_2)$. Find the distribution of X.

***37** Given that $P(X = a) = r$, $P(\max(X, Y) = a) = s$, and $P(\min(X, Y) = a) = t$, show that you can determine $u = P(Y = a)$ in terms of r, s, and t.

38 A fair coin is tossed three times. Let X be the number of heads that turn up on the first two tosses and Y the number of heads that turn up on the third toss. Give the distribution of

(a) the random variables X and Y.

(b) the random variable $Z = X + Y$.

(c) the random variable $W = X - Y$.

39 Assume that the random variables X and Y have the joint distribution given in Table 4.6.

(a) What is $P(X \geq 1 \text{ and } Y \leq 0)$?

(b) What is the conditional probability that $Y \leq 0$ given that $X = 2$?

(c) Are X and Y independent?

(d) What is the distribution of $Z = XY$?

40 In the *problem of points*, discussed in the historical remarks in Section 3.2, two players, A and B, play a series of points in a game with player A winning each point with probability p and player B winning each point with probability $q = 1 - p$. The first player to win N points wins the game. Assume that $N = 3$. Let X be a random variable that has the value 1 if player A wins the

series and 0 otherwise. Let Y be a random variable with value the number of points played in a game. Find the distribution of X and Y when $p = 1/2$. Are X and Y independent in this case? Answer the same questions for the case $p = 2/3$.

41 The letters between Pascal and Fermat, which are often credited with having started probability theory, dealt mostly with the *problem of points* described in Exercise 40. Pascal and Fermat considered the problem of finding a fair division of stakes if the game must be called off when the first player has won r games and the second player has won s games, with $r < N$ and $s < N$. Let $P(r, s)$ be the probability that player A wins the game if he has already won r points and player B has won s points. Then

(a) $P(r, N) = 0$ if $r < N$,

(b) $P(N, s) = 1$ if $s < N$,

(c) $P(r, s) = pP(r + 1, s) + qP(r, s + 1)$ if $r < N$ and $s < N$;

and (1), (2), and (3) determine $P(r, s)$ for $r \leq N$ and $s \leq N$. Pascal used these facts to find $P(r, s)$ by working backward: He first obtained $P(N-1, j)$ for $j = N-1, N-2, \ldots, 0$; then, from these values, he obtained $P(N-2, j)$ for $j = N-1, N-2, \ldots, 0$ and, continuing backward, obtained all the values $P(r, s)$. Write a program to compute $P(r, s)$ for given N, a, b, and p. *Warning*: Follow Pascal and you will be able to run $N = 100$; use recursion and you will not be able to run $N = 20$.

42 Fermat solved the *problem of points* (see Exercise 40) as follows: He realized that the problem was difficult because the possible ways the play might go are not equally likely. For example, when the first player needs two more games and the second needs three to win, two possible ways the series might go for the first player are WLW and LWLW. These sequences are not equally likely. To avoid this difficulty, Fermat extended the play, adding fictitious plays so that the series went the maximum number of games needed (four in this case). He obtained equally likely outcomes and used, in effect, the Pascal triangle to calculate $P(r, s)$. Show that this leads to a *formula* for $P(r, s)$ even for the case $p \neq 1/2$.

43 The Yankees are playing the Dodgers in a world series. The Yankees win each game with probability .6. What is the probability that the Yankees win the series? (The series is won by the first team to win four games.)

44 C. L. Anderson[11] has used Fermat's argument for the *problem of points* to prove the following result due to J. G. Kingston. You are playing the *game of points* (see Exercise 40) but, at each point, when you serve you win with probability p, and when your opponent serves you win with probability $\bar{p}$. You will serve first, but you can choose one of the following two conventions

[11] C. L. Anderson, "Note on the Advantage of First Serve," *Journal of Combinatorial Theory,* Series A, vol. 23 (1977), p. 363.

for serving: for the first convention you alternate service (tennis), and for the second the person serving continues to serve until he loses a point and then the other player serves (racquetball). The first player to win N points wins the game. The problem is to show that the probability of winning the game is the same under either convention.

(a) Show that, under either convention, you will serve at most N points and your opponent at most $N-1$ points.

(b) Extend the number of points to $2N-1$ so that you serve N points and your opponent serves $N-1$. For example, you serve any additional points necessary to make N serves and then your opponent serves any additional points necessary to make him serve $N-1$ points. The winner is now the person, in the extended game, who wins the most points. Show that playing these additional points has not changed the winner.

(c) Show that (a) and (b) prove that you have the same probability of winning the game under either convention.

45 In the previous problem, assume that $p = 1 - \bar{p}$.

(a) Show that under either service convention, the first player will win more often than the second player if and only if $p > .5$.

(b) In volleyball, a team can only win a point while it is serving. Thus, any individual "play" either ends with a point being awarded to the serving team or with the service changing to the other team. The first team to win N points wins the game. (We ignore here the additional restriction that the winning team must be ahead by at least two points at the end of the game.) Assume that each team has the same probability of winning the play when it is serving, i.e., that $p = 1 - \bar{p}$. Show that in this case, the team that serves first will win more than half the time, as long as $p > 0$. (If $p = 0$, then the game never ends.) *Hint*: Define p' to be the probability that a team wins the next point, given that it is serving. If we write $q = 1 - p$, then one can show that

$$p' = \frac{p}{1 - q^2} \ .$$

If one now considers this game in a slightly different way, one can see that the second service convention in the preceding problem can be used, with p replaced by p'.

46 A poker hand consists of 5 cards dealt from a deck of 52 cards. Let X and Y be, respectively, the number of aces and kings in a poker hand. Find the joint distribution of X and Y.

47 Let X_1 and X_2 be independent random variables and let $Y_1 = \phi_1(X_1)$ and $Y_2 = \phi_2(X_2)$.

(a) Show that

$$P(Y_1 = r, Y_2 = s) = \sum_{\substack{\phi_1(a)=r \\ \phi_2(b)=s}} P(X_1 = a, X_2 = b) \ .$$

(b) Using (a), show that $P(Y_1 = r, Y_2 = s) = P(Y_1 = r)P(Y_2 = s)$ so that Y_1 and Y_2 are independent.

48 Let Ω be the sample space of an experiment. Let E be an event with $P(E) > 0$ and define $m_E(\omega)$ by $m_E(\omega) = m(\omega|E)$. Prove that $m_E(\omega)$ is a distribution function on E, that is, that $m_E(\omega) \geq 0$ and that $\sum_{\omega\in\Omega} m_E(\omega) = 1$. The function m_E is called the *conditional distribution given* E.

49 You are given two urns each containing two biased coins. The coins in urn I come up heads with probability p_1, and the coins in urn II come up heads with probability $p_2 \neq p_1$. You are given a choice of (a) choosing an urn at random and tossing the two coins in this urn or (b) choosing one coin from each urn and tossing these two coins. You win a prize if both coins turn up heads. Show that you are better off selecting choice (a).

50 Prove that, if A_1, A_2, ..., A_n are independent events defined on a sample space Ω and if $0 < P(A_j) < 1$ for all j, then Ω must have at least 2^n points.

51 Prove that if

$$P(A|C) \geq P(B|C) \text{ and } P(A|\tilde{C}) \geq P(B|\tilde{C}) \ ,$$

then $P(A) \geq P(B)$.

52 A coin is in one of n boxes. The probability that it is in the ith box is p_i. If you search in the ith box and it is there, you find it with probability a_i. Show that the probability p that the coin is in the jth box, given that you have looked in the ith box and not found it, is

$$p = \begin{cases} p_j/(1 - a_i p_i), & \text{if } j \neq i, \\ (1 - a_i)p_i/(1 - a_i p_i), & \text{if } j = i. \end{cases}$$

53 George Wolford has suggested the following variation on the Linda problem (see Exercise 1.2.25). The registrar is carrying John and Mary's registration cards and drops them in a puddle. When he pickes them up he cannot read the names but on the first card he picked up he can make out Mathematics 23 and Government 35, and on the second card he can make out only Mathematics 23. He asks you if you can help him decide which card belongs to Mary. You know that Mary likes government but does not like mathematics. You know nothing about John and assume that he is just a typical Dartmouth student. From this you estimate:

$$\begin{aligned} P(\text{Mary takes Government 35}) &= .5\ , \\ P(\text{Mary takes Mathematics 23}) &= .1\ , \\ P(\text{John takes Government 35}) &= .3\ , \\ P(\text{John takes Mathematics 23}) &= .2\ . \end{aligned}$$

Assume that their choices for courses are independent events. Show that the card with Mathematics 23 and Government 35 showing is more likely to be Mary's than John's. The conjunction fallacy referred to in the Linda problem would be to assume that the event "Mary takes Mathematics 23 and Government 35" is more likely than the event "Mary takes Mathematics 23." Why are we not making this fallacy here?

54 (Suggested by Eisenberg and Ghosh[12]) A deck of playing cards can be described as a Cartesian product

$$\text{Deck} = \text{Suit} \times \text{Rank}\ ,$$

where Suit $= \{\clubsuit, \diamondsuit, \heartsuit, \spadesuit\}$ and Rank $= \{2, 3, \ldots, 10, \text{J}, \text{Q}, \text{K}, \text{A}\}$. This just means that every card may be thought of as an ordered pair like $(\diamondsuit, 2)$. By a *suit event* we mean any event A contained in Deck which is described in terms of Suit alone. For instance, if A is "the suit is red," then

$$A = \{\diamondsuit, \heartsuit\} \times \text{Rank}\ ,$$

so that A consists of all cards of the form $(\diamondsuit, r)$ or $(\heartsuit, r)$ where r is any rank. Similarly, a *rank event* is any event described in terms of rank alone.

(a) Show that if A is any suit event and B any rank event, then A and B are *independent.* (We can express this briefly by saying that suit and rank are independent.)

(b) Throw away the ace of spades. Show that now no nontrivial (i.e., neither empty nor the whole space) suit event A is independent of any nontrivial rank event B. *Hint*: Here independence comes down to

$$c/51 = (a/51) \cdot (b/51)\ ,$$

where a, b, c are the respective sizes of A, B and $A \cap B$. It follows that 51 must divide ab, hence that 3 must divide one of a and b, and 17 the other. But the possible sizes for suit and rank events preclude this.

(c) Show that the deck in (b) nevertheless does have pairs A, B of nontrivial independent events. *Hint*: Find 2 events A and B of sizes 3 and 17, respectively, which intersect in a single point.

(d) Add a joker to a full deck. Show that now there is no pair A, B of nontrivial independent events. *Hint*: See the hint in (b); 53 is prime.

[12] B. Eisenberg and B. K. Ghosh, "Independent Events in a Discrete Uniform Probability Space," *The American Statistician,* vol. 41, no. 1 (1987), pp. 52–56.

The following problems are suggested by Stanley Gudder in his article "Do Good Hands Attract?"[13] He says that event A *attracts* event B if $P(B|A) > P(B)$ and *repels* B if $P(B|A) < P(B)$.

55 Let R_i be the event that the ith player in a poker game has a royal flush. Show that a royal flush (A,K,Q,J,10 of one suit) attracts another royal flush, that is $P(R_2|R_1) > P(R_2)$. Show that a royal flush repels full houses.

56 Prove that A attracts B if and only if B attracts A. Hence we can say that A and B are *mutually attractive* if A attracts B.

57 Prove that A neither attracts nor repels B if and only if A and B are independent.

58 Prove that A and B are mutually attractive if and only if $P(B|A) > P(B|\tilde{A})$.

59 Prove that if A attracts B, then A repels $\tilde{B}$.

60 Prove that if A attracts both B and C, and A repels $B \cap C$, then A attracts $B \cup C$. Is there any example in which A attracts both B and C and repels $B \cup C$?

61 Prove that if $B_1, B_2, \ldots, B_n$ are mutually disjoint and collectively exhaustive, and if A attracts some B_i, then A must repel some B_j.

62 (a) Suppose that you are looking in your desk for a letter from some time ago. Your desk has eight drawers, and you assess the probability that it is in any particular drawer is 10% (so there is a 20% chance that it is not in the desk at all). Suppose now that you start searching systematically through your desk, one drawer at a time. In addition, suppose that you have not found the letter in the first i drawers, where $0 \leq i \leq 7$. Let p_i denote the probability that the letter will be found in the next drawer, and let q_i denote the probability that the letter will be found in some subsequent drawer (both p_i and q_i are conditional probabilities, since they are based upon the assumption that the letter is not in the first i drawers). Show that the p_i's increase and the q_i's decrease. (This problem is from Falk et al.[14])

(b) The following data appeared in an article in the Wall Street Journal.[15] For the ages 20, 30, 40, 50, and 60, the probability of a woman in the U.S. developing cancer in the next ten years is 0.5%, 1.2%, 3.2%, 6.4%, and 10.8%, respectively. At the same set of ages, the probability of a woman in the U.S. eventually developing cancer is 39.6%, 39.5%, 39.1%,

[13] S. Gudder, "Do Good Hands Attract?" *Mathematics Magazine,* vol. 54, no. 1 (1981), pp. 13–16.

[14] R. Falk, A. Lipson, and C. Konold, "The ups and downs of the hope function in a fruitless search," in *Subjective Probability,* G. Wright and P. Ayton, (eds.) (Chichester: Wiley, 1994), pgs. 353-377.

[15] C. Crossen, "Fright by the numbers: Alarming disease data are frequently flawed," *Wall Street Journal,* 11 April 1996, p. B1.

Chapter 5

Important Distributions and Densities

5.1 Important Distributions

In this chapter, we describe the discrete probability distributions and the continuous probability densities that occur most often in the analysis of experiments. We will also show how one simulates these distributions and densities on a computer.

Discrete Uniform Distribution

In Chapter 1, we saw that in many cases, we assume that all outcomes of an experiment are equally likely. If X is a random variable which represents the outcome of an experiment of this type, then we say that X is uniformly distributed. If the sample space S is of size n, where $0 < n < \infty$, then the distribution function $m(\omega)$ is defined to be $1/n$ for all $\omega \in S$. As is the case with all of the discrete probability distributions discussed in this chapter, this experiment can be simulated on a computer using the program **GeneralSimulation**. However, in this case, a faster algorithm can be used instead. (This algorithm was described in Chapter 1; we repeat the description here for completeness.) The expression

$$1 + \lfloor n\,(rnd) \rfloor$$

takes on as a value each integer between 1 and n with probability $1/n$ (the notation $\lfloor x \rfloor$ denotes the greatest integer not exceeding x). Thus, if the possible outcomes of the experiment are labelled ω_1 ω_2, ..., ω_n, then we use the above expression to represent the subscript of the output of the experiment.

If the sample space is a countably infinite set, such as the set of positive integers, then it is not possible to have an experiment which is uniform on this set (see Exercise 3). If the sample space is an uncountable set, with positive, finite length, such as the interval $[0, 1]$, then we use continuous density functions (see Section 5.2).

Binomial Distribution

The binomial distribution with parameters n, p, and k was defined in Chapter 3. It is the distribution of the random variable which counts the number of heads which occur when a coin is tossed n times, assuming that on any one toss, the probability that a head occurs is p. The distribution function is given by the formula

$$b(n,p,k) = \binom{n}{k} p^k q^{n-k} \ ,$$

where $q = 1 - p$.

One straightforward way to simulate a binomial random variable X is to compute the sum of n independent $0-1$ random variables, each of which take on the value 1 with probability p. This method requires n calls to a random number generator to obtain one value of the random variable. When n is relatively large (say at least 30), the Central Limit Theorem (see Chapter 9) implies that the binomial distribution is well-approximated by the corresponding normal density function (which is defined in Section 5.2) with parameters $\mu = np$ and $\sigma = \sqrt{npq}$. Thus, in this case we can compute a value Y of a normal random variable with these parameters, and if $-1/2 \leq Y < n + 1/2$, we can use the value

$$\lfloor Y + 1/2 \rfloor$$

to represent the random variable X. If $Y < -1/2$ or $Y > n + 1/2$, we reject Y and compute another value. We will see in the next section how we can quickly simulate normal random variables.

Geometric Distribution

Consider a Bernoulli trials process continued for an infinite number of trials; for example, a coin tossed an infinite sequence of times. We showed in Section 2.2 how to assign a probability distribution to the infinite tree. Thus, we can determine the distribution for any random variable X relating to the experiment provided $P(X = a)$ can be computed in terms of a finite number of trials. For example, let T be the number of trials up to and including the first success. Then

$$\begin{aligned} P(T = 1) &= p \ , \\ P(T = 2) &= qp \ , \\ P(T = 3) &= q^2 p \ , \end{aligned}$$

and in general,

$$P(T = n) = q^{n-1} p \ .$$

To show that this is a distribution, we must show that

$$p + qp + q^2 p + \cdots = 1 \ .$$

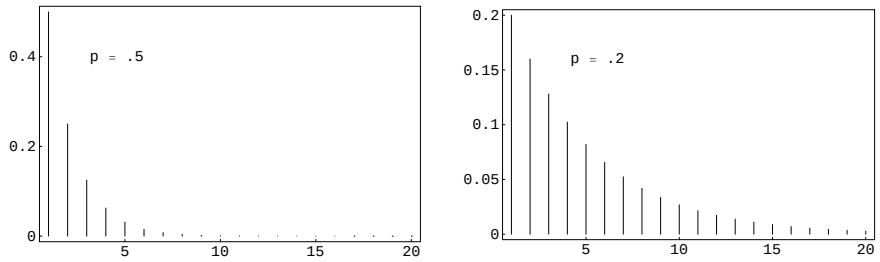


Figure 5.1: Geometric distributions.

The left-hand expression is just a geometric series with first term p and common ratio q, so its sum is

$$\frac{p}{1-q}$$

which equals 1.

In Figure 5.1 we have plotted this distribution using the program **GeometricPlot** for the cases $p = .5$ and $p = .2$. We see that as p decreases we are more likely to get large values for T, as would be expected. In both cases, the most probable value for T is 1. This will always be true since

$$\frac{P(T = j+1)}{P(T = j)} = q < 1 \ .$$

In general, if $0 < p < 1$, and $q = 1 - p$, then we say that the random variable T has a *geometric distribution* if

$$P(T = j) = q^{j-1}p \ ,$$

for $j = 1,\ 2,\ 3,\ \ldots$.

To simulate the geometric distribution with parameter p, we can simply compute a sequence of random numbers in $[0, 1)$, stopping when an entry does not exceed p. However, for small values of p, this is time-consuming (taking, on the average, $1/p$ steps). We now describe a method whose running time does not depend upon the size of p. Define Y to be the smallest integer satisfying the inequality

$$1 - q^Y \geq rnd \ . \tag{5.1}$$

Then we have

$$\begin{aligned} P(Y = j) &= P\Big(1 - q^j \geq rnd > 1 - q^{j-1}\Big) \\ &= q^{j-1} - q^j \\ &= q^{j-1}(1-q) \\ &= q^{j-1}p \ . \end{aligned}$$

Thus, Y is geometrically distributed with parameter p. To generate Y, all we have to do is solve Equation 5.1 for Y. We obtain

$$Y = \left\lceil \frac{\log(1 - rnd)}{\log\ q} \right\rceil ,$$

where the notation $\lceil x \rceil$ means the least integer which is greater than or equal to x. Since $\log(1 - rnd)$ and $\log(rnd)$ are identically distributed, Y can also be generated using the equation

$$Y = \left\lceil \frac{\log\ rnd}{\log\ q} \right\rceil .$$

Example 5.1 The geometric distribution plays an important role in the theory of queues, or waiting lines. For example, suppose a line of customers waits for service at a counter. It is often assumed that, in each small time unit, either 0 or 1 new customers arrive at the counter. The probability that a customer arrives is p and that no customer arrives is $q = 1 - p$. Then the time T until the next arrival has a geometric distribution. It is natural to ask for the probability that no customer arrives in the next k time units, that is, for $P(T > k)$. This is given by

$$\begin{aligned} P(T > k) = \sum_{j=k+1}^{\infty} q^{j-1}p &= q^k(p + qp + q^2p + \cdots) \\ &= q^k . \end{aligned}$$

This probability can also be found by noting that we are asking for no successes (i.e., arrivals) in a sequence of k consecutive time units, where the probability of a success in any one time unit is p. Thus, the probability is just q^k, since arrivals in any two time units are independent events.

It is often assumed that the length of time required to service a customer also has a geometric distribution but with a different value for p. This implies a rather special property of the service time. To see this, let us compute the conditional probability

$$P(T > r + s \,|\, T > r) = \frac{P(T > r + s)}{P(T > r)} = \frac{q^{r+s}}{q^r} = q^s .$$

Thus, the probability that the customer's service takes s more time units is independent of the length of time r that the customer has already been served. Because of this interpretation, this property is called the "memoryless" property, and is also obeyed by the exponential distribution. (Fortunately, not too many service stations have this property.) □

Negative Binomial Distribution

Suppose we are given a coin which has probability p of coming up heads when it is tossed. We fix a positive integer k, and toss the coin until the kth head appears. We

let X represent the number of tosses. When $k = 1$, X is geometrically distributed. For a general k, we say that X has a negative binomial distribution. We now calculate the probability distribution of X. If $X = x$, then it must be true that there were exactly $k - 1$ heads thrown in the first $x - 1$ tosses, and a head must have been thrown on the xth toss. There are

$$\binom{x-1}{k-1}$$

sequences of length x with these properties, and each of them is assigned the same probability, namely

$$p^{k-1}q^{x-k} \ .$$

Therefore, if we define

$$u(x, k, p) = P(X = x) \ ,$$

then

$$u(x, k, p) = \binom{x-1}{k-1} p^k q^{x-k} \ .$$

One can simulate this on a computer by simulating the tossing of a coin. The following algorithm is, in general, much faster. We note that X can be understood as the sum of k outcomes of a geometrically distributed experiment with parameter p. Thus, we can use the following sum as a means of generating X:

$$\sum_{j=1}^{k} \left\lceil \frac{\log \ rnd_j}{\log \ q} \right\rceil \ .$$

Example 5.2 A fair coin is tossed until the second time a head turns up. The distribution for the number of tosses is $u(x, 2, p)$. Thus the probability that x tosses are needed to obtain two heads is found by letting $k = 2$ in the above formula. We obtain

$$u(x, 2, 1/2) = \binom{x-1}{1} \frac{1}{2^x} \ ,$$

for $x = 2, 3, \ldots$.

In Figure 5.2 we give a graph of the distribution for $k = 2$ and $p = .25$. Note that the distribution is quite asymmetric, with a long tail reflecting the fact that large values of x are possible. □

Poisson Distribution

The Poisson distribution arises in many situations. It is safe to say that it is one of the three most important discrete probability distributions (the other two being the uniform and the binomial distributions). The Poisson distribution can be viewed as arising from the binomial distribution or from the exponential density. We shall now explain its connection with the former; its connection with the latter will be explained in the next section.

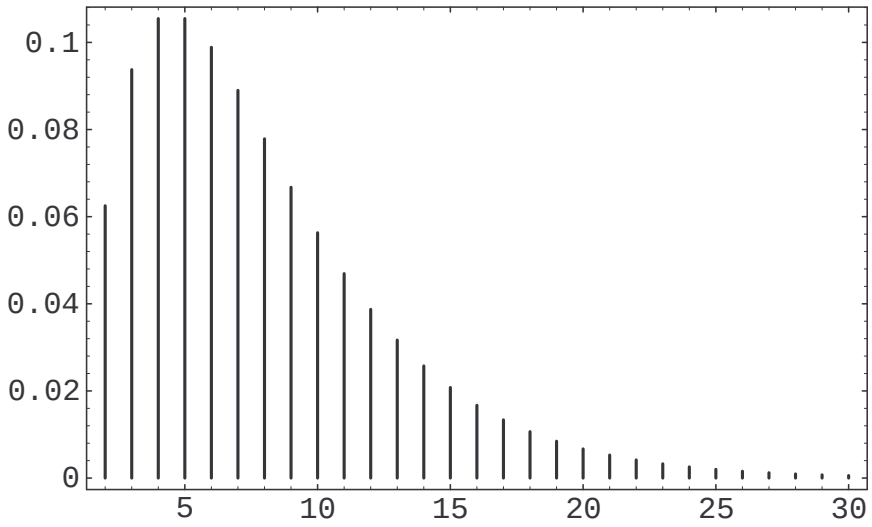


Figure 5.2: Negative binomial distribution with $k = 2$ and $p = .25$.

Suppose that we have a situation in which a certain kind of occurrence happens at random over a period of time. For example, the occurrences that we are interested in might be incoming telephone calls to a police station in a large city. We want to model this situation so that we can consider the probabilities of events such as more than 10 phone calls occurring in a 5-minute time interval. Presumably, in our example, there would be more incoming calls between 6:00 and 7:00 P.M. than between 4:00 and 5:00 A.M., and this fact would certainly affect the above probability. Thus, to have a hope of computing such probabilities, we must assume that the average rate, i.e., the average number of occurrences per minute, is a constant. This rate we will denote by λ. (Thus, in a given 5-minute time interval, we would expect about 5λ occurrences.) This means that if we were to apply our model to the two time periods given above, we would simply use different rates for the two time periods, thereby obtaining two different probabilities for the given event.

Our next assumption is that the number of occurrences in two non-overlapping time intervals are independent. In our example, this means that the events that there are j calls between 5:00 and 5:15 P.M. and k calls between 6:00 and 6:15 P.M. on the same day are independent.

We can use the binomial distribution to model this situation. We imagine that a given time interval is broken up into n subintervals of equal length. If the subintervals are sufficiently short, we can assume that two or more occurrences happen in one subinterval with a probability which is negligible in comparison with the probability of at most one occurrence. Thus, in each subinterval, we are assuming that there is either 0 or 1 occurrence. This means that the sequence of subintervals can be thought of as a sequence of Bernoulli trials, with a success corresponding to an occurrence in the subinterval.

To decide upon the proper value of p, the probability of an occurrence in a given subinterval, we reason as follows. On the average, there are λt occurrences in a time interval of length t. If this time interval is divided into n subintervals, then we would expect, using the Bernoulli trials interpretation, that there should be np occurrences. Thus, we want

$$\lambda t = np \ ,$$

so

$$p = \frac{\lambda t}{n} \ .$$

We now wish to consider the random variable X, which counts the number of occurrences in a given time interval. We want to calculate the distribution of X. For ease of calculation, we will assume that the time interval is of length 1; for time intervals of arbitrary length t, see Exercise 11. We know that

$$P(X = 0) = b(n, p, 0) = (1 - p)^n = \left(1 - \frac{\lambda}{n}\right)^n \ .$$

For large n, this is approximately $e^{-\lambda}$. It is easy to calculate that for any fixed k, we have

$$\frac{b(n, p, k)}{b(n, p, k - 1)} = \frac{\lambda - (k - 1)p}{kq}$$

which, for large n (and therefore small p) is approximately λ/k. Thus, we have

$$P(X = 1) \approx \lambda e^{-\lambda} \ ,$$

and in general,

$$P(X = k) \approx \frac{\lambda^k}{k!} e^{-\lambda} \ . \tag{5.2}$$

The above distribution is the Poisson distribution. We note that it must be checked that the distribution given in Equation 5.2 really *is* a distribution, i.e., that its values are non-negative and sum to 1. (See Exercise 12.)

The Poisson distribution is used as an approximation to the binomial distribution when the parameters n and p are large and small, respectively (see Examples 5.3 and 5.4). However, the Poisson distribution also arises in situations where it may not be easy to interpret or measure the parameters n and p (see Example 5.5).

Example 5.3 A typesetter makes, on the average, one mistake per 1000 words. Assume that he is setting a book with 100 words to a page. Let S_{100} be the number of mistakes that he makes on a single page. Then the exact probability distribution for S_{100} would be obtained by considering S_{100} as a result of 100 Bernoulli trials with $p = 1/1000$. The expected value of S_{100} is $\lambda = 100(1/1000) = .1$. The exact probability that $S_{100} = j$ is $b(100, 1/1000, j)$, and the Poisson approximation is

$$\frac{e^{-.1}(.1)^j}{j!}.$$

In Table 5.1 we give, for various values of n and p, the exact values computed by the binomial distribution and the Poisson approximation. □

j	Poisson $\lambda = .1$	Binomial $n = 100$ $p = .001$	Poisson $\lambda = 1$	Binomial $n = 100$ $p = .01$	Poisson $\lambda = 10$	Binomial $n = 1000$ $p = .01$
0	.9048	.9048	.3679	.3660	.0000	.0000
1	.0905	.0905	.3679	.3697	.0005	.0004
2	.0045	.0045	.1839	.1849	.0023	.0022
3	.0002	.0002	.0613	.0610	.0076	.0074
4	.0000	.0000	.0153	.0149	.0189	.0186
5			.0031	.0029	.0378	.0374
6			.0005	.0005	.0631	.0627
7			.0001	.0001	.0901	.0900
8			.0000	.0000	.1126	.1128
9					.1251	.1256
10					.1251	.1257
11					.1137	.1143
12					.0948	.0952
13					.0729	.0731
14					.0521	.0520
15					.0347	.0345
16					.0217	.0215
17					.0128	.0126
18					.0071	.0069
19					.0037	.0036
20					.0019	.0018
21					.0009	.0009
22					.0004	.0004
23					.0002	.0002
24					.0001	.0001
25					.0000	.0000

Table 5.1: Poisson approximation to the binomial distribution.

Example 5.4 In his book,[1] Feller discusses the statistics of flying bomb hits in the south of London during the Second World War.

Assume that you live in a district of size 10 blocks by 10 blocks so that the total district is divided into 100 small squares. How likely is it that the square in which you live will receive no hits if the total area is hit by 400 bombs?

We assume that a particular bomb will hit your square with probability 1/100. Since there are 400 bombs, we can regard the number of hits that your square receives as the number of *successes* in a Bernoulli trials process with $n = 400$ and $p = 1/100$. Thus we can use the Poisson distribution with $\lambda = 400 \cdot 1/100 = 4$ to approximate the probability that your square will receive j hits. This probability is $p(j) = e^{-4}4^j/j!$. The expected number of squares that receive exactly j hits is then $100 \cdot p(j)$. It is easy to write a program **LondonBombs** to simulate this situation and compare the expected number of squares with j hits with the observed number. In Exercise 26 you are asked to compare the actual observed data with that predicted by the Poisson distribution.

In Figure 5.3, we have shown the simulated hits, together with a spike graph showing both the observed and predicted frequencies. The observed frequencies are shown as squares, and the predicted frequencies are shown as dots. □

If the reader would rather not consider flying bombs, he is invited to instead consider an analogous situation involving cookies and raisins. We assume that we have made enough cookie dough for 500 cookies. We put 600 raisins in the dough, and mix it thoroughly. One way to look at this situation is that we have 500 cookies, and after placing the cookies in a grid on the table, we throw 600 raisins at the cookies. (See Exercise 22.)

Example 5.5 Suppose that in a certain fixed amount A of blood, the average human has 40 white blood cells. Let X be the random variable which gives the number of white blood cells in a random sample of size A from a random individual. We can think of X as binomially distributed with each white blood cell in the body representing a trial. If a given white blood cell turns up in the sample, then the trial corresponding to that blood cell was a success. Then p should be taken as the ratio of A to the total amount of blood in the individual, and n will be the number of white blood cells in the individual. Of course, in practice, neither of these parameters is very easy to measure accurately, but presumably the number 40 is easy to measure. But for the average human, we then have $40 = np$, so we can think of X as being Poisson distributed, with parameter $\lambda = 40$. In this case, it is easier to model the situation using the Poisson distribution than the binomial distribution. □

To simulate a Poisson random variable on a computer, a good way is to take advantage of the relationship between the Poisson distribution and the exponential

[1] ibid., p. 161.

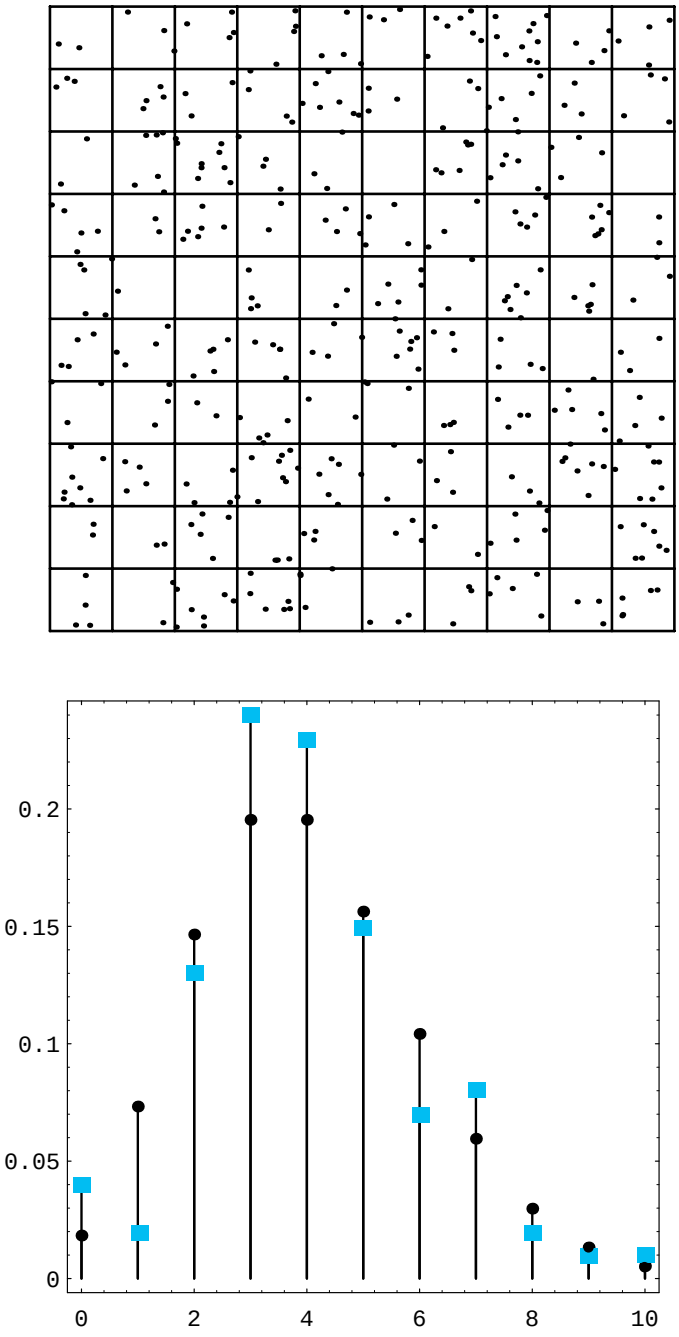


Figure 5.3: Flying bomb hits.

density. This relationship and the resulting simulation algorithm will be described in the next section.

Hypergeometric Distribution

Suppose that we have a set of N balls, of which k are red and $N-k$ are blue. We choose n of these balls, without replacement, and define X to be the number of red balls in our sample. The distribution of X is called the hypergeometric distribution. We note that this distribution depends upon three parameters, namely N, k, and n. There does not seem to be a standard notation for this distribution; we will use the notation $h(N,k,n,x)$ to denote $P(X=x)$. This probability can be found by noting that there are

$$\binom{N}{n}$$

different samples of size n, and the number of such samples with exactly x red balls is obtained by multiplying the number of ways of choosing x red balls from the set of k red balls and the number of ways of choosing $n-x$ blue balls from the set of $N-k$ blue balls. Hence, we have

$$h(N,k,n,x) = \frac{\binom{k}{x}\binom{N-k}{n-x}}{\binom{N}{n}} .$$

This distribution can be generalized to the case where there are more than two types of objects. (See Exercise 40.)

If we let N and k tend to ∞, in such a way that the ratio k/N remains fixed, then the hypergeometric distribution tends to the binomial distribution with parameters n and $p = k/N$. This is reasonable because if N and k are much larger than n, then whether we choose our sample with or without replacement should not affect the probabilities very much, and the experiment consisting of choosing with replacement yields a binomially distributed random variable (see Exercise 44).

An example of how this distribution might be used is given in Exercises 36 and 37. We now give another example involving the hypergeometric distribution. It illustrates a statistical test called Fisher's Exact Test.

Example 5.6 It is often of interest to consider two traits, such as eye color and hair color, and to ask whether there is an association between the two traits. Two traits are associated if knowing the value of one of the traits for a given person allows us to predict the value of the other trait for that person. The stronger the association, the more accurate the predictions become. If there is no association between the traits, then we say that the traits are independent. In this example, we will use the traits of gender and political party, and we will assume that there are only two possible genders, female and male, and only two possible political parties, Democratic and Republican.

Suppose that we have collected data concerning these traits. To test whether there is an association between the traits, we first assume that there is no association between the two traits. This gives rise to an "expected" data set, in which knowledge

	Democrat	Republican	
Female	24	4	28
Male	8	14	22
	32	18	50

Table 5.2: Observed data.

	Democrat	Republican	
Female	s_{11}	s_{12}	t_{11}
Male	s_{21}	s_{22}	t_{12}
	t_{21}	t_{22}	n

Table 5.3: General data table.

of the value of one trait is of no help in predicting the value of the other trait. Our collected data set usually differs from this expected data set. If it differs by quite a bit, then we would tend to reject the assumption of independence of the traits. To nail down what is meant by "quite a bit," we decide which possible data sets differ from the expected data set by at least as much as ours does, and then we compute the probability that any of these data sets would occur under the assumption of independence of traits. If this probability is small, then it is unlikely that the difference between our collected data set and the expected data set is due entirely to chance.

Suppose that we have collected the data shown in Table 5.2. The row and column sums are called marginal totals, or marginals. In what follows, we will denote the row sums by t_{11} and t_{12}, and the column sums by t_{21} and t_{22}. The ijth entry in the table will be denoted by s_{ij}. Finally, the size of the data set will be denoted by n. Thus, a general data table will look as shown in Table 5.3. We now explain the model which will be used to construct the "expected" data set. In the model, we assume that the two traits are independent. We then put t_{21} yellow balls and t_{22} green balls, corresponding to the Democratic and Republican marginals, into an urn. We draw t_{11} balls, without replacement, from the urn, and call these balls females. The t_{12} balls remaining in the urn are called males. In the specific case under consideration, the probability of getting the actual data under this model is given by the expression

$$\frac{\binom{32}{24}\binom{18}{4}}{\binom{50}{28}},$$

i.e., a value of the hypergeometric distribution.

We are now ready to construct the expected data set. If we choose 28 balls out of 50, we should expect to see, on the average, the same percentage of yellow balls in our sample as in the urn. Thus, we should expect to see, on the average, $28(32/50) = 17.92 \approx 18$ yellow balls in our sample. (See Exercise 36.) The other expected values are computed in exactly the same way. Thus, the expected data set is shown in Table 5.4. We note that the value of s_{11} determines the other three values in the table, since the marginals are all fixed. Thus, in considering

	Democrat	Republican	
Female	18	10	28
Male	14	8	22
	32	18	50

Table 5.4: Expected data.

the possible data sets that could appear in this model, it is enough to consider the various possible values of s_{11}. In the specific case at hand, what is the probability of drawing exactly a yellow balls, i.e., what is the probability that $s_{11} = a$? It is

$$\frac{\binom{32}{a}\binom{18}{28-a}}{\binom{50}{28}} \ . \tag{5.3}$$

We are now ready to decide whether our actual data differs from the expected data set by an amount which is greater than could be reasonably attributed to chance alone. We note that the expected number of female Democrats is 18, but the actual number in our data is 24. The other data sets which differ from the expected data set by more than ours correspond to those where the number of female Democrats equals 25, 26, 27, or 28. Thus, to obtain the required probability, we sum the expression in (5.3) from $a = 24$ to $a = 28$. We obtain a value of .000395. Thus, we should reject the hypothesis that the two traits are independent. □

Finally, we turn to the question of how to simulate a hypergeometric random variable X. Let us assume that the parameters for X are N, k, and n. We imagine that we have a set of N balls, labelled from 1 to N. We decree that the first k of these balls are red, and the rest are blue. Suppose that we have chosen m balls, and that j of them are red. Then there are $k - j$ red balls left, and $N - m$ balls left. Thus, our next choice will be red with probability

$$\frac{k-j}{N-m} \ .$$

So at this stage, we choose a random number in $[0, 1]$, and report that a red ball has been chosen if and only if the random number does not exceed the above expression. Then we update the values of m and j, and continue until n balls have been chosen.

Benford Distribution

Our next example of a distribution comes from the study of leading digits in data sets. It turns out that many data sets that occur "in real life" have the property that the first digits of the data are not uniformly distributed over the set $\{1, 2, \ldots, 9\}$. Rather, it appears that the digit 1 is most likely to occur, and that the distribution is monotonically decreasing on the set of possible digits. The Benford distribution appears, in many cases, to fit such data. Many explanations have been given for the occurrence of this distribution. Possibly the most convincing explanation is that this distribution is the only one that is invariant under a change of scale. If one

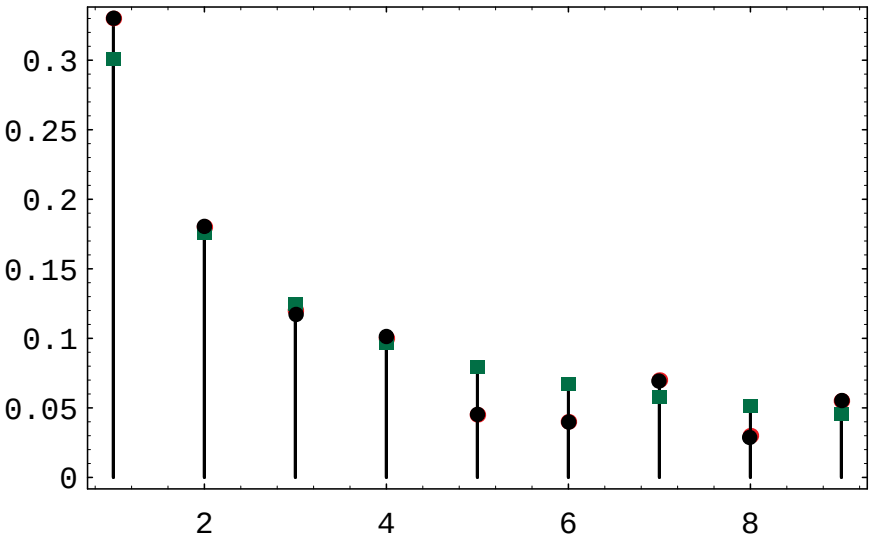


Figure 5.4: Leading digits in President Clinton's tax returns.

thinks of certain data sets as somehow "naturally occurring," then the distribution should be unaffected by which units are chosen in which to represent the data, i.e., the distribution should be invariant under change of scale.

Theodore Hill[2] gives a general description of the Benford distribution, when one considers the first d digits of integers in a data set. We will restrict our attention to the first digit. In this case, the Benford distribution has distribution function

$$f(k) = \log_{10}(k+1) - \log_{10}(k) \ ,$$

for $1 \leq k \leq 9$.

Mark Nigrini[3] has advocated the use of the Benford distribution as a means of testing suspicious financial records such as bookkeeping entries, checks, and tax returns. His idea is that if someone were to "make up" numbers in these cases, the person would probably produce numbers that are fairly uniformly distributed, while if one were to use the actual numbers, the leading digits would roughly follow the Benford distribution. As an example, Nigrini analyzed President Clinton's tax returns for a 13-year period. In Figure 5.4, the Benford distribution values are shown as squares, and the President's tax return data are shown as circles. One sees that in this example, the Benford distribution fits the data very well.

This distribution was discovered by the astronomer Simon Newcomb who stated the following in his paper on the subject: "That the ten digits do not occur with equal frequency must be evident to anyone making use of logarithm tables, and noticing how much faster the first pages wear out than the last ones. The first significant figure is oftener 1 than any other digit, and the frequency diminishes up

[2]T. P. Hill, "The Significant Digit Phenomenon," *American Mathematical Monthly,* vol. 102, no. 4 (April 1995), pgs. 322-327.

[3]M. Nigrini, "Detecting Biases and Irregularities in Tabulated Data," working paper

to 9."[4]

Exercises

1 For which of the following random variables would it be appropriate to assign a uniform distribution?

(a) Let X represent the roll of one die.

(b) Let X represent the number of heads obtained in three tosses of a coin.

(c) A roulette wheel has 38 possible outcomes: 0, 00, and 1 through 36. Let X represent the outcome when a roulette wheel is spun.

(d) Let X represent the birthday of a randomly chosen person.

(e) Let X represent the number of tosses of a coin necessary to achieve a head for the first time.

2 Let n be a positive integer. Let S be the set of integers between 1 and n. Consider the following process: We remove a number from S at random and write it down. We repeat this until S is empty. The result is a permutation of the integers from 1 to n. Let X denote this permutation. Is X uniformly distributed?

3 Let X be a random variable which can take on countably many values. Show that X cannot be uniformly distributed.

4 Suppose we are attending a college which has 3000 students. We wish to choose a subset of size 100 from the student body. Let X represent the subset, chosen using the following possible strategies. For which strategies would it be appropriate to assign the uniform distribution to X? If it is appropriate, what probability should we assign to each outcome?

(a) Take the first 100 students who enter the cafeteria to eat lunch.

(b) Ask the Registrar to sort the students by their Social Security number, and then take the first 100 in the resulting list.

(c) Ask the Registrar for a set of cards, with each card containing the name of exactly one student, and with each student appearing on exactly one card. Throw the cards out of a third-story window, then walk outside and pick up the first 100 cards that you find.

5 Under the same conditions as in the preceding exercise, can you describe a procedure which, if used, would produce each possible outcome with the same probability? Can you describe such a procedure that does not rely on a computer or a calculator?

[4]S. Newcomb, "Note on the frequency of use of the different digits in natural numbers," *American Journal of Mathematics,* vol. 4 (1881), pgs. 39-40.

6 Let X_1, X_2, ..., X_n be n mutually independent random variables, each of which is uniformly distributed on the integers from 1 to k. Let Y denote the minimum of the X_i's. Find the distribution of Y.

7 A die is rolled until the first time T that a six turns up.

(a) What is the probability distribution for T?

(b) Find $P(T > 3)$.

(c) Find $P(T > 6|T > 3)$.

8 If a coin is tossed a sequence of times, what is the probability that the first head will occur after the fifth toss, given that it has not occurred in the first two tosses?

9 A worker for the Department of Fish and Game is assigned the job of estimating the number of trout in a certain lake of modest size. She proceeds as follows: She catches 100 trout, tags each of them, and puts them back in the lake. One month later, she catches 100 more trout, and notes that 10 of them have tags.

(a) Without doing any fancy calculations, give a rough estimate of the number of trout in the lake.

(b) Let N be the number of trout in the lake. Find an expression, in terms of N, for the probability that the worker would catch 10 tagged trout out of the 100 trout that she caught the second time.

(c) Find the value of N which maximizes the expression in part (b). This value is called the *maximum likelihood estimate* for the unknown quantity N. *Hint*: Consider the ratio of the expressions for successive values of N.

10 A census in the United States is an attempt to count everyone in the country. It is inevitable that many people are not counted. The U. S. Census Bureau proposed a way to estimate the number of people who were not counted by the latest census. Their proposal was as follows: In a given locality, let N denote the actual number of people who live there. Assume that the census counted n_1 people living in this area. Now, another census was taken in the locality, and n_2 people were counted. In addition, n_{12} people were counted both times.

(a) Given N, n_1, and n_2, let X denote the number of people counted both times. Find the probability that $X = k$, where k is a fixed positive integer between 0 and n_2.

(b) Now assume that $X = n_{12}$. Find the value of N which maximizes the expression in part (a). *Hint*: Consider the ratio of the expressions for successive values of N.

11 Suppose that X is a random variable which represents the number of calls coming in to a police station in a one-minute interval. In the text, we showed that X could be modelled using a Poisson distribution with parameter λ, where this parameter represents the average number of incoming calls per minute. Now suppose that Y is a random variable which represents the number of incoming calls in an interval of length t. Show that the distribution of Y is given by

$$P(Y = k) = e^{-\lambda t}\frac{(\lambda t)^k}{k!} \ ,$$

i.e., Y is Poisson with parameter λt. *Hint*: Suppose a Martian were to observe the police station. Let us also assume that the basic time interval used on Mars is exactly t Earth minutes. Finally, we will assume that the Martian understands the derivation of the Poisson distribution in the text. What would she write down for the distribution of Y?

12 Show that the values of the Poisson distribution given in Equation 5.2 sum to 1.

13 The Poisson distribution with parameter $\lambda = .3$ has been assigned for the outcome of an experiment. Let X be the outcome function. Find $P(X = 0)$, $P(X = 1)$, and $P(X > 1)$.

14 On the average, only 1 person in 1000 has a particular rare blood type.

(a) Find the probability that, in a city of 10,000 people, no one has this blood type.

(b) How many people would have to be tested to give a probability greater than 1/2 of finding at least one person with this blood type?

15 Write a program for the user to input n, p, j and have the program print out the exact value of $b(n, p, k)$ and the Poisson approximation to this value.

16 Assume that, during each second, a Dartmouth switchboard receives one call with probability .01 and no calls with probability .99. Use the Poisson approximation to estimate the probability that the operator will miss at most one call if she takes a 5-minute coffee break.

17 The probability of a royal flush in a poker hand is $p = 1/649{,}740$. How large must n be to render the probability of having no royal flush in n hands smaller than $1/e$?

18 A baker blends 600 raisins and 400 chocolate chips into a dough mix and, from this, makes 500 cookies.

(a) Find the probability that a randomly picked cookie will have no raisins.

(b) Find the probability that a randomly picked cookie will have exactly two chocolate chips.

(c) Find the probability that a randomly chosen cookie will have at least two bits (raisins or chips) in it.

19 The probability that, in a bridge deal, one of the four hands has all hearts is approximately 6.3×10^{-12}. In a city with about 50,000 bridge players the resident probability expert is called on the average once a year (usually late at night) and told that the caller has just been dealt a hand of all hearts. Should she suspect that some of these callers are the victims of practical jokes?

20 An advertiser drops 10,000 leaflets on a city which has 2000 blocks. Assume that each leaflet has an equal chance of landing on each block. What is the probability that a particular block will receive no leaflets?

21 In a class of 80 students, the professor calls on 1 student chosen at random for a recitation in each class period. There are 32 class periods in a term.

(a) Write a formula for the exact probability that a given student is called upon j times during the term.

(b) Write a formula for the Poisson approximation for this probability. Using your formula estimate the probability that a given student is called upon more than twice.

22 Assume that we are making raisin cookies. We put a box of 600 raisins into our dough mix, mix up the dough, then make from the dough 500 cookies. We then ask for the probability that a randomly chosen cookie will have 0, 1, 2, ... raisins. Consider the cookies as trials in an experiment, and let X be the random variable which gives the number of raisins in a given cookie. Then we can regard the number of raisins in a cookie as the result of $n = 600$ independent trials with probability $p = 1/500$ for success on each trial. Since n is large and p is small, we can use the Poisson approximation with $\lambda = 600(1/500) = 1.2$. Determine the probability that a given cookie will have at least five raisins.

23 For a certain experiment, the Poisson distribution with parameter $\lambda = m$ has been assigned. Show that a most probable outcome for the experiment is the integer value k such that $m - 1 \leq k \leq m$. Under what conditions will there be two most probable values? *Hint*: Consider the ratio of successive probabilities.

24 When John Kemeny was chair of the Mathematics Department at Dartmouth College, he received an average of ten letters each day. On a certain weekday he received no mail and wondered if it was a holiday. To decide this he computed the probability that, in ten years, he would have at least 1 day without any mail. He assumed that the number of letters he received on a given day has a Poisson distribution. What probability did he find? *Hint*: Apply the Poisson distribution twice. First, to find the probability that, in 3000 days, he will have at least 1 day without mail, assuming each year has about 300 days on which mail is delivered.

25 Reese Prosser never puts money in a 10-cent parking meter in Hanover. He assumes that there is a probability of .05 that he will be caught. The first offense costs nothing, the second costs 2 dollars, and subsequent offenses cost 5 dollars each. Under his assumptions, how does the expected cost of parking 100 times without paying the meter compare with the cost of paying the meter each time?

26 Feller[5] discusses the statistics of flying bomb hits in an area in the south of London during the Second World War. The area in question was divided into $24 \times 24 = 576$ small areas. The total number of hits was 537. There were 229 squares with 0 hits, 211 with 1 hit, 93 with 2 hits, 35 with 3 hits, 7 with 4 hits, and 1 with 5 or more. Assuming the hits were purely random, use the Poisson approximation to find the probability that a particular square would have exactly k hits. Compute the expected number of squares that would have 0, 1, 2, 3, 4, and 5 or more hits and compare this with the observed results.

27 Assume that the probability that there is a significant accident in a nuclear power plant during one year's time is .001. If a country has 100 nuclear plants, estimate the probability that there is at least one such accident during a given year.

28 An airline finds that 4 percent of the passengers that make reservations on a particular flight will not show up. Consequently, their policy is to sell 100 reserved seats on a plane that has only 98 seats. Find the probability that every person who shows up for the flight will find a seat available.

29 The king's coinmaster boxes his coins 500 to a box and puts 1 counterfeit coin in each box. The king is suspicious, but, instead of testing all the coins in 1 box, he tests 1 coin chosen at random out of each of 500 boxes. What is the probability that he finds at least one fake? What is it if the king tests 2 coins from each of 250 boxes?

30 (From Kemeny[6]) Show that, if you make 100 bets on the number 17 at roulette at Monte Carlo (see Example 6.13), you will have a probability greater than 1/2 of coming out ahead. What is your expected winning?

31 In one of the first studies of the Poisson distribution, von Bortkiewicz[7] considered the frequency of deaths from kicks in the Prussian army corps. From the study of 14 corps over a 20-year period, he obtained the data shown in Table 5.5. Fit a Poisson distribution to this data and see if you think that the Poisson distribution is appropriate.

32 It is often assumed that the auto traffic that arrives at the intersection during a unit time period has a Poisson distribution with expected value m. Assume

[5] ibid., p. 161.
[6] Private communication.
[7] L. von Bortkiewicz, *Das Gesetz der Kleinen Zahlen* (Leipzig: Teubner, 1898), p. 24.

Number of deaths	Number of corps with x deaths in a given year
0	144
1	91
2	32
3	11
4	2

Table 5.5: Mule kicks.

that the number of cars X that arrive at an intersection from the north in unit time has a Poisson distribution with parameter $\lambda = m$ and the number Y that arrive from the west in unit time has a Poisson distribution with parameter $\lambda = \bar{m}$. If X and Y are independent, show that the total number $X + Y$ that arrive at the intersection in unit time has a Poisson distribution with parameter $\lambda = m + \bar{m}$.

33 Cars coming along Magnolia Street come to a fork in the road and have to choose either Willow Street or Main Street to continue. Assume that the number of cars that arrive at the fork in unit time has a Poisson distribution with parameter $\lambda = 4$. A car arriving at the fork chooses Main Street with probability 3/4 and Willow Street with probability 1/4. Let X be the random variable which counts the number of cars that, in a given unit of time, pass by Joe's Barber Shop on Main Street. What is the distribution of X?

34 In the appeal of the *People v. Collins* case (see Exercise 4.1.28), the counsel for the defense argued as follows: Suppose, for example, there are 5,000,000 couples in the Los Angeles area and the probability that a randomly chosen couple fits the witnesses' description is 1/12,000,000. Then the probability that there are two such couples given that there is at least one is not at all small. Find this probability. (The California Supreme Court overturned the initial guilty verdict.)

35 A manufactured lot of brass turnbuckles has S items of which D are defective. A sample of s items is drawn without replacement. Let X be a random variable that gives the number of defective items in the sample. Let $p(d) = P(X = d)$.

(a) Show that

$$p(d) = \frac{\binom{D}{d}\binom{S-D}{s-d}}{\binom{S}{s}} .$$

Thus, X is hypergeometric.

(b) Prove the following identity, known as *Euler's formula*:

$$\sum_{d=0}^{\min(D,s)} \binom{D}{d}\binom{S-D}{s-d} = \binom{S}{s} .$$

36 A bin of 1000 turnbuckles has an unknown number D of defectives. A sample of 100 turnbuckles has 2 defectives. The *maximum likelihood estimate* for D is the number of defectives which gives the highest probability for obtaining the number of defectives observed in the sample. Guess this number D and then write a computer program to verify your guess.

37 There are an unknown number of moose on Isle Royale (a National Park in Lake Superior). To estimate the number of moose, 50 moose are captured and tagged. Six months later 200 moose are captured and it is found that 8 of these were tagged. Estimate the number of moose on Isle Royale from these data, and then verify your guess by computer program (see Exercise 36).

38 A manufactured lot of buggy whips has 20 items, of which 5 are defective. A random sample of 5 items is chosen to be inspected. Find the probability that the sample contains exactly one defective item

(a) if the sampling is done with replacement.

(b) if the sampling is done without replacement.

39 Suppose that N and k tend to ∞ in such a way that k/N remains fixed. Show that

$$h(N, k, n, x) \to b(n, k/N, x) \ .$$

40 A bridge deck has 52 cards with 13 cards in each of four suits: spades, hearts, diamonds, and clubs. A hand of 13 cards is dealt from a shuffled deck. Find the probability that the hand has

(a) a distribution of suits 4, 4, 3, 2 (for example, four spades, four hearts, three diamonds, two clubs).

(b) a distribution of suits 5, 3, 3, 2.

41 Write a computer algorithm that simulates a hypergeometric random variable with parameters N, k, and n.

42 You are presented with four different dice. The first one has two sides marked 0 and four sides marked 4. The second one has a 3 on every side. The third one has a 2 on four sides and a 6 on two sides, and the fourth one has a 1 on three sides and a 5 on three sides. You allow your friend to pick any of the four dice he wishes. Then you pick one of the remaining three and you each roll your die. The person with the largest number showing wins a dollar. Show that you can choose your die so that you have probability 2/3 of winning no matter which die your friend picks. (See Tenney and Foster.[8])

43 The students in a certain class were classified by hair color and eye color. The conventions used were: Brown and black hair were considered dark, and red and blonde hair were considered light; black and brown eyes were considered

[8] R. L. Tenney and C. C. Foster, *Non-transitive Dominance*, Math. Mag. 49 (1976) no. 3, pgs. 115-120.

	Dark Eyes	Light Eyes	
Dark Hair	28	15	43
Light Hair	9	23	32
	37	38	75

Table 5.6: Observed data.

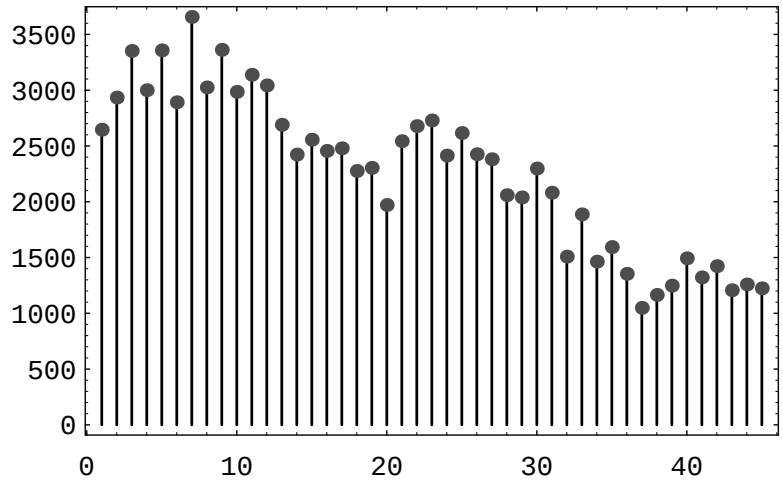


Figure 5.5: Distribution of choices in the Powerball lottery.

dark, and blue and green eyes were considered light. They collected the data shown in Table 5.6. Are these traits independent? (See Example 5.6.)

44 Suppose that in the hypergeometric distribution, we let N and k tend to ∞ in such a way that the ratio k/N approaches a real number p between 0 and 1. Show that the hypergeometric distribution tends to the binomial distribution with parameters n and p.

45 (a) Compute the leading digits of the first 100 powers of 2, and see how well these data fit the Benford distribution.

(b) Multiply each number in the data set of part (a) by 3, and compare the distribution of the leading digits with the Benford distribution.

46 In the Powerball lottery, contestants pick 5 different integers between 1 and 45, and in addition, pick a bonus integer from the same range (the bonus integer can equal one of the first five integers chosen). Some contestants choose the numbers themselves, and others let the computer choose the numbers. The data shown in Table 5.7 are the contestant-chosen numbers in a certain state on May 3, 1996. A spike graph of the data is shown in Figure 5.5.

The goal of this problem is to check the hypothesis that the chosen numbers are uniformly distributed. To do this, compute the value v of the random variable χ^2 given in Example 5.6. In the present case, this random variable has

Chapter 6

Expected Value and Variance

6.1 Expected Value of Discrete Random Variables

When a large collection of numbers is assembled, as in a census, we are usually interested not in the individual numbers, but rather in certain descriptive quantities such as the average or the median. In general, the same is true for the probability distribution of a numerically-valued random variable. In this and in the next section, we shall discuss two such descriptive quantities: the *expected value* and the *variance.* Both of these quantities apply only to numerically-valued random variables, and so we assume, in these sections, that all random variables have numerical values. To give some intuitive justification for our definition, we consider the following game.

Average Value

A die is rolled. If an odd number turns up, we win an amount equal to this number; if an even number turns up, we lose an amount equal to this number. For example, if a two turns up we lose 2, and if a three comes up we win 3. We want to decide if this is a reasonable game to play. We first try simulation. The program **Die** carries out this simulation.

The program prints the frequency and the relative frequency with which each outcome occurs. It also calculates the average winnings. We have run the program twice. The results are shown in Table 6.1.

In the first run we have played the game 100 times. In this run our average gain is $-.57$. It looks as if the game is unfavorable, and we wonder how unfavorable it really is. To get a better idea, we have played the game 10,000 times. In this case our average gain is $-.4949$.

We note that the relative frequency of each of the six possible outcomes is quite close to the probability 1/6 for this outcome. This corresponds to our frequency interpretation of probability. It also suggests that for very large numbers of plays, our average gain should be

$$\mu = 1\left(\frac{1}{6}\right) - 2\left(\frac{1}{6}\right) + 3\left(\frac{1}{6}\right) - 4\left(\frac{1}{6}\right) + 5\left(\frac{1}{6}\right) - 6\left(\frac{1}{6}\right)$$

Winning	n = 100		n = 10000	
	Frequency	Relative Frequency	Frequency	Relative Frequency
1	17	.17	1681	.1681
-2	17	.17	1678	.1678
3	16	.16	1626	.1626
-4	18	.18	1696	.1696
5	16	.16	1686	.1686
-6	16	.16	1633	.1633

Table 6.1: Frequencies for dice game.

$$= \quad \frac{9}{6} - \frac{12}{6} = -\frac{3}{6} = -.5 \ .$$

This agrees quite well with our average gain for 10,000 plays.

We note that the value we have chosen for the average gain is obtained by taking the possible outcomes, multiplying by the probability, and adding the results. This suggests the following definition for the expected outcome of an experiment.

Expected Value

Definition 6.1 Let X be a numerically-valued discrete random variable with sample space Ω and distribution function $m(x)$. The *expected value* $E(X)$ is defined by

$$E(X) = \sum_{x \in \Omega} xm(x) \ ,$$

provided this sum converges absolutely. We often refer to the expected value as the *mean,* and denote $E(X)$ by μ for short. If the above sum does not converge absolutely, then we say that X does not have an expected value. □

Example 6.1 Let an experiment consist of tossing a fair coin three times. Let X denote the number of heads which appear. Then the possible values of X are $0, 1, 2$ and 3. The corresponding probabilities are $1/8, 3/8, 3/8$, and $1/8$. Thus, the expected value of X equals

$$0\left(\frac{1}{8}\right) + 1\left(\frac{3}{8}\right) + 2\left(\frac{3}{8}\right) + 3\left(\frac{1}{8}\right) = \frac{3}{2} \ .$$

Later in this section we shall see a quicker way to compute this expected value, based on the fact that X can be written as a sum of simpler random variables. □

Example 6.2 Suppose that we toss a fair coin until a head first comes up, and let X represent the number of tosses which were made. Then the possible values of X are $1, 2, \ldots$, and the distribution function of X is defined by

$$m(i) = \frac{1}{2^i} \ .$$

(This is just the geometric distribution with parameter 1/2.) Thus, we have

$$\begin{aligned} E(X) &= \sum_{i=1}^{\infty} i \frac{1}{2^i} \\ &= \sum_{i=1}^{\infty} \frac{1}{2^i} + \sum_{i=2}^{\infty} \frac{1}{2^i} + \cdots \\ &= 1 + \frac{1}{2} + \frac{1}{2^2} + \cdots \\ &= 2 \ . \end{aligned}$$

□

Example 6.3 (Example 6.2 continued) Suppose that we flip a coin until a head first appears, and if the number of tosses equals n, then we are paid 2^n dollars. What is the expected value of the payment?

We let Y represent the payment. Then,

$$P(Y = 2^n) = \frac{1}{2^n} \ ,$$

for $n \geq 1$. Thus,

$$E(Y) = \sum_{n=1}^{\infty} 2^n \frac{1}{2^n} \ ,$$

which is a divergent sum. Thus, Y has no expectation. This example is called the *St. Petersburg Paradox*. The fact that the above sum is infinite suggests that a player should be willing to pay any fixed amount per game for the privilege of playing this game. The reader is asked to consider how much he or she would be willing to pay for this privilege. It is unlikely that the reader's answer is more than 10 dollars; therein lies the paradox.

In the early history of probability, various mathematicians gave ways to resolve this paradox. One idea (due to G. Cramer) consists of assuming that the amount of money in the world is finite. He thus assumes that there is some fixed value of n such that if the number of tosses equals or exceeds n, the payment is 2^n dollars. The reader is asked to show in Exercise 20 that the expected value of the payment is now finite.

Daniel Bernoulli and Cramer also considered another way to assign value to the payment. Their idea was that the value of a payment is some function of the payment; such a function is now called a utility function. Examples of reasonable

utility functions might include the square-root function or the logarithm function. In both cases, the value of $2n$ dollars is less than twice the value of n dollars. It can easily be shown that in both cases, the expected utility of the payment is finite (see Exercise 20). □

Example 6.4 Let T be the time for the first success in a Bernoulli trials process. Then we take as sample space Ω the integers $1, 2, \ldots$ and assign the geometric distribution

$$m(j) = P(T = j) = q^{j-1}p \ .$$

Thus,

$$\begin{aligned} E(T) &= 1 \cdot p + 2qp + 3q^2p + \cdots \\ &= p(1 + 2q + 3q^2 + \cdots) \ . \end{aligned}$$

Now if $|x| < 1$, then

$$1 + x + x^2 + x^3 + \cdots = \frac{1}{1-x} \ .$$

Differentiating this formula, we get

$$1 + 2x + 3x^2 + \cdots = \frac{1}{(1-x)^2} \ ,$$

so

$$E(T) = \frac{p}{(1-q)^2} = \frac{p}{p^2} = \frac{1}{p} \ .$$

In particular, we see that if we toss a fair coin a sequence of times, the expected time until the first heads is $1/(1/2) = 2$. If we roll a die a sequence of times, the expected number of rolls until the first six is $1/(1/6) = 6$. □

Interpretation of Expected Value

In statistics, one is frequently concerned with the average value of a set of data. The following example shows that the ideas of average value and expected value are very closely related.

Example 6.5 The heights, in inches, of the women on the Swarthmore basketball team are 5' 9", 5' 9", 5' 6", 5' 8", 5' 11", 5' 5", 5' 7", 5' 6", 5' 6", 5' 7", 5' 10", and 6' 0".

A statistician would compute the average height (in inches) as follows:

$$\frac{69 + 69 + 66 + 68 + 71 + 65 + 67 + 66 + 66 + 67 + 70 + 72}{12} = 67.9 \ .$$

One can also interpret this number as the expected value of a random variable. To see this, let an experiment consist of choosing one of the women at random, and let X denote her height. Then the expected value of X equals 67.9. □

Of course, just as with the frequency interpretation of probability, to interpret expected value as an average outcome requires further justification. We know that for any finite experiment the average of the outcomes is not predictable. However, we shall eventually prove that the average will usually be close to $E(X)$ if we repeat the experiment a large number of times. We first need to develop some properties of the expected value. Using these properties, and those of the concept of the variance to be introduced in the next section, we shall be able to prove the *Law of Large Numbers.* This theorem will justify mathematically both our frequency concept of probability and the interpretation of expected value as the average value to be expected in a large number of experiments.

Expectation of a Function of a Random Variable

Suppose that X is a discrete random variable with sample space Ω, and $\phi(x)$ is a real-valued function with domain Ω. Then $\phi(X)$ is a real-valued random variable. One way to determine the expected value of $\phi(X)$ is to first determine the distribution function of this random variable, and then use the definition of expectation. However, there is a better way to compute the expected value of $\phi(X)$, as demonstrated in the next example.

Example 6.6 Suppose a coin is tossed 9 times, with the result

$$HHHTTTTHT\ .$$

The first set of three heads is called a *run*. There are three more runs in this sequence, namely the next four tails, the next head, and the next tail. We do not consider the first two tosses to constitute a run, since the third toss has the same value as the first two.

Now suppose an experiment consists of tossing a fair coin three times. Find the expected number of runs. It will be helpful to think of two random variables, X and Y, associated with this experiment. We let X denote the sequence of heads and tails that results when the experiment is performed, and Y denote the number of runs in the outcome X. The possible outcomes of X and the corresponding values of Y are shown in Table 6.2.

To calculate $E(Y)$ using the definition of expectation, we first must find the distribution function $m(y)$ of Y i.e., we group together those values of X with a common value of Y and add their probabilities. In this case, we calculate that the distribution function of Y is: $m(1) = 1/4,\ m(2) = 1/2$, and $m(3) = 1/4$. One easily finds that $E(Y) = 2$.

Now suppose we didn't group the values of X with a common Y-value, but instead, for each X-value x, we multiply the probability of x and the corresponding value of Y, and add the results. We obtain

$$1\left(\frac{1}{8}\right) + 2\left(\frac{1}{8}\right) + 3\left(\frac{1}{8}\right) + 2\left(\frac{1}{8}\right) + 2\left(\frac{1}{8}\right) + 3\left(\frac{1}{8}\right) + 2\left(\frac{1}{8}\right) + 1\left(\frac{1}{8}\right)\ ,$$

X	Y
HHH	1
HHT	2
HTH	3
HTT	2
THH	2
THT	3
TTH	2
TTT	1

Table 6.2: Tossing a coin three times.

which equals 2.

This illustrates the following general principle. If X and Y are two random variables, and Y can be written as a function of X, then one can compute the expected value of Y using the distribution function of X. □

Theorem 6.1 If X is a discrete random variable with sample space Ω and distribution function $m(x)$, and if $\phi : \Omega \to \mathrm{R}$ is a function, then

$$E(\phi(X)) = \sum_{x \in \Omega} \phi(x) m(x) \ ,$$

provided the series converges absolutely. □

The proof of this theorem is straightforward, involving nothing more than grouping values of X with a common Y-value, as in Example 6.6.

The Sum of Two Random Variables

Many important results in probability theory concern sums of random variables. We first consider what it means to add two random variables.

Example 6.7 We flip a coin and let X have the value 1 if the coin comes up heads and 0 if the coin comes up tails. Then, we roll a die and let Y denote the face that comes up. What does $X + Y$ mean, and what is its distribution? This question is easily answered in this case, by considering, as we did in Chapter 4, the joint random variable $Z = (X, Y)$, whose outcomes are ordered pairs of the form (x, y), where $0 \leq x \leq 1$ and $1 \leq y \leq 6$. The description of the experiment makes it reasonable to assume that X and Y are independent, so the distribution function of Z is uniform, with $1/12$ assigned to each outcome. Now it is an easy matter to find the set of outcomes of $X + Y$, and its distribution function. □

In Example 6.1, the random variable X denoted the number of heads which occur when a fair coin is tossed three times. It is natural to think of X as the sum of the random variables X_1, X_2, X_3, where X_i is defined to be 1 if the ith toss comes up heads, and 0 if the ith toss comes up tails. The expected values of the X_i's are extremely easy to compute. It turns out that the expected value of X can be obtained by simply adding the expected values of the X_i's. This fact is stated in the following theorem.

Theorem 6.2 Let X and Y be random variables with finite expected values. Then

$$E(X+Y) = E(X) + E(Y) \ ,$$

and if c is any constant, then

$$E(cX) = cE(X) \ .$$

Proof. Let the sample spaces of X and Y be denoted by Ω_X and Ω_Y, and suppose that

$$\Omega_X = \{x_1, x_2, \ldots\}$$

and

$$\Omega_Y = \{y_1, y_2, \ldots\} \ .$$

Then we can consider the random variable $X+Y$ to be the result of applying the function $\phi(x,y) = x+y$ to the joint random variable (X,Y). Then, by Theorem 6.1, we have

$$\begin{aligned} E(X+Y) &= \sum_j \sum_k (x_j + y_k) P(X = x_j,\ Y = y_k) \\ &= \sum_j \sum_k x_j P(X = x_j,\ Y = y_k) + \sum_j \sum_k y_k P(X = x_j,\ Y = y_k) \\ &= \sum_j x_j P(X = x_j) + \sum_k y_k P(Y = y_k) \ . \end{aligned}$$

The last equality follows from the fact that

$$\sum_k P(X = x_j,\ Y = y_k) \quad = \quad P(X = x_j)$$

and

$$\sum_j P(X = x_j,\ Y = y_k) \quad = \quad P(Y = y_k) \ .$$

Thus,

$$E(X+Y) = E(X) + E(Y) \ .$$

	X		Y
a	b	c	3
a	c	b	1
b	a	c	1
b	c	a	0
c	a	b	0
c	b	a	1

Table 6.3: Number of fixed points.

If c is any constant,

$$\begin{aligned} E(cX) &= \sum_j cx_j P(X = x_j) \\ &= c\sum_j x_j P(X = x_j) \\ &= cE(X) \ . \end{aligned}$$

□

It is easy to prove by mathematical induction that *the expected value of the sum of any finite number of random variables is the sum of the expected values of the individual random variables.*

It is important to note that mutual independence of the summands was not needed as a hypothesis in the Theorem 6.2 and its generalization. The fact that expectations add, whether or not the summands are mutually independent, is sometimes referred to as the First Fundamental Mystery of Probability.

Example 6.8 Let Y be the number of fixed points in a random permutation of the set $\{a, b, c\}$. To find the expected value of Y, it is helpful to consider the basic random variable associated with this experiment, namely the random variable X which represents the random permutation. There are six possible outcomes of X, and we assign to each of them the probability 1/6 see Table 6.3. Then we can calculate $E(Y)$ using Theorem 6.1, as

$$3\Big(\frac{1}{6}\Big) + 1\Big(\frac{1}{6}\Big) + 1\Big(\frac{1}{6}\Big) + 0\Big(\frac{1}{6}\Big) + 0\Big(\frac{1}{6}\Big) + 1\Big(\frac{1}{6}\Big) = 1 \ .$$

We now give a very quick way to calculate the average number of fixed points in a random permutation of the set $\{1, 2, 3, \ldots, n\}$. Let Z denote the random permutation. For each i, $1 \leq i \leq n$, let X_i equal 1 if Z fixes i, and 0 otherwise. So if we let F denote the number of fixed points in Z, then

$$F = X_1 + X_2 + \cdots + X_n \ .$$

Therefore, Theorem 6.2 implies that

$$E(F) = E(X_1) + E(X_2) + \cdots + E(X_n) \ .$$

But it is easy to see that for each i,

$$E(X_i) = \frac{1}{n} \ ,$$

so

$$E(F) = 1 \ .$$

This method of calculation of the expected value is frequently very useful. It applies whenever the random variable in question can be written as a sum of simpler random variables. We emphasize again that it is not necessary that the summands be mutually independent. □

Bernoulli Trials

Theorem 6.3 Let S_n be the number of successes in n Bernoulli trials with probability p for success on each trial. Then the expected number of successes is np. That is,

$$E(S_n) = np \ .$$

Proof. Let X_j be a random variable which has the value 1 if the jth outcome is a success and 0 if it is a failure. Then, for each X_j,

$$E(X_j) = 0 \cdot (1 - p) + 1 \cdot p = p \ .$$

Since

$$S_n = X_1 + X_2 + \cdots + X_n \ ,$$

and the expected value of the sum is the sum of the expected values, we have

$$\begin{aligned} E(S_n) &= E(X_1) + E(X_2) + \cdots + E(X_n) \\ &= np \ . \end{aligned}$$

□

Poisson Distribution

Recall that the Poisson distribution with parameter λ was obtained as a limit of binomial distributions with parameters n and p, where it was assumed that $np = \lambda$, and $n \to \infty$. Since for each n, the corresponding binomial distribution has expected value λ, it is reasonable to guess that the expected value of a Poisson distribution with parameter λ also has expectation equal to λ. This is in fact the case, and the reader is invited to show this (see Exercise 21).

Independence

If X and Y are two random variables, it is not true in general that $E(X \cdot Y) = E(X)E(Y)$. However, this is true if X and Y are *independent.*

Theorem 6.4 If X and Y are independent random variables, then

$$E(X \cdot Y) = E(X)E(Y) \ .$$

Proof. Suppose that

$$\Omega_X = \{x_1, x_2, \ldots\}$$

and

$$\Omega_Y = \{y_1, y_2, \ldots\}$$

are the sample spaces of X and Y, respectively. Using Theorem 6.1, we have

$$E(X \cdot Y) = \sum_j \sum_k x_j y_k P(X = x_j, \ Y = y_k) \ .$$

But if X and Y are independent,

$$P(X = x_j, Y = y_k) = P(X = x_j)P(Y = y_k) \ .$$

Thus,

$$\begin{aligned} E(X \cdot Y) &= \sum_j \sum_k x_j y_k P(X = x_j)P(Y = y_k) \\ &= \left(\sum_j x_j P(X = x_j) \right) \left(\sum_k y_k P(Y = y_k) \right) \\ &= E(X)E(Y) \ . \end{aligned}$$

□

Example 6.9 A coin is tossed twice. $X_i = 1$ if the ith toss is heads and 0 otherwise. We know that X_1 and X_2 are independent. They each have expected value 1/2. Thus $E(X_1 \cdot X_2) = E(X_1)E(X_2) = (1/2)(1/2) = 1/4$. □

We next give a simple example to show that the expected values need not multiply if the random variables are not independent.

Example 6.10 Consider a single toss of a coin. We define the random variable X to be 1 if heads turns up and 0 if tails turns up, and we set $Y = 1 - X$. Then $E(X) = E(Y) = 1/2$. But $X \cdot Y = 0$ for either outcome. Hence, $E(X \cdot Y) = 0 \neq E(X)E(Y)$. □

We return to our records example of Section 3.1 for another application of the result that the expected value of the sum of random variables is the sum of the expected values of the individual random variables.

Records

Example 6.11 We start keeping snowfall records this year and want to find the expected number of records that will occur in the next n years. The first year is necessarily a record. The second year will be a record if the snowfall in the second year is greater than that in the first year. By symmetry, this probability is 1/2. More generally, let X_j be 1 if the jth year is a record and 0 otherwise. To find $E(X_j)$, we need only find the probability that the jth year is a record. But the record snowfall for the first j years is equally likely to fall in any one of these years, so $E(X_j) = 1/j$. Therefore, if S_n is the total number of records observed in the first n years,

$$E(S_n) = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} \ .$$

This is the famous *divergent harmonic series.* It is easy to show that

$$E(S_n) \sim \log n$$

as $n \to \infty$. A more accurate approximation to $E(S_n)$ is given by the expression

$$\log n + \gamma + \frac{1}{2n} \ ,$$

where γ denotes Euler's constant, and is approximately equal to .5772.

Therefore, in ten years the expected number of records is approximately 2.9298; the exact value is the sum of the first ten terms of the harmonic series which is 2.9290. □

Craps

Example 6.12 In the game of craps, the player makes a bet and rolls a pair of dice. If the sum of the numbers is 7 or 11 the player wins, if it is 2, 3, or 12 the player loses. If any other number results, say r, then r becomes the player's point and he continues to roll until either r or 7 occurs. If r comes up first he wins, and if 7 comes up first he loses. The program **Craps** simulates playing this game a number of times.

We have run the program for 1000 plays in which the player bets 1 dollar each time. The player's average winnings were $-.006$. The game of craps would seem to be only slightly unfavorable. Let us calculate the expected winnings on a single play and see if this is the case. We construct a two-stage tree measure as shown in Figure 6.1.

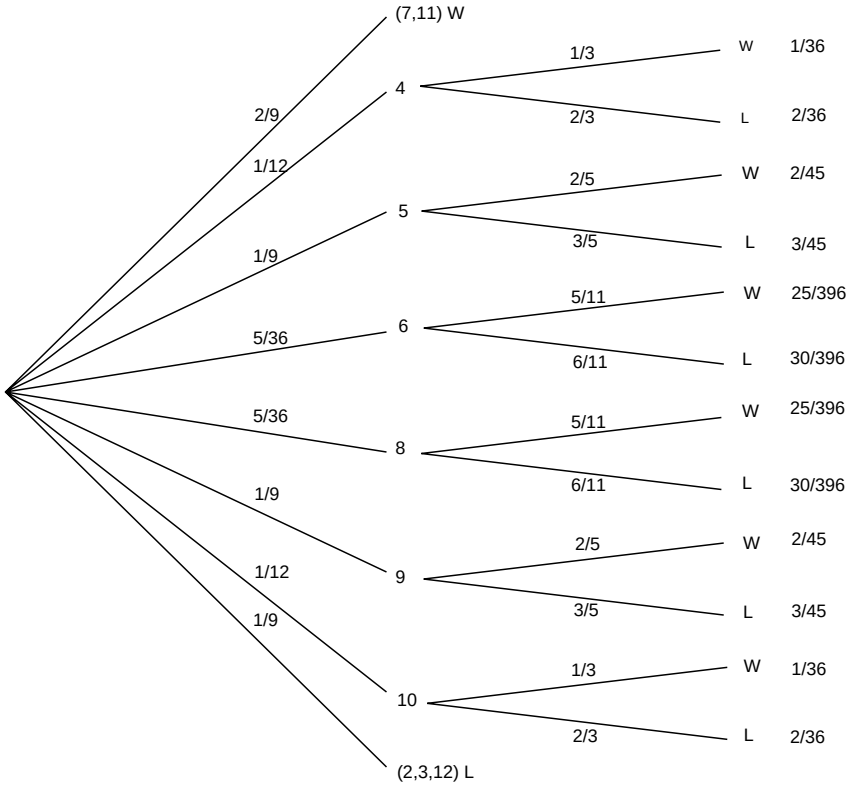


Figure 6.1: Tree measure for craps.

The first stage represents the possible sums for his first roll. The second stage represents the possible outcomes for the game if it has not ended on the first roll. In this stage we are representing the possible outcomes of a sequence of rolls required to determine the final outcome. The branch probabilities for the first stage are computed in the usual way assuming all 36 possibilites for outcomes for the pair of dice are equally likely. For the second stage we assume that the game will eventually end, and we compute the conditional probabilities for obtaining either the point or a 7. For example, assume that the player's point is 6. Then the game will end when one of the eleven pairs, $(1,5)$, $(2,4)$, $(3,3)$, $(4,2)$, $(5,1)$, $(1,6)$, $(2,5)$, $(3,4)$, $(4,3)$, $(5,2)$, $(6,1)$, occurs. We assume that each of these possible pairs has the same probability. Then the player wins in the first five cases and loses in the last six. Thus the probability of winning is $5/11$ and the probability of losing is $6/11$. From the path probabilities, we can find the probability that the player wins 1 dollar; it is $244/495$. The probability of losing is then $251/495$. Thus if X is his winning for a dollar bet,

$$\begin{aligned} E(X) &= 1\Big(\frac{244}{495}\Big) + (-1)\Big(\frac{251}{495}\Big) \\ &= -\frac{7}{495} \approx -.0141 \ . \end{aligned}$$

The game is unfavorable, but only slightly. The player's expected gain in n plays is $-n(.0141)$. If n is not large, this is a small expected loss for the player. The casino makes a large number of plays and so can afford a small average gain per play and still expect a large profit. □

Roulette

Example 6.13 In Las Vegas, a roulette wheel has 38 slots numbered 0, 00, 1, 2, ..., 36. The 0 and 00 slots are green, and half of the remaining 36 slots are red and half are black. A croupier spins the wheel and throws an ivory ball. If you bet 1 dollar on red, you win 1 dollar if the ball stops in a red slot, and otherwise you lose a dollar. We wish to calculate the expected value of your winnings, if you bet 1 dollar on red.

Let X be the random variable which denotes your winnings in a 1 dollar bet on red in Las Vegas roulette. Then the distribution of X is given by

$$m_X = \begin{pmatrix} -1 & 1 \\ 20/38 & 18/38 \end{pmatrix},$$

and one can easily calculate (see Exercise 5) that

$$E(X) \approx -.0526 \ .$$

We now consider the roulette game in Monte Carlo, and follow the treatment of Sagan.[1] In the roulette game in Monte Carlo there is only one 0. If you bet 1

[1] H. Sagan, *Markov Chains in Monte Carlo,* Math. Mag., vol. 54, no. 1 (1981), pp. 3-10.

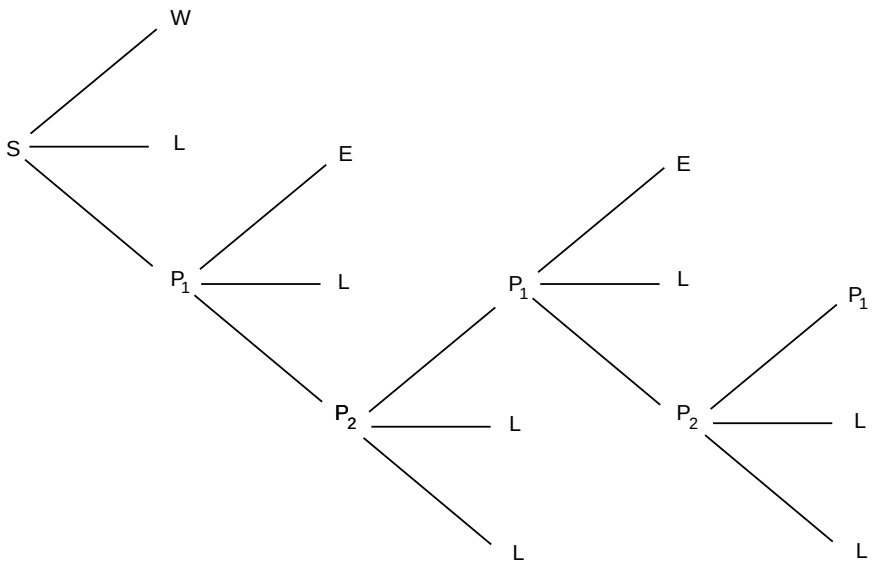


Figure 6.2: Tree for 2-prison Monte Carlo roulette.

franc on red and a 0 turns up, then, depending upon the casino, one or more of the following options may be offered:

(a) You get 1/2 of your bet back, and the casino gets the other half of your bet.

(b) Your bet is put "in prison," which we will denote by P_1. If red comes up on the next turn, you get your bet back (but you don't win any money). If black or 0 comes up, you lose your bet.

(c) Your bet is put in prison P_1, as before. If red comes up on the next turn, you get your bet back, and if black comes up on the next turn, then you lose your bet. If a 0 comes up on the next turn, then your bet is put into double prison, which we will denote by P_2. If your bet is in double prison, and if red comes up on the next turn, then your bet is moved back to prison P_1 and the game proceeds as before. If your bet is in double prison, and if black or 0 come up on the next turn, then you lose your bet. We refer the reader to Figure 6.2, where a tree for this option is shown. In this figure, S is the starting position, W means that you win your bet, L means that you lose your bet, and E means that you break even.

It is interesting to compare the expected winnings of a 1 franc bet on red, under each of these three options. We leave the first two calculations as an exercise (see Exercise 37). Suppose that you choose to play alternative (c). The calculation for this case illustrates the way that the early French probabilists worked problems like this.

Suppose you bet on red, you choose alternative (c), and a 0 comes up. Your possible future outcomes are shown in the tree diagram in Figure 6.3. Assume that your money is in the first prison and let x be the probability that you lose your

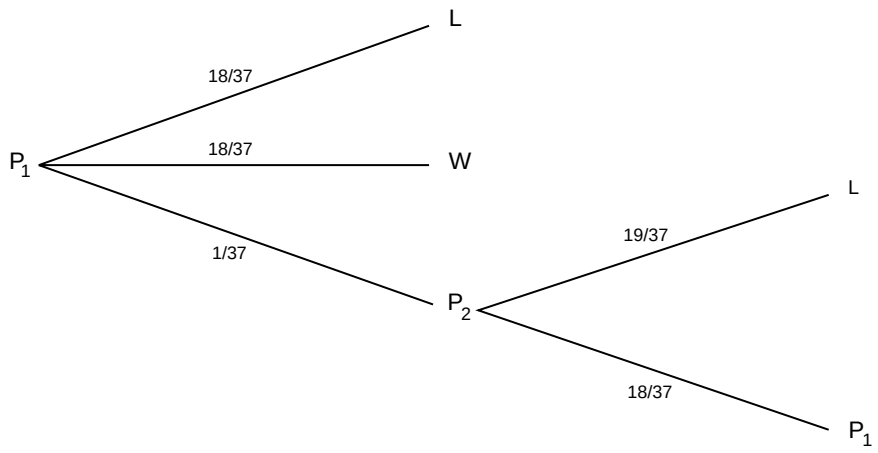


Figure 6.3: Your money is put in prison.

franc. From the tree diagram we see that

$$x = \frac{18}{37} + \frac{1}{37}P(\text{you lose your franc} \mid \text{your franc is in } P_2) \ .$$

Also,

$$P(\text{you lose your franc} \mid \text{your franc is in } P_2) = \frac{19}{37} + \frac{18}{37}x \ .$$

So, we have

$$x = \frac{18}{37} + \frac{1}{37}\Big(\frac{19}{37} + \frac{18}{37}x\Big) \ .$$

Solving for x, we obtain $x = 685/1351$. Thus, starting at S, the probability that you lose your bet equals

$$\frac{18}{37} + \frac{1}{37}x = \frac{25003}{49987} \ .$$

To find the probability that you win when you bet on red, note that you can only win if red comes up on the first turn, and this happens with probability 18/37. Thus your expected winnings are

$$1 \cdot \frac{18}{37} - 1 \cdot \frac{25003}{49987} = -\frac{687}{49987} \approx -.0137 \ .$$

It is interesting to note that the more romantic option (c) is less favorable than option (a) (see Exercise 37).

If you bet 1 dollar on the number 17, then the distribution function for your winnings X is

$$P_X = \begin{pmatrix} -1 & 35 \\ 36/37 & 1/37 \end{pmatrix} \ ,$$

and the expected winnings are

$$-1 \cdot \frac{36}{37} + 35 \cdot \frac{1}{37} = -\frac{1}{37} \approx -.027 \ .$$

Thus, at Monte Carlo different bets have different expected values. In Las Vegas almost all bets have the same expected value of $-2/38 = -.0526$ (see Exercises 4 and 5). □

Conditional Expectation

Definition 6.2 If F is any event and X is a random variable with sample space $\Omega = \{x_1, x_2, \ldots\}$, then the *conditional expectation given* F is defined by

$$E(X|F) = \sum_j x_j P(X = x_j|F) \ .$$

Conditional expectation is used most often in the form provided by the following theorem. □

Theorem 6.5 Let X be a random variable with sample space Ω. If F_1, F_2, ..., F_r are events such that $F_i \cap F_j = \emptyset$ for $i \neq j$ and $\Omega = \cup_j F_j$, then

$$E(X) = \sum_j E(X|F_j)P(F_j) \ .$$

Proof. We have

$$\begin{aligned}
\sum_j E(X|F_j)P(F_j) &= \sum_j \sum_k x_k P(X = x_k|F_j)P(F_j) \\
&= \sum_j \sum_k x_k P(X = x_k \text{ and } F_j \text{ occurs}) \\
&= \sum_k \sum_j x_k P(X = x_k \text{ and } F_j \text{ occurs}) \\
&= \sum_k x_k P(X = x_k) \\
&= E(X) \ .
\end{aligned}$$

□

Example 6.14 (Example 6.12 continued) Let T be the number of rolls in a single play of craps. We can think of a single play as a two-stage process. The first stage consists of a single roll of a pair of dice. The play is over if this roll is a 2, 3, 7, 11, or 12. Otherwise, the player's point is established, and the second stage begins. This second stage consists of a sequence of rolls which ends when either the player's point or a 7 is rolled. We record the outcomes of this two-stage experiment using

the random variables X and S, where X denotes the first roll, and S denotes the number of rolls in the second stage of the experiment (of course, S is sometimes equal to 0). Note that $T = S + 1$. Then by Theorem 6.5

$$E(T) = \sum_{j=2}^{12} E(T|X = j)P(X = j) \ .$$

If $j = 7$, 11 or 2, 3, 12, then $E(T|X = j) = 1$. If $j = 4, 5, 6, 8, 9$, or 10, we can use Example 6.4 to calculate the expected value of S. In each of these cases, we continue rolling until we get either a j or a 7. Thus, S is geometrically distributed with parameter p, which depends upon j. If $j = 4$, for example, the value of p is $3/36 + 6/36 = 1/4$. Thus, in this case, the expected number of additional rolls is $1/p = 4$, so $E(T|X = 4) = 1 + 4 = 5$. Carrying out the corresponding calculations for the other possible values of j and using Theorem 6.5 gives

$$\begin{aligned} E(T) &= 1\Big(\frac{12}{36}\Big) + \Big(1 + \frac{36}{3+6}\Big)\Big(\frac{3}{36}\Big) + \Big(1 + \frac{36}{4+6}\Big)\Big(\frac{4}{36}\Big) \\ &\quad + \Big(1 + \frac{36}{5+6}\Big)\Big(\frac{5}{36}\Big) + \Big(1 + \frac{36}{5+6}\Big)\Big(\frac{5}{36}\Big) \\ &\quad + \Big(1 + \frac{36}{4+6}\Big)\Big(\frac{4}{36}\Big) + \Big(1 + \frac{36}{3+6}\Big)\Big(\frac{3}{36}\Big) \\ &= \frac{557}{165} \\ &\approx 3.375\ldots \ . \end{aligned}$$

□

Martingales

We can extend the notion of fairness to a player playing a sequence of games by using the concept of conditional expectation.

Example 6.15 Let $S_1, S_2, \ldots, S_n$ be Peter's accumulated fortune in playing heads or tails (see Example 1.4). Then

$$E(S_n|S_{n-1} = a, \ldots, S_1 = r) = \frac{1}{2}(a + 1) + \frac{1}{2}(a - 1) = a \ .$$

We note that Peter's expected fortune after the next play is equal to his present fortune. When this occurs, we say the game is *fair*. A fair game is also called a *martingale*. If the coin is biased and comes up heads with probability p and tails with probability $q = 1 - p$, then

$$E(S_n|S_{n-1} = a, \ldots, S_1 = r) = p(a + 1) + q(a - 1) = a + p - q \ .$$

Thus, if $p < q$, this game is unfavorable, and if $p > q$, it is favorable. □

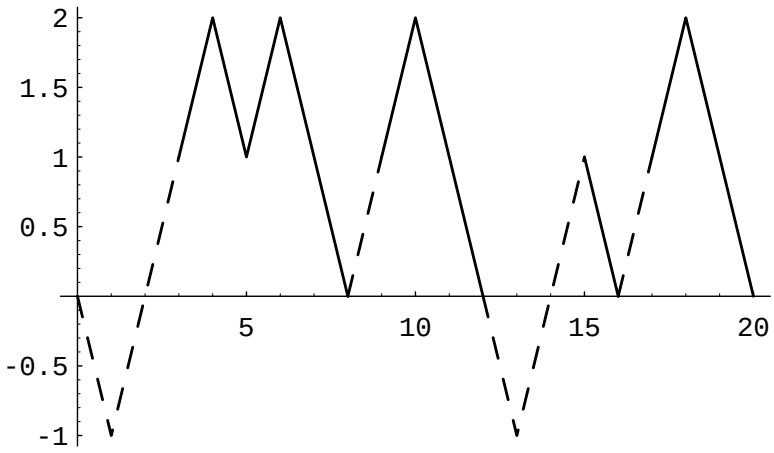


Figure 6.4: Mr. Ace's system.

If you are in a casino, you will see players adopting elaborate *systems* of play to try to make unfavorable games favorable. Two such systems, the martingale doubling system and the more conservative Labouchere system, were described in Exercises 1.1.9 and 1.1.10. Unfortunately, such systems cannot change even a fair game into a favorable game.

Even so, it is a favorite pastime of many people to develop systems of play for gambling games and for other games such as the stock market. We close this section with a simple illustration of such a system.

Stock Prices

Example 6.16 Let us assume that a stock increases or decreases in value each day by 1 dollar, each with probability 1/2. Then we can identify this simplified model with our familiar game of heads or tails. We assume that a buyer, Mr. Ace, adopts the following strategy. He buys the stock on the first day at its price V. He then waits until the price of the stock increases by one to $V + 1$ and sells. He then continues to watch the stock until its price falls back to V. He buys again and waits until it goes up to $V + 1$ and sells. Thus he holds the stock in intervals during which it increases by 1 dollar. In each such interval, he makes a profit of 1 dollar. However, we assume that he can do this only for a finite number of trading days. Thus he can lose if, in the last interval that he holds the stock, it does not get back up to $V + 1$; and this is the only way he can lose. In Figure 6.4 we illustrate a typical history if Mr. Ace must stop in twenty days. Mr. Ace holds the stock under his system during the days indicated by broken lines. We note that for the history shown in Figure 6.4, his system nets him a gain of 4 dollars.

We have written a program **StockSystem** to simulate the fortune of Mr. Ace if he uses his sytem over an n-day period. If one runs this program a large number

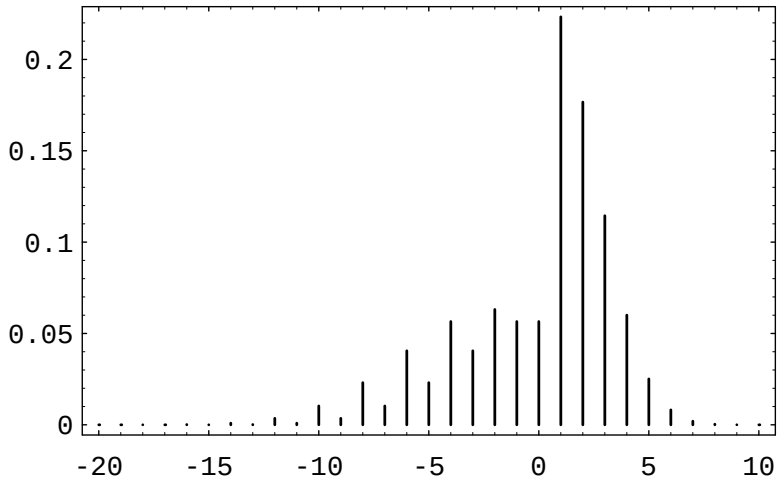


Figure 6.5: Winnings distribution for $n = 20$.

of times, for $n = 20$, say, one finds that his expected winnings are very close to 0, but the probability that he is ahead after 20 days is significantly greater than 1/2. For small values of n, the exact distribution of winnings can be calculated. The distribution for the case $n = 20$ is shown in Figure 6.5. Using this distribution, it is easy to calculate that the expected value of his winnings is exactly 0. This is another instance of the fact that a fair game (a martingale) remains fair under quite general systems of play.

Although the expected value of his winnings is 0, the probability that Mr. Ace is ahead after 20 days is about .610. Thus, he would be able to tell his friends that his system gives him a better chance of being ahead than that of someone who simply buys the stock and holds it, if our simple random model is correct. There have been a number of studies to determine how random the stock market is. □

Historical Remarks

With the Law of Large Numbers to bolster the frequency interpretation of probability, we find it natural to justify the definition of expected value in terms of the average outcome over a large number of repetitions of the experiment. The concept of expected value was used before it was formally defined; and when it was used, it was considered not as an average value but rather as the appropriate value for a gamble. For example recall, from the Historical Remarks section of Chapter 1, Section 1.2, Pascal's way of finding the value of a three-game series that had to be called off before it is finished.

Pascal first observed that if each player has only one game to win, then the stake of 64 pistoles should be divided evenly. Then he considered the case where one player has won two games and the other one.

> Then consider, Sir, if the first man wins, he gets 64 pistoles, if he loses he gets 32. Thus if they do not wish to risk this last game, but wish

> to separate without playing it, the first man must say: "I am certain to get 32 pistoles, even if I lose I still get them; but as for the other 32 pistoles, perhaps I will get them, perhaps you will get them, the chances are equal. Let us then divide these 32 pistoles in half and give one half to me as well as my 32 which are mine for sure." He will then have 48 pistoles and the other 16.[2]

Note that Pascal reduced the problem to a symmetric bet in which each player gets the same amount and takes it as obvious that in this case the stakes should be divided equally.

The first systematic study of expected value appears in Huygens' book. Like Pascal, Huygens find the value of a gamble by assuming that the answer is obvious for certain symmetric situations and uses this to deduce the expected for the general situation. He does this in steps. His first proposition is

> Prop. I. If I expect a or b, either of which, with equal probability, may fall to me, then my Expectation is worth $(a+b)/2$, that is, the half Sum of a and b.[3]

Huygens proved this as follows: Assume that two player A and B play a game in which each player puts up a stake of $(a+b)/2$ with an equal chance of winning the total stake. Then the value of the game to each player is $(a+b)/2$. For example, if the game had to be called off clearly each player should just get back his original stake. Now, by symmetry, this value is not changed if we add the condition that the winner of the game has to pay the loser an amount b as a consolation prize. Then for player A the value is still $(a+b)/2$. But what are his possible outcomes for the modified game? If he wins he gets the total stake $a+b$ and must pay B an amount b so ends up with a. If he loses he gets an amount b from player B. Thus player A wins a or b with equal chances and the value to him is $(a+b)/2$.

Huygens illustrated this proof in terms of an example. If you are offered a game in which you have an equal chance of winning 2 or 8, the expected value is 5, since this game is equivalent to the game in which each player stakes 5 and agrees to pay the loser 3 — a game in which the value is obviously 5.

Huygens' second proposition is

> Prop. II. If I expect a, b, or c, either of which, with equal facility, may happen, then the Value of my Expectation is $(a+b+c)/3$, or the third of the Sum of a, b, and c.[4]

His argument here is similar. Three players, A, B, and C, each stake

$$(a+b+c)/3$$

[2]Quoted in F. N. David, *Games, Gods and Gambling* (London: Griffin, 1962), p. 231.

[3]C. Huygens, *Calculating in Games of Chance,* translation attributed to John Arbuthnot (London, 1692), p. 34.

[4]ibid., p. 35.

in a game they have an equal chance of winning. The value of this game to player A is clearly the amount he has staked. Further, this value is not changed if A enters into an agreement with B that if one of them wins he pays the other a consolation prize of b and with C that if one of them wins he pays the other a consolation prize of c. By symmetry these agreements do not change the value of the game. In this modified game, if A wins he wins the total stake $a + b + c$ minus the consolation prizes $b + c$ giving him a final winning of a. If B wins, A wins b and if C wins, A wins c. Thus A finds himself in a game with value $(a + b + c)/3$ and with outcomes a, b, and c occurring with equal chance. This proves Proposition II.

More generally, this reasoning shows that if there are n outcomes

$$a_1,\ a_2,\ \ldots,\ a_n\ ,$$

all occurring with the same probability, the expected value is

$$\frac{a_1 + a_2 + \cdots + a_n}{n}\ .$$

In his third proposition Huygens considered the case where you win a or b but with unequal probabilities. He assumed there are p chances of winning a, and q chances of winning b, all having the same probability. He then showed that the expected value is

$$E = \frac{p}{p+q} \cdot a + \frac{q}{p+q} \cdot b\ .$$

This follows by considering an equivalent gamble with $p + q$ outcomes all occurring with the same probability and with a payoff of a in p of the outcomes and b in q of the outcomes. This allowed Huygens to compute the expected value for experiments with unequal probabilities, at least when these probablities are rational numbers.

Thus, instead of defining the expected value as a weighted average, Huygens assumed that the expected value of certain symmetric gambles are known and deduced the other values from these. Although this requires a good deal of clever manipulation, Huygens ended up with values that agree with those given by our modern definition of expected value. One advantage of this method is that it gives a justification for the expected value in cases where it is not reasonable to assume that you can repeat the experiment a large number of times, as for example, in betting that at least two presidents died on the same day of the year. (In fact, three did; all were signers of the Declaration of Independence, and all three died on July 4.)

In his book, Huygens calculated the expected value of games using techniques similar to those which we used in computing the expected value for roulette at Monte Carlo. For example, his proposition XIV is:

> Prop. XIV. If I were playing with another by turns, with two Dice, on this Condition, that if I throw 7 I gain, and if he throws 6 he gains allowing him the first Throw: To find the proportion of my Hazard to his.[5]

[5] ibid., p. 47.

	God does not exist	God exists
	p	$1-p$
believe	$-u$	v
not believe	0	$-x$

Table 6.4: Payoffs.

A modern description of this game is as follows. Huygens and his opponent take turns rolling a die. The game is over if Huygens rolls a 7 or his opponent rolls a 6. His opponent rolls first. What is the probability that Huygens wins the game?

To solve this problem Huygens let x be his chance of winning when his opponent threw first and y his chance of winning when he threw first. Then on the first roll his opponent wins on 5 out of the 36 possibilities. Thus,

$$x = \frac{31}{36} \cdot y \ .$$

But when Huygens rolls he wins on 6 out of the 36 possible outcomes, and in the other 30, he is led back to where his chances are x. Thus

$$y = \frac{6}{36} + \frac{30}{36} \cdot x \ .$$

From these two equations Huygens found that $x = 31/61$.

Another early use of expected value appeared in Pascal's argument to show that a rational person should believe in the existence of God.[6] Pascal said that we have to make a wager whether to believe or not to believe. Let p denote the probability that God does not exist. His discussion suggests that we are playing a game with two strategies, believe and not believe, with payoffs as shown in Table 6.4.

Here $-u$ represents the cost to you of passing up some worldly pleasures as a consequence of believing that God exists. If you do not believe, and God is a vengeful God, you will lose x. If God exists and you do believe you will gain v. Now to determine which strategy is best you should compare the two expected values

$$p(-u) + (1-p)v \qquad \text{and} \qquad p0 + (1-p)(-x),$$

and choose the larger of the two. In general, the choice will depend upon the value of p. But Pascal assumed that the value of v is infinite and so the strategy of believing is best no matter what probability you assign for the existence of God. This example is considered by some to be the beginning of decision theory. Decision analyses of this kind appear today in many fields, and, in particular, are an important part of medical diagnostics and corporate business decisions.

[6] Quoted in I. Hacking, *The Emergence of Probability* (Cambridge: Cambridge Univ. Press, 1975).

Age	Survivors
0	100
6	64
16	40
26	25
36	16
46	10
56	6
66	3
76	1

Table 6.5: Graunt's mortality data.

Another early use of expected value was to decide the price of annuities. The study of statistics has its origins in the use of the bills of mortality kept in the parishes in London from 1603. These records kept a weekly tally of christenings and burials. From these John Graunt made estimates for the population of London and also provided the first mortality data,[7] shown in Table 6.5.

As Hacking observes, Graunt apparently constructed this table by assuming that after the age of 6 there is a constant probability of about 5/8 of surviving for another decade.[8] For example, of the 64 people who survive to age 6, 5/8 of 64 or 40 survive to 16, 5/8 of these 40 or 25 survive to 26, and so forth. Of course, he rounded off his figures to the nearest whole person.

Clearly, a constant mortality rate cannot be correct throughout the whole range, and later tables provided by Halley were more realistic in this respect.[9]

A *terminal annuity* provides a fixed amount of money during a period of n years. To determine the price of a terminal annuity one needs only to know the appropriate interest rate. A *life annuity* provides a fixed amount during each year of the buyer's life. The appropriate price for a life annuity is the expected value of the terminal annuity evaluated for the random lifetime of the buyer. Thus, the work of Huygens in introducing expected value and the work of Graunt and Halley in determining mortality tables led to a more rational method for pricing annuities. This was one of the first serious uses of probability theory outside the gambling houses.

Although expected value plays a role now in every branch of science, it retains its importance in the casino. In 1962, Edward Thorp's book *Beat the Dealer*[10] provided the reader with a strategy for playing the popular casino game of blackjack that would assure the player a positive expected winning. This book forevermore changed the belief of the casinos that they could not be beat.

[7] ibid., p. 108.
[8] ibid., p. 109.
[9] E. Halley, "An Estimate of The Degrees of Mortality of Mankind," *Phil. Trans. Royal. Soc.,* vol. 17 (1693), pp. 596–610; 654–656.
[10] E. Thorp, *Beat the Dealer* (New York: Random House, 1962).

Exercises

1 A card is drawn at random from a deck consisting of cards numbered 2 through 10. A player wins 1 dollar if the number on the card is odd and loses 1 dollar if the number if even. What is the expected value of his winnings?

2 A card is drawn at random from a deck of playing cards. If it is red, the player wins 1 dollar; if it is black, the player loses 2 dollars. Find the expected value of the game.

3 In a class there are 20 students: 3 are 5' 6", 5 are 5'8", 4 are 5'10", 4 are 6', and 4 are 6' 2". A student is chosen at random. What is the student's expected height?

4 In Las Vegas the roulette wheel has a 0 and a 00 and then the numbers 1 to 36 marked on equal slots; the wheel is spun and a ball stops randomly in one slot. When a player bets 1 dollar on a number, he receives 36 dollars if the ball stops on this number, for a net gain of 35 dollars; otherwise, he loses his dollar bet. Find the expected value for his winnings.

5 In a second version of roulette in Las Vegas, a player bets on red or black. Half of the numbers from 1 to 36 are red, and half are black. If a player bets a dollar on black, and if the ball stops on a black number, he gets his dollar back and another dollar. If the ball stops on a red number or on 0 or 00 he loses his dollar. Find the expected winnings for this bet.

6 A die is rolled twice. Let X denote the sum of the two numbers that turn up, and Y the difference of the numbers (specifically, the number on the first roll minus the number on the second). Show that $E(XY) = E(X)E(Y)$. Are X and Y independent?

***7** Show that, if X and Y are random variables taking on only two values each, and if $E(XY) = E(X)E(Y)$, then X and Y are independent.

8 A royal family has children until it has a boy or until it has three children, whichever comes first. Assume that each child is a boy with probability 1/2. Find the expected number of boys in this royal family and the expected number of girls.

9 If the first roll in a game of craps is neither a natural nor craps, the player can make an additional bet, equal to his original one, that he will make his point before a seven turns up. If his point is four or ten he is paid off at $2:1$ odds; if it is a five or nine he is paid off at odds $3:2$; and if it is a six or eight he is paid off at odds $6:5$. Find the player's expected winnings if he makes this additional bet when he has the opportunity.

10 In Example 6.16 assume that Mr. Ace decides to buy the stock and hold it until it goes up 1 dollar and then sell and not buy again. Modify the program

StockSystem to find the distribution of his profit under this system after a twenty-day period. Find the expected profit and the probability that he comes out ahead.

11 On September 26, 1980, the *New York Times* reported that a mysterious stranger strode into a Las Vegas casino, placed a single bet of 777,000 dollars on the "don't pass" line at the crap table, and walked away with more than 1.5 million dollars. In the "don't pass" bet, the bettor is essentially betting with the house. An exception occurs if the roller rolls a 12 on the first roll. In this case, the roller loses and the "don't pass" better just gets back the money bet instead of winning. Show that the "don't pass" bettor has a more favorable bet than the roller.

12 Recall that in the *martingale doubling system* (see Exercise 1.1.10), the player doubles his bet each time he loses. Suppose that you are playing roulette in a *fair casino* where there are no 0's, and you bet on red each time. You then win with probability 1/2 each time. Assume that you enter the casino with 100 dollars, start with a 1-dollar bet and employ the martingale system. You stop as soon as you have won one bet, or in the unlikely event that black turns up six times in a row so that you are down 63 dollars and cannot make the required 64-dollar bet. Find your expected winnings under this system of play.

13 You have 80 dollars and play the following game. An urn contains two white balls and two black balls. You draw the balls out one at a time without replacement until all the balls are gone. On each draw, you bet half of your present fortune that you will draw a white ball. What is your expected final fortune?

14 In the hat check problem (see Example 3.12), it was assumed that N people check their hats and the hats are handed back at random. Let $X_j = 1$ if the jth person gets his or her hat and 0 otherwise. Find $E(X_j)$ and $E(X_j \cdot X_k)$ for j not equal to k. Are X_j and X_k independent?

15 A box contains two gold balls and three silver balls. You are allowed to choose successively balls from the box at random. You win 1 dollar each time you draw a gold ball and lose 1 dollar each time you draw a silver ball. After a draw, the ball is not replaced. Show that, if you draw until you are ahead by 1 dollar or until there are no more gold balls, this is a favorable game.

16 Gerolamo Cardano in his book, *The Gambling Scholar,* written in the early 1500s, considers the following carnival game. There are six dice. Each of the dice has five blank sides. The sixth side has a number between 1 and 6—a different number on each die. The six dice are rolled and the player wins a prize depending on the total of the numbers which turn up.

(a) Find, as Cardano did, the expected total without finding its distribution.

(b) Large prizes were given for large totals with a modest fee to play the game. Explain why this could be done.

17 Let X be the first time that a *failure* occurs in an infinite sequence of Bernoulli trials with probability p for success. Let $p_k = P(X = k)$ for $k = 1, 2, \ldots$. Show that $p_k = p^{k-1}q$ where $q = 1 - p$. Show that $\sum_k p_k = 1$. Show that $E(X) = 1/q$. What is the expected number of tosses of a coin required to obtain the first tail?

18 Exactly one of six similar keys opens a certain door. If you try the keys, one after another, what is the expected number of keys that you will have to try before success?

19 A multiple choice exam is given. A problem has four possible answers, and exactly one answer is correct. The student is allowed to choose a subset of the four possible answers as his answer. If his chosen subset contains the correct answer, the student receives three points, but he loses one point for each wrong answer in his chosen subset. Show that if he just guesses a subset uniformly and randomly his expected score is zero.

20 You are offered the following game to play: a fair coin is tossed until heads turns up for the first time (see Example 6.3). If this occurs on the first toss you receive 2 dollars, if it occurs on the second toss you receive $2^2 = 4$ dollars and, in general, if heads turns up for the first time on the nth toss you receive 2^n dollars.

(a) Show that the expected value of your winnings does not exist (i.e., is given by a divergent sum) for this game. Does this mean that this game is favorable no matter how much you pay to play it?

(b) Assume that you only receive 2^{10} dollars if any number greater than or equal to ten tosses are required to obtain the first head. Show that your expected value for this modified game is finite and find its value.

(c) Assume that you pay 10 dollars for each play of the original game. Write a program to simulate 100 plays of the game and see how you do.

(d) Now assume that the utility of n dollars is $\sqrt{n}$. Write an expression for the expected utility of the payment, and show that this expression has a finite value. Estimate this value. Repeat this exercise for the case that the utility function is $\log(n)$.

21 Let X be a random variable which is Poisson distributed with parameter λ. Show that $E(X) = \lambda$. *Hint*: Recall that

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots .$$

22 Recall that in Exercise 1.1.14, we considered a town with two hospitals. In the large hospital about 45 babies are born each day, and in the smaller

hospital about 15 babies are born each day. We were interested in guessing which hospital would have on the average the largest number of days with the property that more than 60 percent of the children born on that day are boys. For each hospital find the expected number of days in a year that have the property that more than 60 percent of the children born on that day were boys.

23 An insurance company has 1,000 policies on men of age 50. The company estimates that the probability that a man of age 50 dies within a year is .01. Estimate the number of claims that the company can expect from beneficiaries of these men within a year.

24 Using the life table for 1981 in Appendix C, write a program to compute the expected lifetime for males and females of each possible age from 1 to 85. Compare the results for males and females. Comment on whether life insurance should be priced differently for males and females.

***25** A deck of ESP cards consists of 20 cards each of two types: say ten stars, ten circles (normally there are five types). The deck is shuffled and the cards turned up one at a time. You, the alleged percipient, are to name the symbol on each card *before* it is turned up.

Suppose that you are really just guessing at the cards. If you do not get to see each card after you have made your guess, then it is easy to calculate the expected number of correct guesses, namely ten.

If, on the other hand, you are guessing with information, that is, if you see each card after your guess, then, of course, you might expect to get a higher score. This is indeed the case, but calculating the correct expectation is no longer easy.

But it is easy to do a computer simulation of this guessing with information, so we can get a good idea of the expectation by simulation. (This is similar to the way that skilled blackjack players make blackjack into a favorable game by observing the cards that have already been played. See Exercise 29.)

(a) First, do a simulation of guessing without information, repeating the experiment at least 1000 times. Estimate the expected number of correct answers and compare your result with the theoretical expectation.

(b) What is the best strategy for guessing with information?

(c) Do a simulation of guessing with information, using the strategy in (b). Repeat the experiment at least 1000 times, and estimate the expectation in this case.

(d) Let S be the number of stars and C the number of circles in the deck. Let $h(S, C)$ be the expected winnings using the optimal guessing strategy in (b). Show that $h(S, C)$ satisfies the recursion relation

$$h(S,C) = \frac{S}{S+C}h(S-1,C) + \frac{C}{S+C}h(S,C-1) + \frac{\max(S,C)}{S+C} \ ,$$

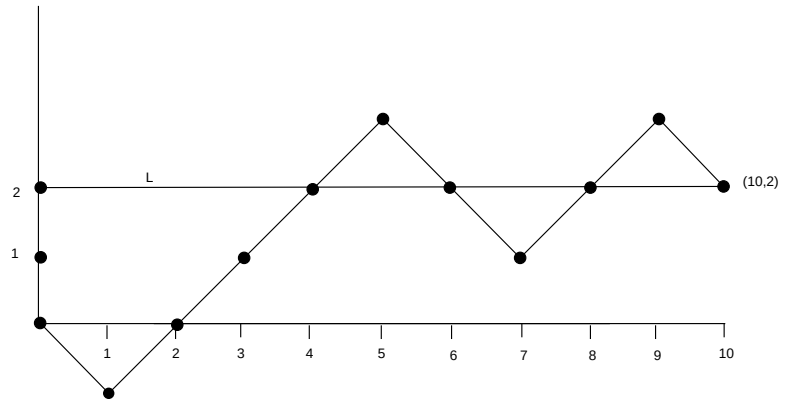


Figure 6.6: Random walk for ESP.

and $h(0,0) = h(-1,0) = h(0,-1) = 0$. Using this relation, write a program to compute $h(S,C)$ and find $h(10,10)$. Compare the computed value of $h(10,10)$ with the result of your simulation in (c). For more about this exercise and Exercise 26 see Diaconis and Graham.[11]

***26** Consider the ESP problem as described in Exercise 25. You are again guessing with information, and you are using the optimal guessing strategy of guessing *star* if the remaining deck has more stars, *circle* if more circles, and tossing a coin if the number of stars and circles are equal. Assume that $S \geq C$, where S is the number of stars and C the number of circles.

We can plot the results of a typical game on a graph, where the horizontal axis represents the number of steps and the vertical axis represents the *difference* between the number of stars and the number of circles that have been turned up. A typical game is shown in Figure 6.6. In this particular game, the order in which the cards were turned up is (C,S,S,S,S,C,C,S,S,C). Thus, in this particular game, there were six stars and four circles in the deck. This means, in particular, that every game played with this deck would have a graph which ends at the point $(10,2)$. We define the line L to be the horizontal line which goes through the ending point on the graph (so its vertical coordinate is just the difference between the number of stars and circles in the deck).

(a) Show that, when the random walk is below the line L, the player guesses right when the graph goes up (star is turned up) and, when the walk is above the line, the player guesses right when the walk goes down (circle turned up). Show from this property that the subject is sure to have at least S correct guesses.

(b) When the walk is at a point (x,x) *on* the line L the number of stars and circles remaining is the same, and so the subject tosses a coin. Show that

[11] P. Diaconis and R. Graham, "The Analysis of Sequential Experiments with Feedback to Subjects," *Annals of Statistics,* vol. 9 (1981), pp. 3–23.

the probability that the walk reaches (x, x) is

$$\frac{\binom{S}{x}\binom{C}{x}}{\binom{S+C}{2x}} .$$

Hint: The outcomes of $2x$ cards is a hypergeometric distribution (see Section 5.1).

(c) Using the results of (a) and (b) show that the expected number of correct guesses under intelligent guessing is

$$S + \sum_{x=1}^{C} \frac{1}{2} \frac{\binom{S}{x}\binom{C}{x}}{\binom{S+C}{2x}} .$$

27 It has been said[12] that a Dr. B. Muriel Bristol declined a cup of tea stating that she preferred a cup into which milk had been poured first. The famous statistician R. A. Fisher carried out a test to see if she could tell whether milk was put in before or after the tea. Assume that for the test Dr. Bristol was given eight cups of tea—four in which the milk was put in before the tea and four in which the milk was put in after the tea.

(a) What is the expected number of correct guesses the lady would make if she had no information after each test and was just guessing?

(b) Using the result of Exercise 26 find the expected number of correct guesses if she was told the result of each guess and used an optimal guessing strategy.

28 In a popular computer game the computer picks an integer from 1 to n at random. The player is given k chances to guess the number. After each guess the computer responds "correct," "too small," or "too big."

(a) Show that if $n \leq 2^k - 1$, then there is a strategy that guarantees you will correctly guess the number in k tries.

(b) Show that if $n \geq 2^k - 1$, there is a strategy that assures you of identifying one of $2^k - 1$ numbers and hence gives a probability of $(2^k - 1)/n$ of winning. Why is this an optimal strategy? Illustrate your result in terms of the case $n = 9$ and $k = 3$.

29 In the casino game of blackjack the dealer is dealt two cards, one face up and one face down, and each player is dealt two cards, both face down. If the dealer is showing an ace the player can look at his down cards and then make a bet called an *insurance* bet. (Expert players will recognize why it is called insurance.) If you make this bet you will win the bet if the dealer's second card is a *ten card*: namely, a ten, jack, queen, or king. If you win, you are paid twice your insurance bet; otherwise you lose this bet. Show that, if the

[12] J. F. Box, *R. A. Fisher, The Life of a Scientist* (New York: John Wiley and Sons, 1978).

only cards you can see are the dealer's ace and your two cards and if your cards are not ten cards, then the insurance bet is an unfavorable bet. Show, however, that if you are playing two hands simultaneously, and you have no ten cards, then it is a favorable bet. (Thorp[13] has shown that the game of blackjack is favorable to the player if he or she can keep good enough track of the cards that have been played.)

30 Assume that, every time you buy a box of Wheaties, you receive a picture of one of the n players for the New York Yankees (see Exercise 3.2.34). Let X_k be the number of additional boxes you have to buy, after you have obtained $k-1$ different pictures, in order to obtain the next new picture. Thus $X_1 = 1$, X_2 is the number of boxes bought after this to obtain a picture different from the first pictured obtained, and so forth.

(a) Show that X_k has a geometric distribution with $p = (n-k+1)/n$.

(b) Simulate the experiment for a team with 26 players (25 would be more accurate but we want an even number). Carry out a number of simulations and estimate the expected time required to get the first 13 players and the expected time to get the second 13. How do these expectations compare?

(c) Show that, if there are $2n$ players, the expected time to get the first half of the players is

$$2n\left(\frac{1}{2n} + \frac{1}{2n-1} + \cdots + \frac{1}{n+1}\right) ,$$

and the expected time to get the second half is

$$2n\left(\frac{1}{n} + \frac{1}{n-1} + \cdots + 1\right) .$$

(d) In Example 6.11 we stated that

$$1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} \sim \log n + .5772 + \frac{1}{2n} .$$

Use this to estimate the expression in (c). Compare these estimates with the exact values and also with your estimates obtained by simulation for the case $n = 26$.

***31** (Feller[14]) A large number, N, of people are subjected to a blood test. This can be administered in two ways: (1) Each person can be tested separately, in this case N test are required, (2) the blood samples of k persons can be pooled and analyzed together. If this test is *negative,* this one test suffices for the k people. If the test is *positive,* each of the k persons must be tested

[13]E. Thorp, *Beat the Dealer* (New York: Random House, 1962).

[14]W. Feller, *Introduction to Probability Theory and Its Applications,* 3rd ed., vol. 1 (New York: John Wiley and Sons, 1968), p. 240.

separately, and in all, $k+1$ tests are required for the k people. Assume that the probability p that a test is positive is the same for all people and that these events are independent.

(a) Find the probability that the test for a pooled sample of k people will be positive.

(b) What is the expected value of the number X of tests necessary under plan (2)? (Assume that N is divisible by k.)

(c) For small p, show that the value of k which will minimize the expected number of tests under the second plan is approximately $1/\sqrt{p}$.

32 Write a program to add random numbers chosen from $[0,1]$ until the first time the sum is greater than one. Have your program repeat this experiment a number of times to estimate the expected number of selections necessary in order that the sum of the chosen numbers first exceeds 1. On the basis of your experiments, what is your estimate for this number?

***33** The following related discrete problem also gives a good clue for the answer to Exercise 32. Randomly select with replacement t_1, t_2, ..., t_r from the set $(1/n, 2/n, \ldots, n/n)$. Let X be the smallest value of r satisfying

$$t_1 + t_2 + \cdots + t_r > 1 \ .$$

Then $E(X) = (1+1/n)^n$. To prove this, we can just as well choose t_1, t_2, ..., t_r randomly with replacement from the set $(1, 2, \ldots, n)$ and let X be the smallest value of r for which

$$t_1 + t_2 + \cdots + t_r > n \ .$$

(a) Use Exercise 3.2.36 to show that

$$P(X \geq j+1) = \binom{n}{j}\left(\frac{1}{n}\right)^j \ .$$

(b) Show that

$$E(X) = \sum_{j=0}^{n} P(X \geq j+1) \ .$$

(c) From these two facts, find an expression for $E(X)$. This proof is due to Harris Schultz.[15]

***34** (Banach's Matchbox[16]) A man carries in each of his two front pockets a box of matches originally containing N matches. Whenever he needs a match, he chooses a pocket at random and removes one from that box. One day he reaches into a pocket and finds the box empty.

[15] H. Schultz, "An Expected Value Problem," *Two-Year Mathematics Journal,* vol. 10, no. 4 (1979), pp. 277–78.

[16] W. Feller, *Introduction to Probability Theory,* vol. 1, p. 166.

(a) Let p_r denote the probability that the other pocket contains r matches. Define a sequence of *counter* random variables as follows: Let $X_i = 1$ if the ith draw is from the left pocket, and 0 if it is from the right pocket. Interpret p_r in terms of $S_n = X_1 + X_2 + \cdots + X_n$. Find a binomial expression for p_r.

(b) Write a computer program to compute the p_r, as well as the probability that the other pocket contains at least r matches, for $N = 100$ and r from 0 to 50.

(c) Show that $(N - r)p_r = (1/2)(2N + 1)p_{r+1} - (1/2)(r + 1)p_{r+1}$.

(d) Evaluate $\sum_r p_r$.

(e) Use (c) and (d) to determine the expectation E of the distribution $\{p_r\}$.

(f) Use Stirling's formula to obtain an approximation for E. How many matches must each box contain to ensure a value of about 13 for the expectation E? (Take $\pi = 22/7$.)

35 A coin is tossed until the first time a head turns up. If this occurs on the nth toss and n is odd you win $2^n/n$, but if n is even then you lose $2^n/n$. Then if your expected winnings exist they are given by the convergent series

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$$

called the alternating *harmonic series.* It is tempting to say that this should be the expected value of the experiment. Show that if we were to do this, the expected value of an experiment would depend upon the order in which the outcomes are listed.

36 Suppose we have an urn containing c yellow balls and d green balls. We draw k balls, without replacement, from the urn. Find the expected number of yellow balls drawn. *Hint*: Write the number of yellow balls drawn as the sum of c random variables.

37 The reader is referred to Example 6.13 for an explanation of the various options available in Monte Carlo roulette.

(a) Compute the expected winnings of a 1 franc bet on red under option (a).

(b) Repeat part (a) for option (b).

(c) Compare the expected winnings for all three options.

***38** (from Pittel[17]) Telephone books, n in number, are kept in a stack. The probability that the book numbered i (where $1 \leq i \leq n$) is consulted for a given phone call is $p_i > 0$, where the p_i's sum to 1. After a book is used, it is placed at the top of the stack. Assume that the calls are independent and evenly spaced, and that the system has been employed indefinitely far into the past. Let d_i be the average depth of book i in the stack. Show that

[17]B. Pittel, Problem #1195, *Mathematics Magazine,* vol. 58, no. 3 (May 1985), pg. 183.

$d_i \leq d_j$ whenever $p_i \geq p_j$. Thus, on the average, the more popular books have a tendency to be closer to the top of the stack. *Hint*: Let p_{ij} denote the probability that book i is above book j. Show that $p_{ij} = p_{ij}(1 - p_j) + p_{ji}p_i$.

***39** (from Propp[18]) In the previous problem, let P be the probability that at the present time, each book is in its proper place, i.e., book i is ith from the top. Find a formula for P in terms of the p_i's. In addition, find the least upper bound on P, if the p_i's are allowed to vary. *Hint*: First find the probability that book 1 is in the right place. Then find the probability that book 2 is in the right place, given that book 1 is in the right place. Continue.

***40** (from H. Shultz and B. Leonard[19]) A sequence of random numbers in $[0, 1)$ is generated until the sequence is no longer monotone increasing. The numbers are chosen according to the uniform distribution. What is the expected length of the sequence? (In calculating the length, the term that destroys monotonicity is included.) *Hint*: Let $a_1,\ a_2,\ \ldots$ be the sequence and let X denote the length of the sequence. Then

$$P(X > k) = P(a_1 < a_2 < \cdots < a_k) \ ,$$

and the probability on the right-hand side is easy to calculate. Furthermore, one can show that

$$E(X) = 1 + P(X > 1) + P(X > 2) + \cdots \ .$$

41 Let T be the random variable that counts the number of 2-unshuffles performed on an n-card deck until all of the labels on the cards are distinct. This random variable was discussed in Section 3.3. Using Equation 3.4 in that section, together with the formula

$$E(T) = \sum_{s=0}^{\infty} P(T > s)$$

that was proved in Exercise 33, show that

$$E(T) = \sum_{s=0}^{\infty} \left(1 - \binom{2^s}{n} \frac{n!}{2^{sn}} \right) \ .$$

Show that for $n = 52$, this expression is approximately equal to 11.7. (As was stated in Chapter 3, this means that on the average, almost 12 riffle shuffles of a 52-card deck are required in order for the process to be considered random.)

[18] J. Propp, Problem #1159, *Mathematics Magazine* vol. 57, no. 1 (Feb. 1984), pg. 50.

[19] H. Shultz and B. Leonard, "Unexpected Occurrences of the Number e," *Mathematics Magazine* vol. 62, no. 4 (October, 1989), pp. 269-271.

6.2 Variance of Discrete Random Variables

The usefulness of the expected value as a prediction for the outcome of an experiment is increased when the outcome is not likely to deviate too much from the expected value. In this section we shall introduce a measure of this deviation, called the variance.

Variance

Definition 6.3 Let X be a numerically valued random variable with expected value $\mu = E(X)$. Then the *variance* of X, denoted by $V(X)$, is

$$V(X) = E((X - \mu)^2) \ .$$

□

Note that, by Theorem 6.1, $V(X)$ is given by

$$V(X) = \sum_x (x - \mu)^2 m(x) \ , \tag{6.1}$$

where m is the distribution function of X.

Standard Deviation

The *standard deviation* of X, denoted by $D(X)$, is $D(X) = \sqrt{V(X)}$. We often write σ for $D(X)$ and σ^2 for $V(X)$.

Example 6.17 Consider one roll of a die. Let X be the number that turns up. To find $V(X)$, we must first find the expected value of X. This is

$$\begin{aligned} \mu &= E(X) = 1\Big(\frac{1}{6}\Big) + 2\Big(\frac{1}{6}\Big) + 3\Big(\frac{1}{6}\Big) + 4\Big(\frac{1}{6}\Big) + 5\Big(\frac{1}{6}\Big) + 6\Big(\frac{1}{6}\Big) \\ &= \frac{7}{2} \ . \end{aligned}$$

To find the variance of X, we form the new random variable $(X - \mu)^2$ and compute its expectation. We can easily do this using the following table.

From this table we find $E((X - \mu)^2)$ is

$$\begin{aligned} V(X) &= \frac{1}{6}\left(\frac{25}{4} + \frac{9}{4} + \frac{1}{4} + \frac{1}{4} + \frac{9}{4} + \frac{25}{4}\right) \\ &= \frac{35}{12} \ , \end{aligned}$$

and the standard deviation $D(X) = \sqrt{35/12} \approx 1.707$. □

x	$m(x)$	$(x-7/2)^2$
1	1/6	25/4
2	1/6	9/4
3	1/6	1/4
4	1/6	1/4
5	1/6	9/4
6	1/6	25/4

Table 6.6: Variance calculation.

Calculation of Variance

We next prove a theorem that gives us a useful alternative form for computing the variance.

Theorem 6.6 If X is any random variable with $E(X) = \mu$, then

$$V(X) = E(X^2) - \mu^2 \ .$$

Proof. We have

$$\begin{aligned} V(X) &= E((X-\mu)^2) = E(X^2 - 2\mu X + \mu^2) \\ &= E(X^2) - 2\mu E(X) + \mu^2 = E(X^2) - \mu^2 \ . \end{aligned}$$

□

Using Theorem 6.6, we can compute the variance of the outcome of a roll of a die by first computing

$$\begin{aligned} E(X^2) &= 1\Big(\frac{1}{6}\Big) + 4\Big(\frac{1}{6}\Big) + 9\Big(\frac{1}{6}\Big) + 16\Big(\frac{1}{6}\Big) + 25\Big(\frac{1}{6}\Big) + 36\Big(\frac{1}{6}\Big) \\ &= \frac{91}{6} \ , \end{aligned}$$

and,

$$V(X) = E(X^2) - \mu^2 = \frac{91}{6} - \Big(\frac{7}{2}\Big)^2 = \frac{35}{12} \ ,$$

in agreement with the value obtained directly from the definition of $V(X)$.

Properties of Variance

The variance has properties very different from those of the expectation. If c is any constant, $E(cX) = cE(X)$ and $E(X + c) = E(X) + c$. These two statements imply that the expectation is a linear function. However, the variance is not linear, as seen in the next theorem.

Theorem 6.7 If X is any random variable and c is any constant, then

$$V(cX) = c^2 V(X)$$

and

$$V(X + c) = V(X) \ .$$

Proof. Let $\mu = E(X)$. Then $E(cX) = c\mu$, and

$$\begin{aligned} V(cX) &= E((cX - c\mu)^2) = E(c^2(X - \mu)^2) \\ &= c^2 E((X - \mu)^2) = c^2 V(X) \ . \end{aligned}$$

To prove the second assertion, we note that, to compute $V(X + c)$, we would replace x by $x+c$ and μ by $\mu+c$ in Equation 6.1. Then the c's would cancel, leaving $V(X)$. □

We turn now to some general properties of the variance. Recall that if X and Y are any two random variables, $E(X+Y) = E(X)+E(Y)$. This is not always true for the case of the variance. For example, let X be a random variable with $V(X) \neq 0$, and define $Y = -X$. Then $V(X) = V(Y)$, so that $V(X) + V(Y) = 2V(X)$. But $X + Y$ is always 0 and hence has variance 0. Thus $V(X + Y) \neq V(X) + V(Y)$.

In the important case of mutually independent random variables, however, *the variance of the sum is the sum of the variances.*

Theorem 6.8 Let X and Y be two *independent* random variables. Then

$$V(X + Y) = V(X) + V(Y) \ .$$

Proof. Let $E(X) = a$ and $E(Y) = b$. Then

$$\begin{aligned} V(X + Y) &= E((X + Y)^2) - (a + b)^2 \\ &= E(X^2) + 2E(XY) + E(Y^2) - a^2 - 2ab - b^2 \ . \end{aligned}$$

Since X and Y are independent, $E(XY) = E(X)E(Y) = ab$. Thus,

$$V(X + Y) = E(X^2) - a^2 + E(Y^2) - b^2 = V(X) + V(Y) \ .$$

□

It is easy to extend this proof, by mathematical induction, to show that *the variance of the sum of any number of mutually independent random variables is the sum of the individual variances.* Thus we have the following theorem.

Theorem 6.9 Let $X_1, X_2, \ldots, X_n$ be an independent trials process with $E(X_j) = \mu$ and $V(X_j) = \sigma^2$. Let

$$S_n = X_1 + X_2 + \cdots + X_n$$

be the sum, and

$$A_n = \frac{S_n}{n}$$

be the average. Then

$$\begin{aligned} E(S_n) &= n\mu \ , \\ V(S_n) &= n\sigma^2 \ , \\ \sigma(S_n) &= \sigma\sqrt{n} \ , \\ E(A_n) &= \mu \ , \\ V(A_n) &= \frac{\sigma^2}{n}, \\ \sigma(A_n) &= \frac{\sigma}{\sqrt{n}} \ . \end{aligned}$$

Proof. Since all the random variables X_j have the same expected value, we have

$$E(S_n) = E(X_1) + \cdots + E(X_n) = n\mu \ ,$$

$$V(S_n) = V(X_1) + \cdots + V(X_n) = n\sigma^2 \ ,$$

and

$$\sigma(S_n) = \sigma\sqrt{n} \ .$$

We have seen that, if we multiply a random variable X with mean μ and variance σ^2 by a constant c, the new random variable has expected value $c\mu$ and variance $c^2\sigma^2$. Thus,

$$E(A_n) = E\left(\frac{S_n}{n}\right) = \frac{n\mu}{n} = \mu \ ,$$

and

$$V(A_n) = V\left(\frac{S_n}{n}\right) = \frac{V(S_n)}{n^2} = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n} \ .$$

Finally, the standard deviation of A_n is given by

$$\sigma(A_n) = \frac{\sigma}{\sqrt{n}} \ .$$

□

The last equation in the above theorem implies that in an independent trials process, if the individual summands have finite variance, then the standard deviation of the average goes to 0 as $n \to \infty$. Since the standard deviation tells us something about the spread of the distribution around the mean, we see that for large values of n, the value of A_n is usually very close to the mean of A_n, which equals μ, as shown above. This statement is made precise in Chapter 8, where it

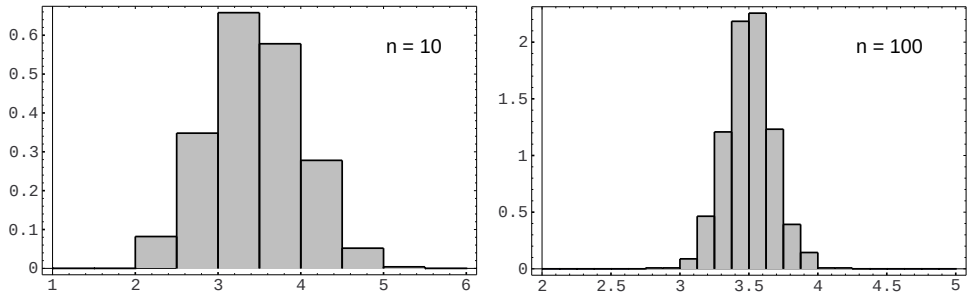


Figure 6.7: Empirical distribution of A_n.

is called the Law of Large Numbers. For example, let X represent the roll of a fair die. In Figure 6.7, we show the distribution of a random variable A_n corresponding to X, for $n = 10$ and $n = 100$.

Example 6.18 Consider n rolls of a die. We have seen that, if X_j is the outcome if the jth roll, then $E(X_j) = 7/2$ and $V(X_j) = 35/12$. Thus, if S_n is the sum of the outcomes, and $A_n = S_n/n$ is the average of the outcomes, we have $E(A_n) = 7/2$ and $V(A_n) = (35/12)/n$. Therefore, as n increases, the expected value of the average remains constant, but the variance tends to 0. If the variance is a measure of the expected deviation from the mean this would indicate that, for large n, we can expect the average to be very near the expected value. This is in fact the case, and we shall justify it in Chapter 8. □

Bernoulli Trials

Consider next the general Bernoulli trials process. As usual, we let $X_j = 1$ if the jth outcome is a success and 0 if it is a failure. If p is the probability of a success, and $q = 1 - p$, then

$$\begin{aligned} E(X_j) &= 0q + 1p = p \ , \\ E(X_j^2) &= 0^2q + 1^2p = p \ , \end{aligned}$$

and

$$V(X_j) = E(X_j^2) - (E(X_j))^2 = p - p^2 = pq \ .$$

Thus, for Bernoulli trials, if $S_n = X_1 + X_2 + \cdots + X_n$ is the number of successes, then $E(S_n) = np$, $V(S_n) = npq$, and $D(S_n) = \sqrt{npq}$. If $A_n = S_n/n$ is the average number of successes, then $E(A_n) = p$, $V(A_n) = pq/n$, and $D(A_n) = \sqrt{pq/n}$. We see that the expected proportion of successes remains p and the variance tends to 0. This suggests that the frequency interpretation of probability is a correct one. We shall make this more precise in Chapter 8.

Example 6.19 Let T denote the number of trials until the first success in a Bernoulli trials process. Then T is geometrically distributed. What is the variance of T? In Example 4.15, we saw that

$$m_T = \begin{pmatrix} 1 & 2 & 3 & \cdots \\ p & qp & q^2p & \cdots \end{pmatrix}.$$

In Example 6.4, we showed that

$$E(T) = 1/p \ .$$

Thus,

$$V(T) = E(T^2) - 1/p^2 \ ,$$

so we need only find

$$\begin{aligned} E(T^2) &= 1p + 4qp + 9q^2p + \cdots \\ &= p(1 + 4q + 9q^2 + \cdots) \ . \end{aligned}$$

To evaluate this sum, we start again with

$$1 + x + x^2 + \cdots = \frac{1}{1-x} \ .$$

Differentiating, we obtain

$$1 + 2x + 3x^2 + \cdots = \frac{1}{(1-x)^2} \ .$$

Multiplying by x,

$$x + 2x^2 + 3x^3 + \cdots = \frac{x}{(1-x)^2} \ .$$

Differentiating again gives

$$1 + 4x + 9x^2 + \cdots = \frac{1+x}{(1-x)^3} \ .$$

Thus,

$$E(T^2) = p\frac{1+q}{(1-q)^3} = \frac{1+q}{p^2}$$

and

$$\begin{aligned} V(T) &= E(T^2) - (E(T))^2 \\ &= \frac{1+q}{p^2} - \frac{1}{p^2} = \frac{q}{p^2} \ . \end{aligned}$$

For example, the variance for the number of tosses of a coin until the first head turns up is $(1/2)/(1/2)^2 = 2$. The variance for the number of rolls of a die until the first six turns up is $(5/6)/(1/6)^2 = 30$. Note that, as p decreases, the variance increases rapidly. This corresponds to the increased spread of the geometric distribution as p decreases (noted in Figure 5.1). □

Poisson Distribution

Just as in the case of expected values, it is easy to guess the variance of the Poisson distribution with parameter λ. We recall that the variance of a binomial distribution with parameters n and p equals npq. We also recall that the Poisson distribution could be obtained as a limit of binomial distributions, if n goes to ∞ and p goes to 0 in such a way that their product is kept fixed at the value λ. In this case, $npq = \lambda q$ approaches λ, since q goes to 1. So, given a Poisson distribution with parameter λ, we should guess that its variance is λ. The reader is asked to show this in Exercise 29.

Exercises

1 A number is chosen at random from the set $S = \{-1, 0, 1\}$. Let X be the number chosen. Find the expected value, variance, and standard deviation of X.

2 A random variable X has the distribution

$$p_X = \begin{pmatrix} 0 & 1 & 2 & 4 \\ 1/3 & 1/3 & 1/6 & 1/6 \end{pmatrix} .$$

Find the expected value, variance, and standard deviation of X.

3 You place a 1-dollar bet on the number 17 at Las Vegas, and your friend places a 1-dollar bet on black (see Exercises 1.1.6 and 1.1.7). Let X be your winnings and Y be her winnings. Compare $E(X)$, $E(Y)$, and $V(X)$, $V(Y)$. What do these computations tell you about the nature of your winnings if you and your friend make a sequence of bets, with you betting each time on a number and your friend betting on a color?

4 X is a random variable with $E(X) = 100$ and $V(X) = 15$. Find

(a) $E(X^2)$.

(b) $E(3X + 10)$.

(c) $E(-X)$.

(d) $V(-X)$.

(e) $D(-X)$.

5 In a certain manufacturing process, the (Fahrenheit) temperature never varies by more than 2° from 62°. The temperature is, in fact, a random variable F with distribution

$$P_F = \begin{pmatrix} 60 & 61 & 62 & 63 & 64 \\ 1/10 & 2/10 & 4/10 & 2/10 & 1/10 \end{pmatrix} .$$

(a) Find $E(F)$ and $V(F)$.

(b) Define $T = F - 62$. Find $E(T)$ and $V(T)$, and compare these answers with those in part (a).

(c) It is decided to report the temperature readings on a Celsius scale, that is, $C = (5/9)(F - 32)$. What is the expected value and variance for the readings now?

6 Write a computer program to calculate the mean and variance of a distribution which you specify as data. Use the program to compare the variances for the following densities, both having expected value 0:

$$p_X = \begin{pmatrix} -2 & -1 & 0 & 1 & 2 \\ 3/11 & 2/11 & 1/11 & 2/11 & 3/11 \end{pmatrix} ;$$

$$p_Y = \begin{pmatrix} -2 & -1 & 0 & 1 & 2 \\ 1/11 & 2/11 & 5/11 & 2/11 & 1/11 \end{pmatrix} .$$

7 A coin is tossed three times. Let X be the number of heads that turn up. Find $V(X)$ and $D(X)$.

8 A random sample of 2400 people are asked if they favor a government proposal to develop new nuclear power plants. If 40 percent of the people in the country are in favor of this proposal, find the expected value and the standard deviation for the number S_{2400} of people in the sample who favored the proposal.

9 A die is loaded so that the probability of a face coming up is proportional to the number on that face. The die is rolled with outcome X. Find $V(X)$ and $D(X)$.

10 Prove the following facts about the standard deviation.

(a) $D(X + c) = D(X)$.

(b) $D(cX) = |c|D(X)$.

11 A number is chosen at random from the integers 1, 2, 3, ..., n. Let X be the number chosen. Show that $E(X) = (n+1)/2$ and $V(X) = (n-1)(n+1)/12$. *Hint*: The following identity may be useful:

$$1^2 + 2^2 + \cdots + n^2 = \frac{(n)(n+1)(2n+1)}{6} .$$

12 Let X be a random variable with $\mu = E(X)$ and $\sigma^2 = V(X)$. Define $X^* = (X-\mu)/\sigma$. The random variable X^* is called the *standardized random variable* associated with X. Show that this standardized random variable has expected value 0 and variance 1.

13 Peter and Paul play Heads or Tails (see Example 1.4). Let W_n be Peter's winnings after n matches. Show that $E(W_n) = 0$ and $V(W_n) = n$.

14 Find the expected value and the variance for the number of boys and the number of girls in a royal family that has children until there is a boy or until there are three children, whichever comes first.

15 Suppose that n people have their hats returned at random. Let $X_i = 1$ if the ith person gets his or her own hat back and 0 otherwise. Let $S_n = \sum_{i=1}^{n} X_i$. Then S_n is the total number of people who get their own hats back. Show that

(a) $E(X_i^2) = 1/n$.

(b) $E(X_i \cdot X_j) = 1/n(n-1)$ for $i \neq j$.

(c) $E(S_n^2) = 2$ (using (a) and (b)).

(d) $V(S_n) = 1$.

16 Let S_n be the number of successes in n independent trials. Use the program **BinomialProbabilities** (Section 3.2) to compute, for given n, p, and j, the probability

$$P(-j\sqrt{npq} < S_n - np < j\sqrt{npq}) \ .$$

(a) Let $p = .5$, and compute this probability for $j = 1, 2, 3$ and $n = 10, 30, 50$. Do the same for $p = .2$.

(b) Show that the *standardized random variable* $S_n^* = (S_n - np)/\sqrt{npq}$ has expected value 0 and variance 1. What do your results from (a) tell you about this standardized quantity S_n^*?

17 Let X be the outcome of a chance experiment with $E(X) = \mu$ and $V(X) = \sigma^2$. When μ and σ^2 are unknown, the statistician often estimates them by repeating the experiment n times with outcomes x_1, x_2, ..., x_n, estimating μ by the *sample mean*

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \ ,$$

and σ^2 by the *sample variance*

$$s^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2 \ .$$

Then s is the *sample standard deviation.* These formulas should remind the reader of the definitions of the theoretical mean and variance. (Many statisticians define the sample variance with the coefficient $1/n$ replaced by $1/(n-1)$. If this alternative definition is used, the expected value of s^2 is equal to σ^2. See Exercise 18, part (d).)

Write a computer program that will roll a die n times and compute the sample mean and sample variance. Repeat this experiment several times for $n = 10$ and $n = 1000$. How well do the sample mean and sample variance estimate the true mean 7/2 and variance 35/12?

18 Show that, for the sample mean $\bar{x}$ and sample variance s^2 as defined in Exercise 17,

(a) $E(\bar{x}) = \mu$.

(b) $E\big((\bar{x}-\mu)^2\big) = \sigma^2/n$.

(c) $E(s^2) = \frac{n-1}{n}\sigma^2$. *Hint*: For (c) write

$$\begin{aligned}\sum_{i=1}^{n}(x_i-\bar{x})^2 &= \sum_{i=1}^{n}\big((x_i-\mu)-(\bar{x}-\mu)\big)^2 \\ &= \sum_{i=1}^{n}(x_i-\mu)^2 - 2(\bar{x}-\mu)\sum_{i=1}^{n}(x_i-\mu) + n(\bar{x}-\mu)^2 \\ &= \sum_{i=1}^{n}(x_i-\mu)^2 - n(\bar{x}-\mu)^2,\end{aligned}$$

and take expectations of both sides, using part (b) when necessary.

(d) Show that if, in the definition of s^2 in Exercise 17, we replace the coefficient $1/n$ by the coefficient $1/(n-1)$, then $E(s^2) = \sigma^2$. (This shows why many statisticians use the coefficient $1/(n-1)$. The number s^2 is used to estimate the unknown quantity σ^2. If an estimator has an average value which equals the quantity being estimated, then the estimator is said to be *unbiased*. Thus, the statement $E(s^2) = \sigma^2$ says that s^2 is an unbiased estimator of σ^2.)

19 Let X be a random variable taking on values $a_1, a_2, \ldots, a_r$ with probabilities $p_1, p_2, \ldots, p_r$ and with $E(X) = \mu$. Define the *spread* of X as follows:

$$\bar{\sigma} = \sum_{i=1}^{r}|a_i-\mu|p_i \ .$$

This, like the standard deviation, is a way to quantify the amount that a random variable is spread out around its mean. Recall that the variance of a sum of mutually independent random variables is the sum of the individual variances. The square of the spread corresponds to the variance in a manner similar to the correspondence between the spread and the standard deviation. Show by an example that it is not necessarily true that the square of the spread of the sum of two independent random variables is the sum of the squares of the individual spreads.

20 We have two instruments that measure the distance between two points. The measurements given by the two instruments are random variables X_1 and X_2 that are independent with $E(X_1) = E(X_2) = \mu$, where μ is the true distance. From experience with these instruments, we know the values of the variances σ_1^2 and σ_2^2. These variances are not necessarily the same. From two measurements, we estimate μ by the weighted average $\bar{\mu} = wX_1 + (1-w)X_2$. Here w is chosen in $[0,1]$ to minimize the variance of $\bar{\mu}$.

(a) What is $E(\bar{\mu})$?

(b) How should w be chosen in $[0,1]$ to minimize the variance of $\bar{\mu}$?

21 Let X be a random variable with $E(X) = \mu$ and $V(X) = \sigma^2$. Show that the function $f(x)$ defined by

$$f(x) = \sum_{\omega}(X(\omega) - x)^2 p(\omega)$$

has its minimum value when $x = \mu$.

22 Let X and Y be two random variables defined on the finite sample space Ω. Assume that X, Y, $X + Y$, and $X - Y$ all have the same distribution. Prove that $P(X = Y = 0) = 1$.

23 If X and Y are any two random variables, then the *covariance* of X and Y is defined by $\text{Cov}(X, Y) = E((X - E(X))(Y - E(Y)))$. Note that $\text{Cov}(X, X) = V(X)$. Show that, if X and Y are independent, then $\text{Cov}(X, Y) = 0$; and show, by an example, that we can have $\text{Cov}(X, Y) = 0$ and X and Y not independent.

***24** A professor wishes to make up a true-false exam with n questions. She assumes that she can design the problems in such a way that a student will answer the jth problem correctly with probability p_j, and that the answers to the various problems may be considered independent experiments. Let S_n be the number of problems that a student will get correct. The professor wishes to choose p_j so that $E(S_n) = .7n$ and so that the variance of S_n is as large as possible. Show that, to achieve this, she should choose $p_j = .7$ for all j; that is, she should make all the problems have the same difficulty.

25 (Lamperti[20]) An urn contains exactly 5000 balls, of which an unknown number X are white and the rest red, where X is a random variable with a probability distribution on the integers 0, 1, 2, ..., 5000.

(a) Suppose we know that $E(X) = \mu$. Show that this is enough to allow us to calculate the probability that a ball drawn at random from the urn will be white. What is this probability?

(b) We draw a ball from the urn, examine its color, replace it, and then draw another. Under what conditions, if any, are the results of the two drawings independent; that is, does

$$P(\text{white}, \text{white}) = P(\text{white})^2 \ ?$$

(c) Suppose the variance of X is σ^2. What is the probability of drawing two white balls in part (b)?

26 For a sequence of Bernoulli trials, let X_1 be the number of trials until the first success. For $j \geq 2$, let X_j be the number of trials after the $(j-1)$st success until the jth success. It can be shown that $X_1, X_2, \ldots$ is an independent trials process.

[20] Private communication.